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(12) **United States Patent**
Fung

(10) **Patent No.: US 12,399,564 B2**
(45) **Date of Patent: *Aug. 26, 2025**

(54) **TREATMENT OF NEUROLOGICAL FUNCTIONING AND CONDITIONS USING COMPUTER-ASSISTED DUAL-TASK METHODOLOGY**

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(72) Inventor: **Coleman Fung**, Austin, TX (US)

(73) Assignee: **BLUE GOJI LLC**, Austin, TX (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 5 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/506,761**

(22) Filed: **Nov. 10, 2023**

(65) **Prior Publication Data**

US 2024/0152209 A1 May 9, 2024

Related U.S. Application Data

(63) Continuation-in-part of application No. 18/188,388, filed on Mar. 22, 2023, now Pat. No. 11,791,026, and (Continued)

(51) **Int. Cl.**
G06F 3/01 (2006.01)
A61B 5/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC .... ***G06F 3/015*** (2013.01); ***A61B 5/165*** (2013.01); ***A61B 5/4082*** (2013.01); (Continued)

(58) **Field of Classification Search**
CPC .... G06F 3/015; G06F 1/163; G06F 3/011; G06F 3/016; A63F 13/212; A63F 13/214; (Continued)

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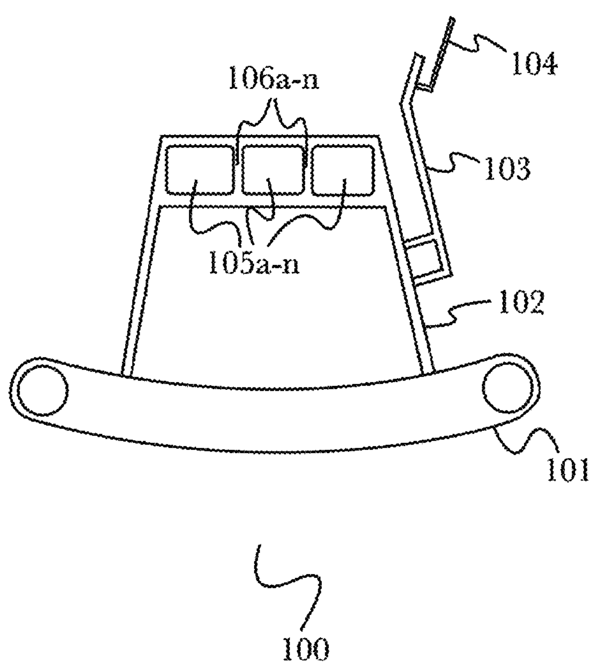
Primary Examiner — Phong H Dang

(74) *Attorney, Agent, or Firm* — Galvin Patent Law LLC; Brian R Galvin

(57) **ABSTRACT**

A system and method for targeted treatment of human brain function using a combination of physical and mental activity to cause neurogenesis and neuroplasticity in targeted regions of the brain using computer-enhanced dual-task analysis and treatment. The system and method involve having a subject engage in physical and mental activities at levels of intensity or stress associated with increased neurogenesis and neuroplasticity, the mental activities being associated with certain brain regions intended to be targeted by the treatment, wherein the physical and mental activities are assigned based on dual-task analyses of the subject's brain function and machine learning algorithms trained to optimize the stress levels associated with the physical and mental activities to maximize neurogenesis and neuroplasticity. The physical and mental activities may be continuously adjusted to ensure that the subject remains in appropriate ranges for neurogenesis and neuroplasticity for the brain region or regions being targeted for treatment.

18 Claims, 43 Drawing Sheets



Related U.S. Application Data

a continuation-in-part of application No. 18/171,330, filed on Feb. 18, 2023, now Pat. No. 11,914,776, said application No. 18/188,388 is a continuation of application No. 18/080,343, filed on Dec. 13, 2022, now Pat. No. 11,823,781, and a continuation of application No. 17/888,449, filed on Aug. 15, 2022, now Pat. No. 12,053,670, which is a continuation of application No. 17/575,600, filed on Jan. 13, 2022, now Pat. No. 11,465,013, said application No. 18/080,343 is a continuation-in-part of application No. 17/574,540, filed on Jan. 12, 2022, now Pat. No. 11,524,209, said application No. 17/575,600 is a continuation-in-part of application No. 16/951,281, filed on Nov. 18, 2020, now Pat. No. 11,123,604, said application No. 17/030,233 is a continuation-in-part of application No. 17/030,195, filed on Sep. 23, 2020, now abandoned, said application No. 16/951,281 is a continuation of application No. 17/030,195, filed on Sep. 23, 2020, now abandoned, said application No. 17/575,600 is a continuation-in-part of application No. 17/030,233, filed on Sep. 23, 2020, now Pat. No. 11,662,818, said application No. 18/171,330 is a continuation of application No. 17/030,233, filed on Sep. 23, 2020, now Pat. No. 11,662,818, which is a continuation-in-part of application No. 16/927,704, filed on Jul. 13, 2020, now abandoned, which is a continuation of application No. 16/867,238, filed on May 5, 2020, now abandoned, which is a continuation-in-part of application No. 16/793,915, filed on Feb. 18, 2020, now abandoned, said application No. 17/030,195 is a continuation-in-part of application No. 16/781,663, filed on Feb. 4, 2020, now Pat. No. 11,191,996, which is a continuation-in-part of application No. 16/354,374, filed on Mar. 15, 2019, now Pat. No. 10,549,153, said application No. 16/793,915 is a continuation-in-part of application No. 16/255,641, filed on Jan. 23, 2019, now Pat. No. 10,561,900, which is a continuation of application No. 16/223,034, filed on Jan. 23, 2019, now Pat. No. 10,688,341, said application No. 16/354,374 is a continuation-in-part of application No. 16/176,511, filed on Oct. 31, 2018, now Pat. No. 10,960,264, said application No. 16/223,034 is a continuation-in-part of application No. 16/176,511, filed on Oct. 31, 2018, now Pat. No. 10,960,264, which is a continuation-in-part of application No. 16/011,394, filed on Jun. 18, 2018, now Pat. No. 10,155,133, which is a continuation-in-part of application No. 15/853,746, filed on Dec. 23, 2017, now Pat. No. 10,265,578, which is a continuation of application No. 15/219,115, filed on Jul. 25, 2016, now Pat. No. 9,849,333, which is a continuation-in-part of application No. 15/193,112, filed on Jun. 27, 2016, now abandoned, which is a continuation-in-part of application No. 15/187,787, filed on Jun. 21, 2016, now Pat. No. 10,124,255, which is a continuation-in-part of application No. 15/175,043, filed on Jun. 7, 2016, now Pat. No. 9,766,696, said application No. 15/187,787 is a continuation-in-part of application No. 14/846,966, filed on Sep. 7, 2015, now Pat. No. 10,080,958, and a continuation-in-part of application No. 14/012,879, filed on Aug. 28, 2013, now Pat. No. 10,737,175.

(60) Provisional application No. 62/330,602, filed on May 2, 2016, provisional application No. 62/330,642, filed on May 2, 2016, provisional application No. 62/310,568, filed on Mar. 18, 2016, provisional application No. 61/696,068, filed on Aug. 31, 2012, provisional application No. 62/697,973, filed on Jul. 13, 2018.

(51) **Int. Cl.**

A61B 5/16	(2006.01)
A63B 22/00	(2006.01)
A63B 22/02	(2006.01)
A63B 22/06	(2006.01)
A63B 23/04	(2006.01)
A63F 13/212	(2014.01)
A63F 13/214	(2014.01)
A63F 13/40	(2014.01)
A63F 13/65	(2014.01)
G06F 1/16	(2006.01)
G06T 19/00	(2011.01)
A61B 5/11	(2006.01)
A63B 24/00	(2006.01)
G06F 3/03	(2006.01)
H04W 84/18	(2009.01)

(52) **U.S. Cl.**

CPC ***A61B 5/4088*** (2013.01); ***A61B 5/4842*** (2013.01); ***A63B 22/0046*** (2013.01); ***A63B 22/0285*** (2013.01); ***A63B 22/0292*** (2015.10); ***A63B 22/06*** (2013.01); ***A63B 23/04*** (2013.01); ***A63F 13/212*** (2014.09); ***A63F 13/214*** (2014.09); ***A63F 13/40*** (2014.09); ***A63F 13/65*** (2014.09); ***G06F 1/163*** (2013.01); ***G06F 3/011*** (2013.01); ***G06F 3/016*** (2013.01); ***G06T 19/006*** (2013.01); *A61B 5/1124* (2013.01); *A63B 24/0003* (2013.01); *G06F 3/0304* (2013.01); *G06F 2203/012* (2013.01); *H04W 84/18* (2013.01)

(58) **Field of Classification Search**

CPC A63F 13/40; A63F 13/65; A63B 22/0292; A63B 22/0046; A63B 22/0285; A63B 22/06; A63B 23/04; A61B 5/165; A61B 5/4082; A61B 5/4088; A61B 5/4842; G06T 19/006

See application file for complete search history.

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2020/0402643	A1	12/2020	Trees et al.	
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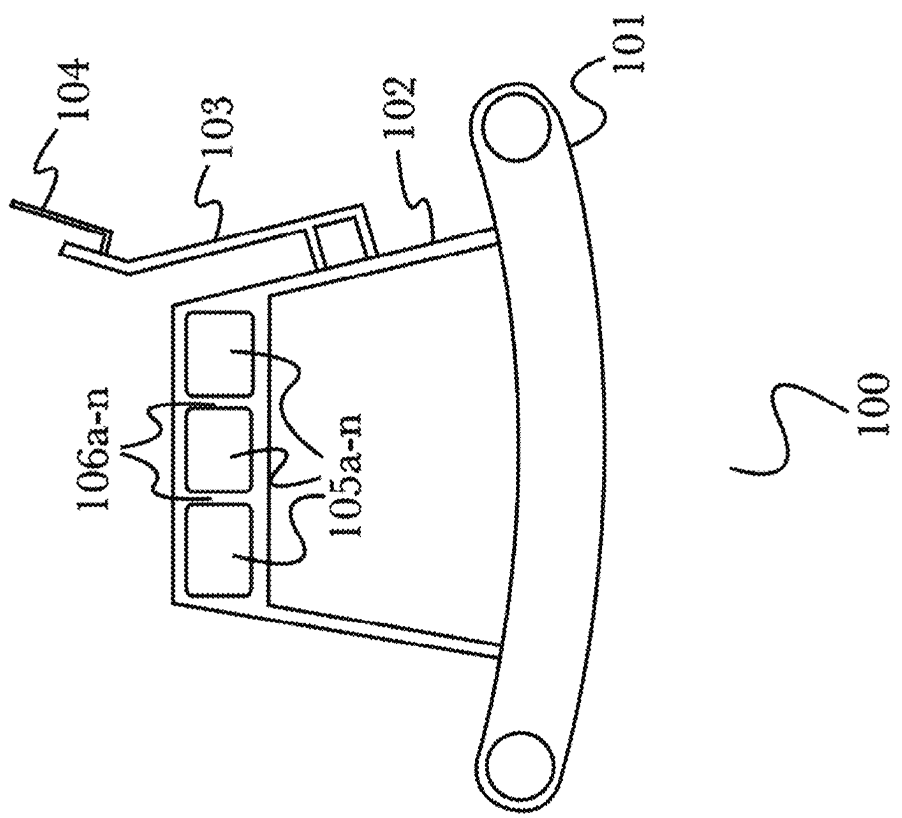


Fig. 1

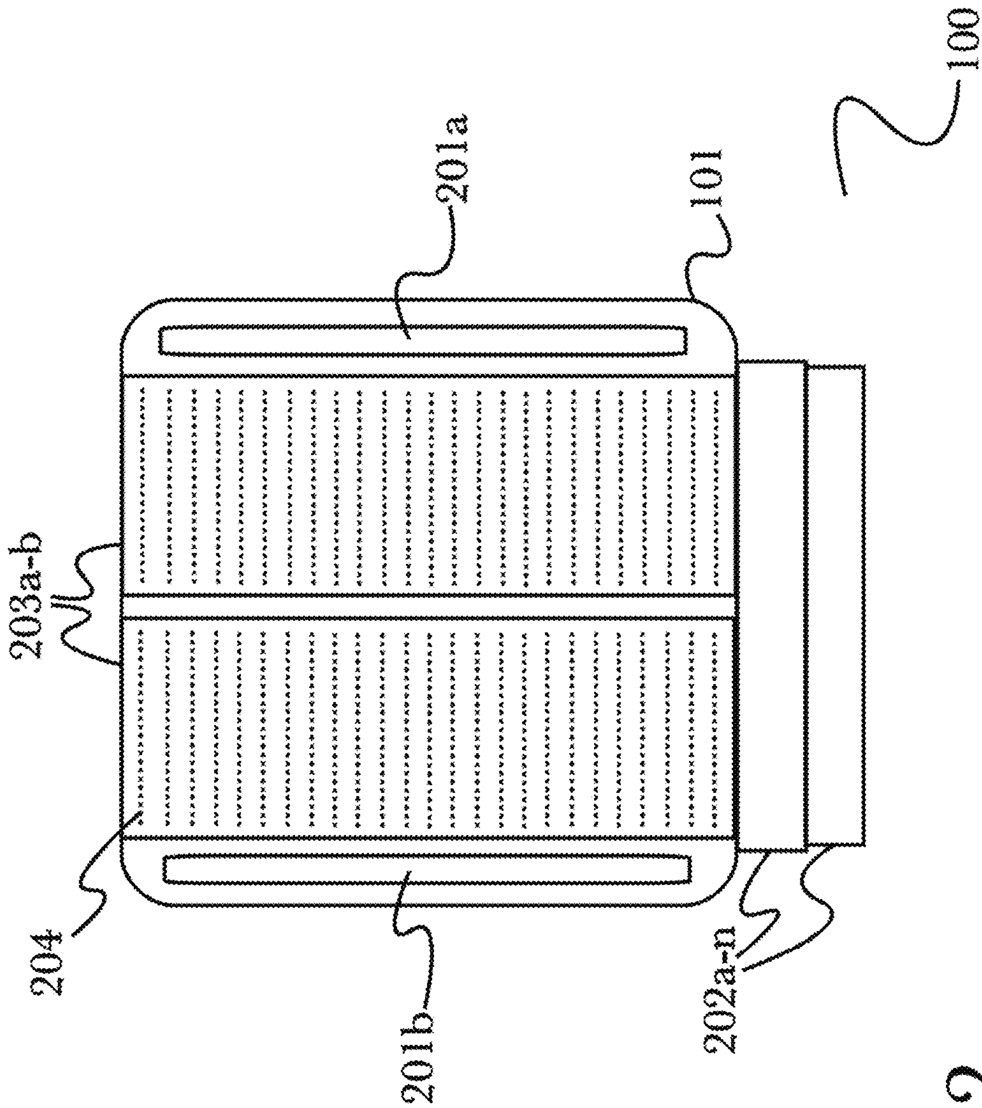


Fig. 2

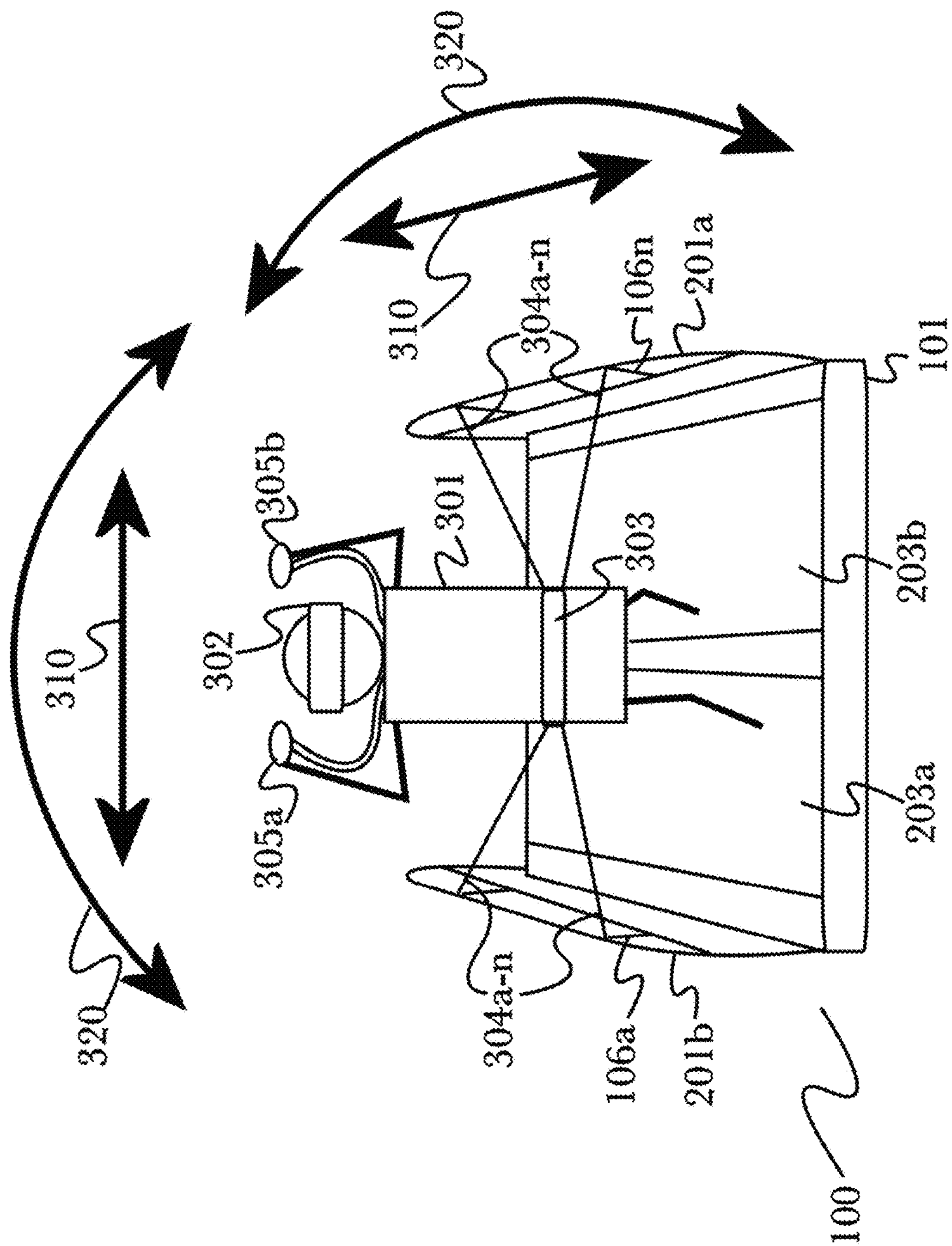

320
310
304a-n
106n
201a
101
305b
301
303
203b
302
310
305a
203a
320
304a-n
106a
201b
100

Fig. 3

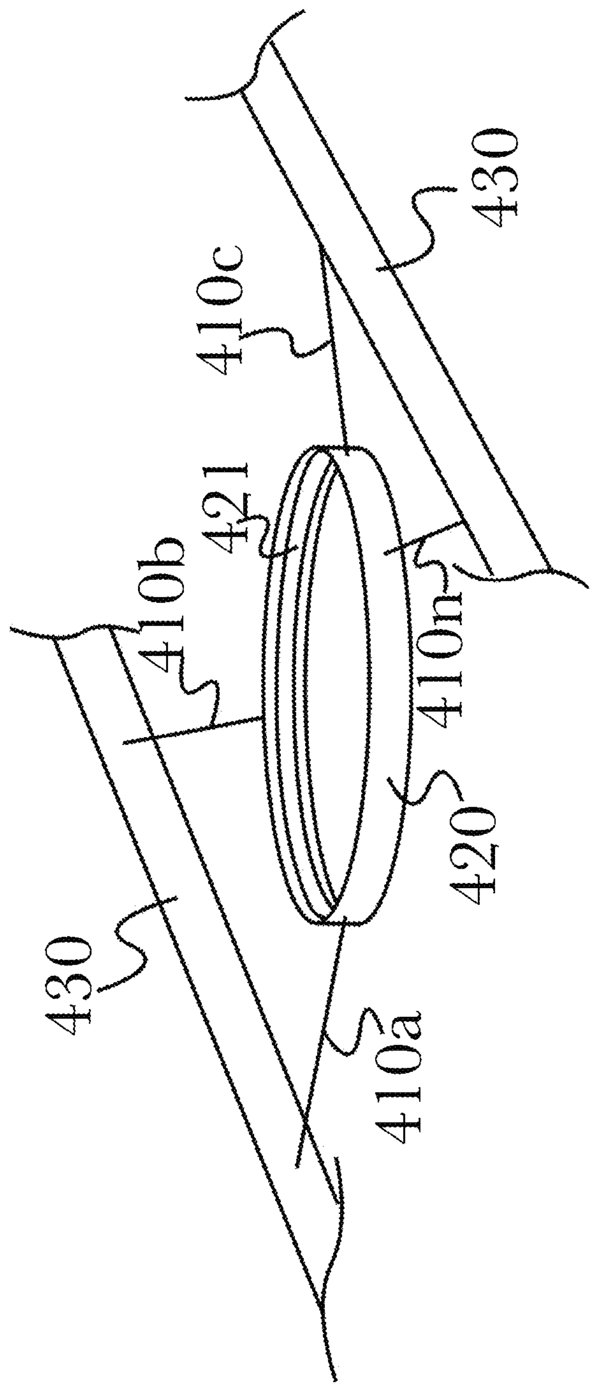


Fig. 4

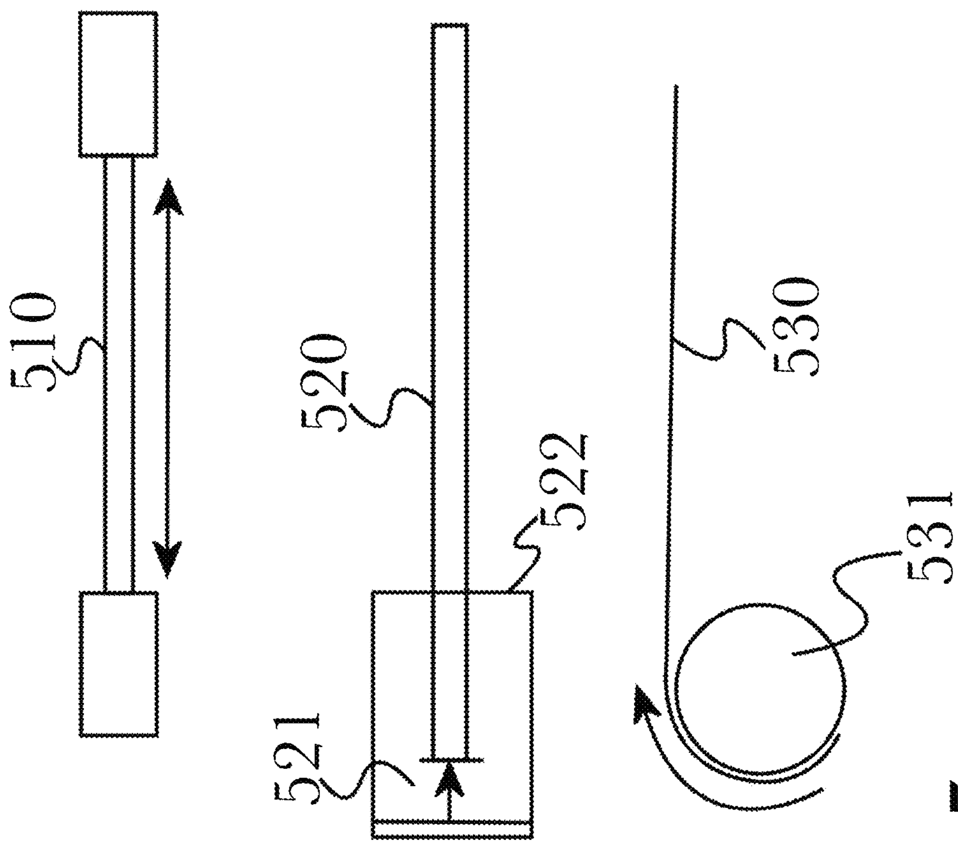


Fig. 5

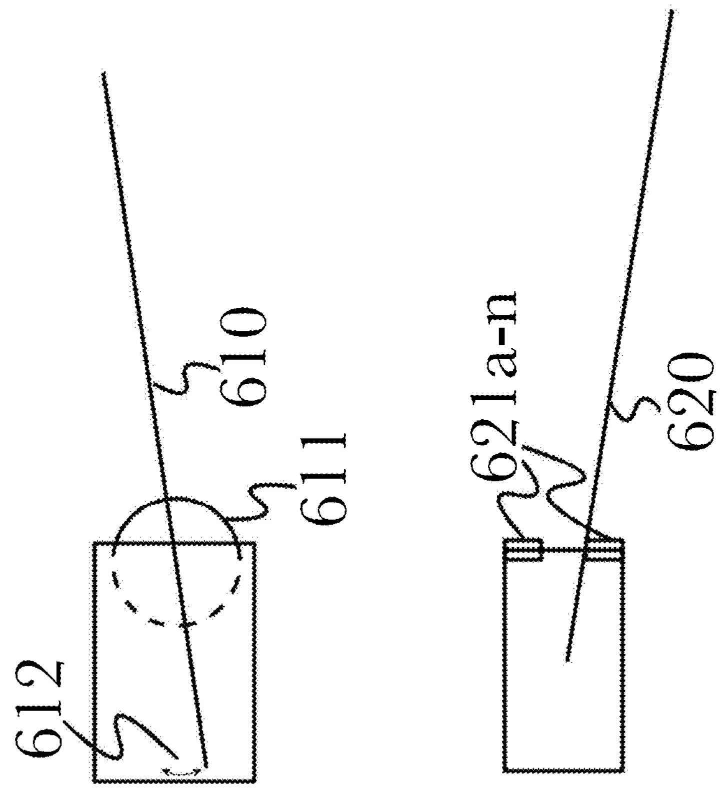


Fig. 6

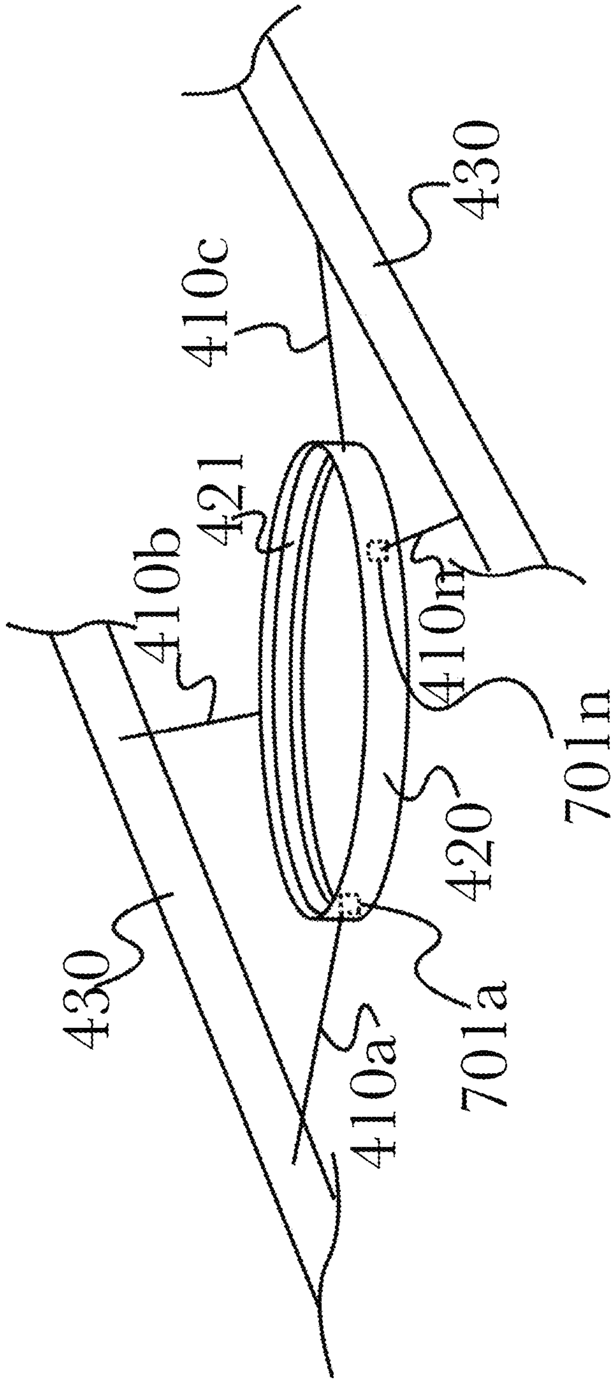


Fig. 7

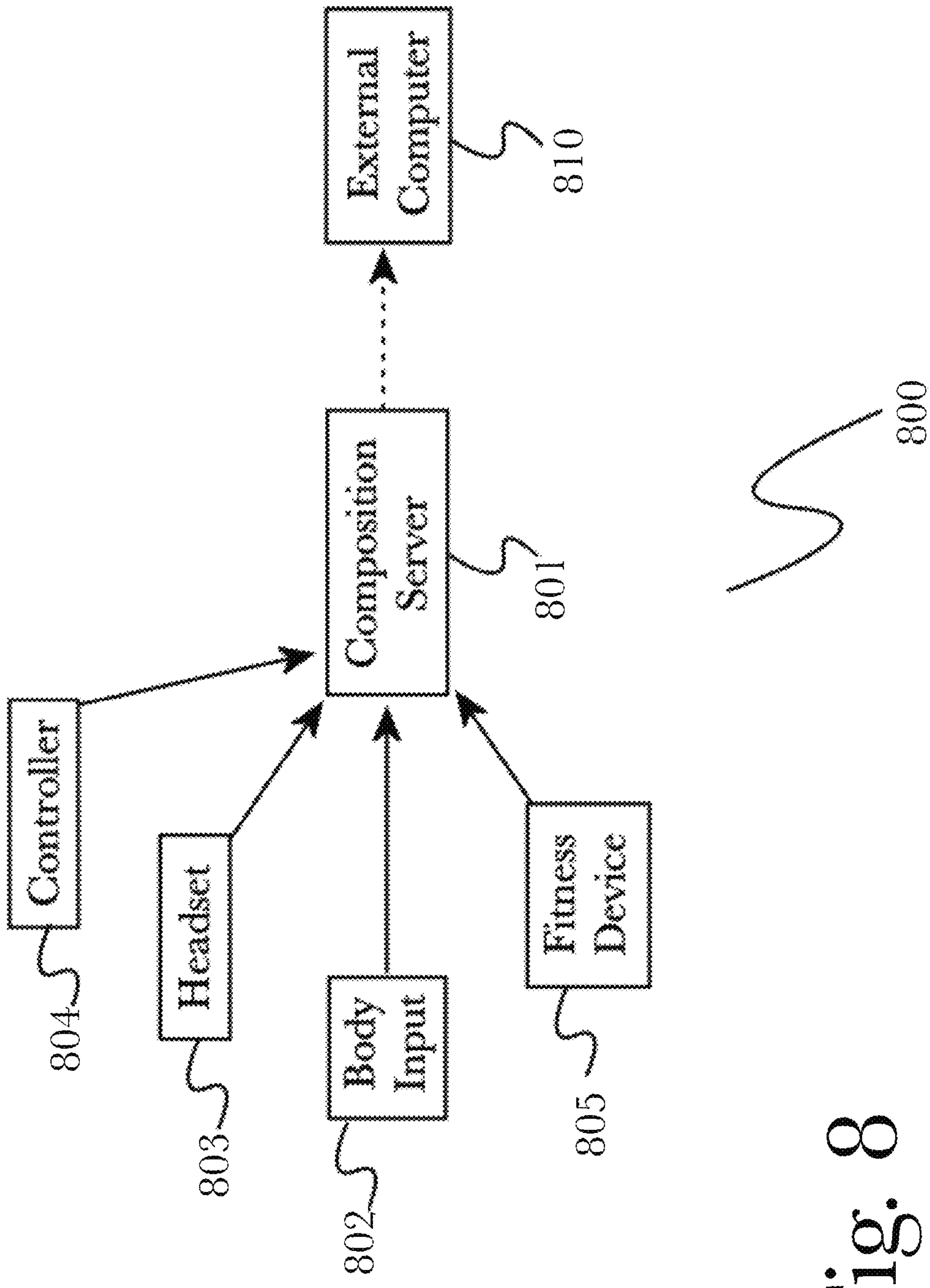

External Computer
810
Composition Server
801
800
Controller
804
Headset
803
Body Input
802
Fitness Device
805

Fig. 8

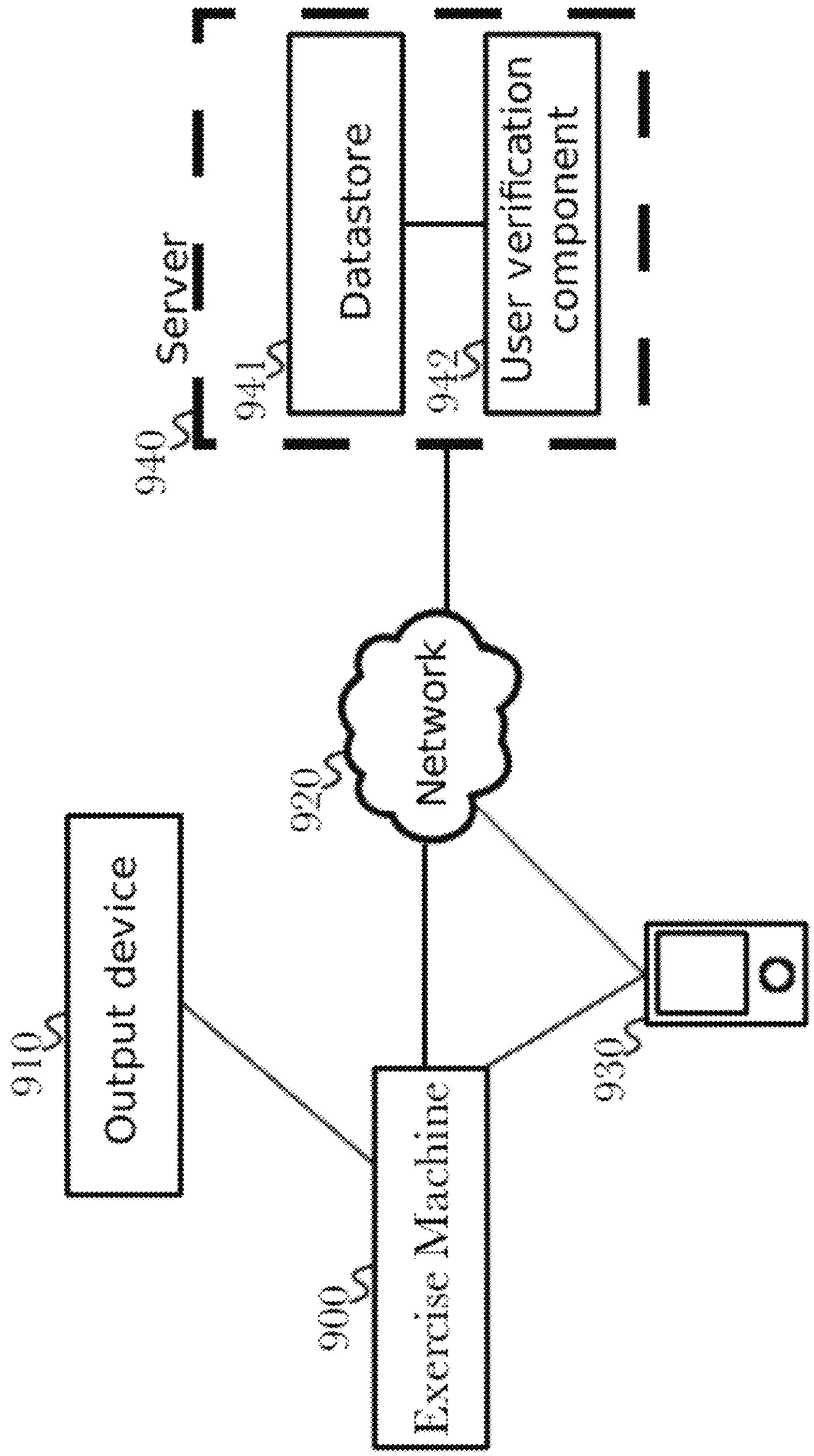

Server
940
Datastore
941
User verification component
942
Network
920
Output device
910
Exercise Machine
900
930

Fig. 9

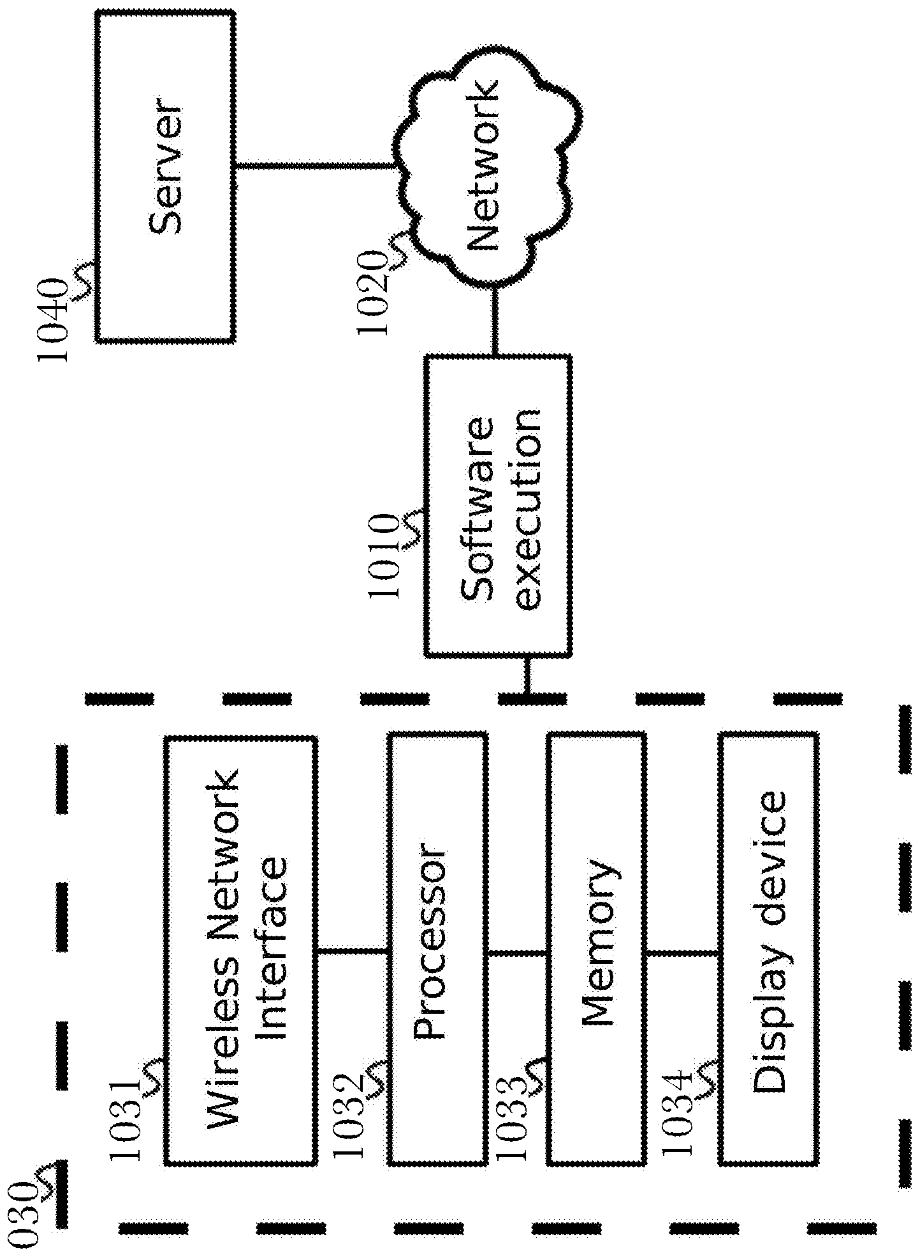

1040
Server
1020
Network
1010
Software execution
1030
1031
Wireless Network Interface
1032
Processor
1033
Memory
1034
Display device

Fig. 10

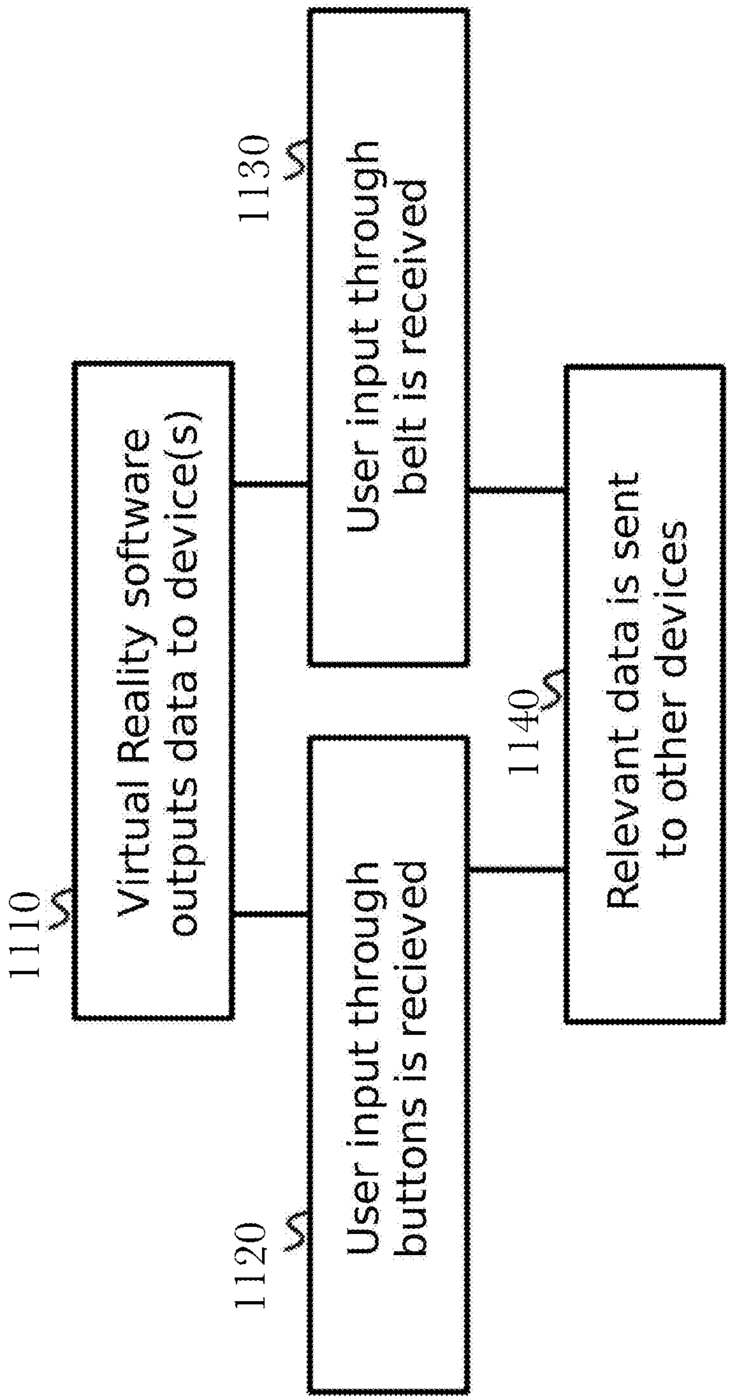

1110
Virtual Reality software outputs data to device(s)
1120
User input through buttons is recieved
1130
User input through belt is received
1140
Relevant data is sent to other devices

Fig. 11

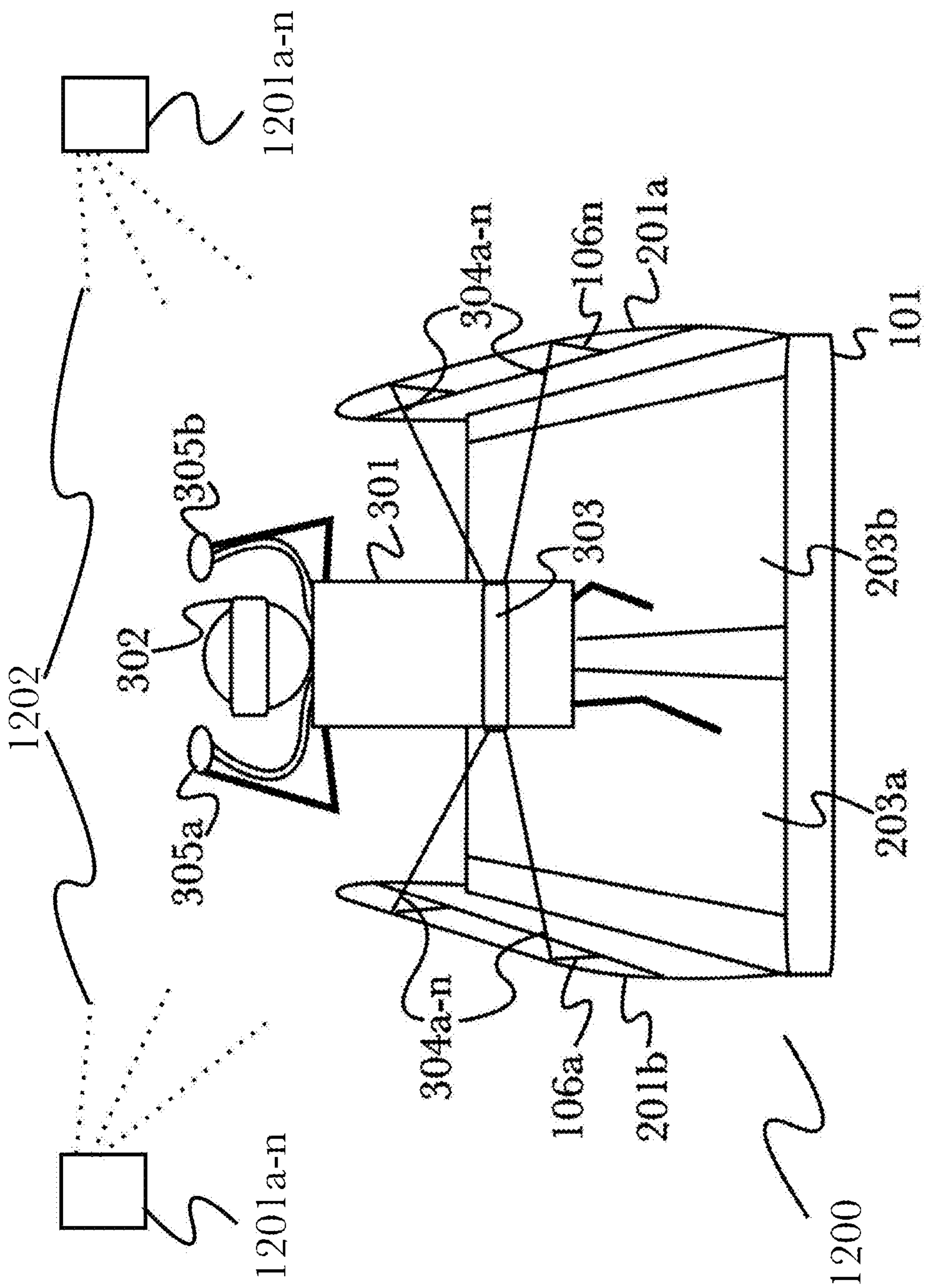

1201a-n
1202
305b
301
302
303
305a
304a-n
106n
201a
101
203b
203a
304a-n
106a
201b
1200
1201a-n

Fig. 12

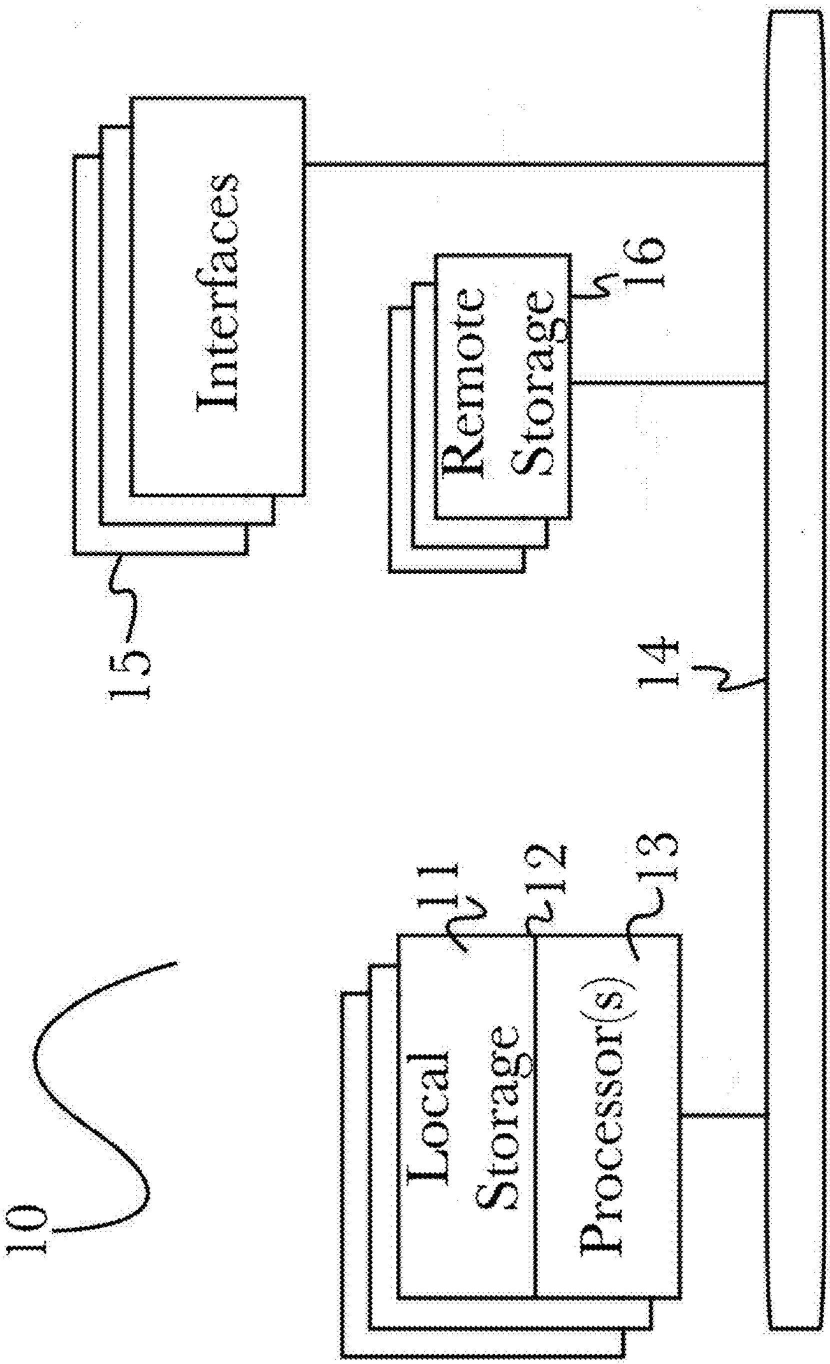
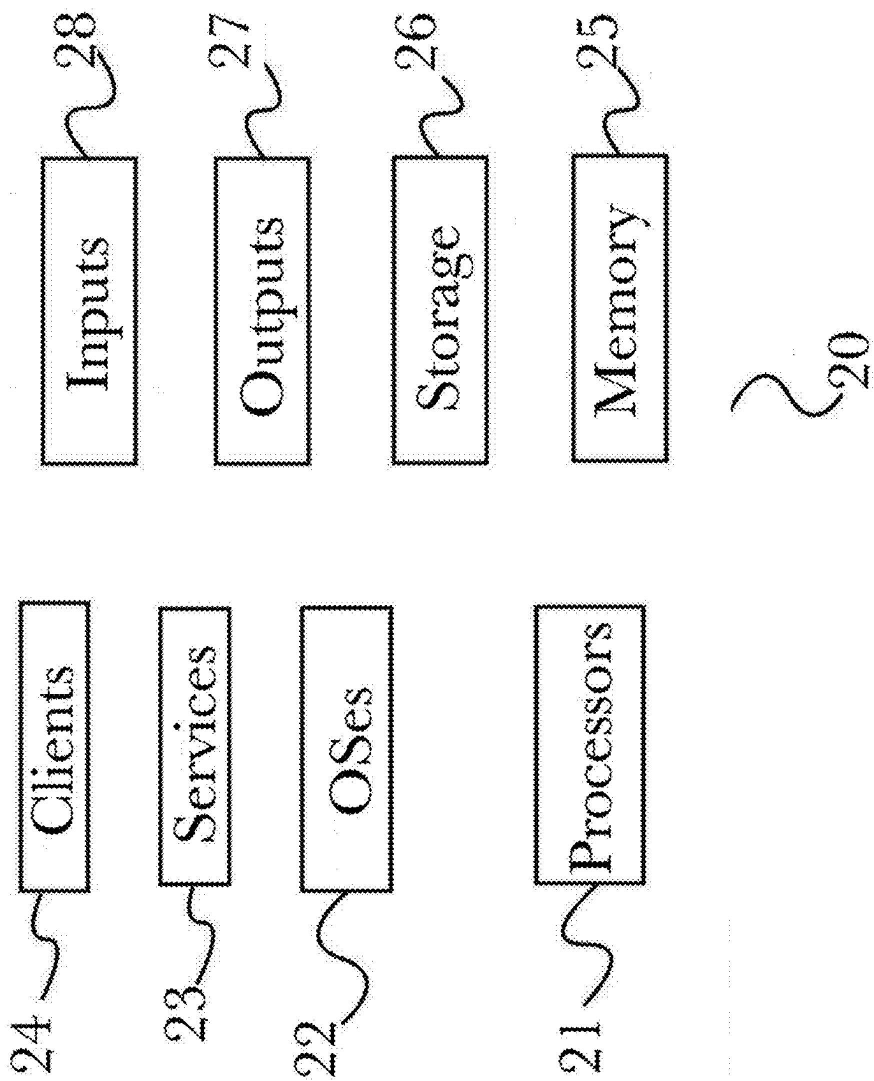

10
Local
Storage
Processor(s)
11
12
13
14
Remote
Storage
16
Interfaces
15

Fig. 13



Inputs
28
Outputs
27
Storage
26
Memory
25
20
Clients
24
Services
23
OSes
22
Processors
21

Fig. 14

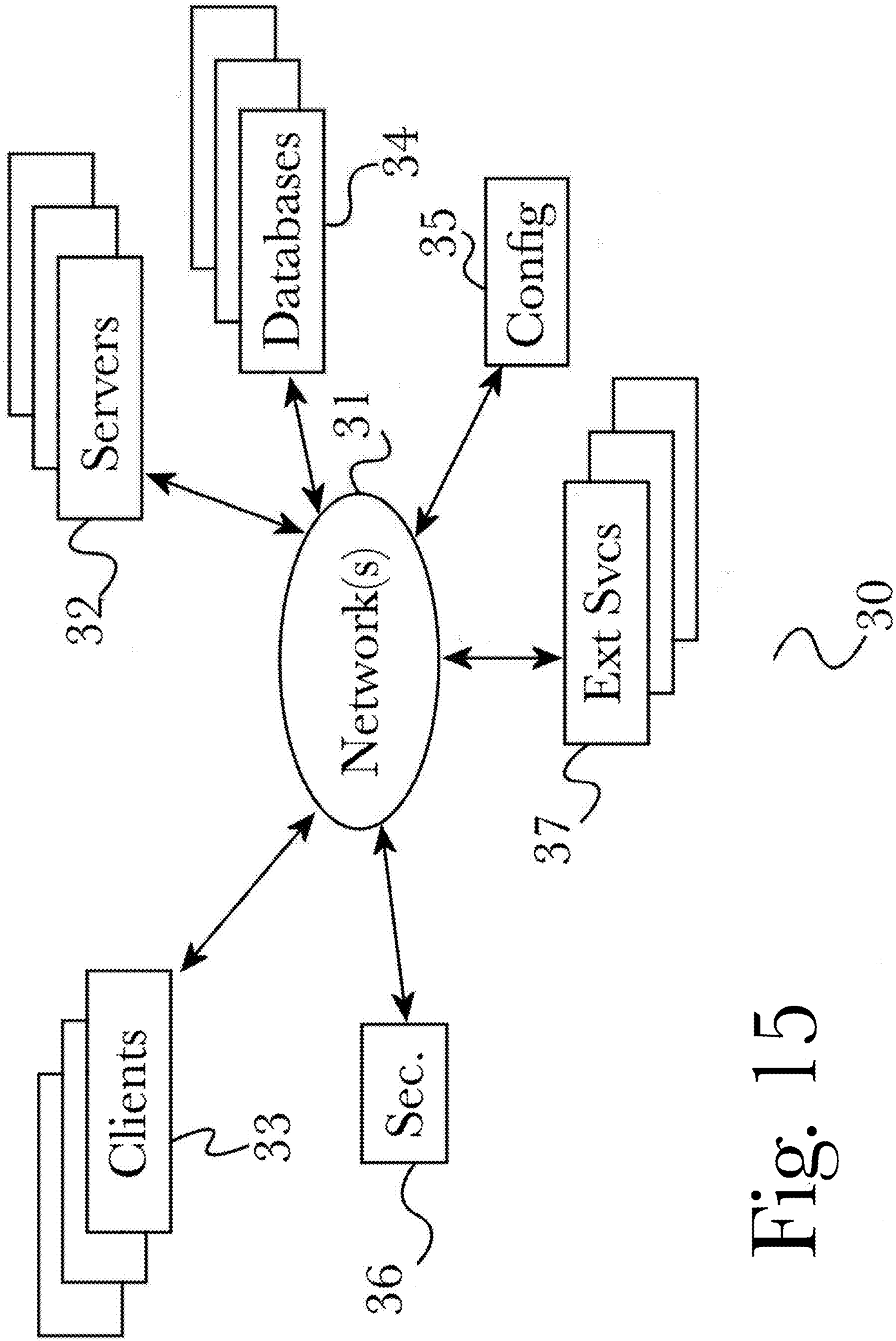

Servers
32
Databases
34
Config
35
31
Network(s)
Ext Svcs
37
30
Clients
33
Sec.
36

Fig. 15

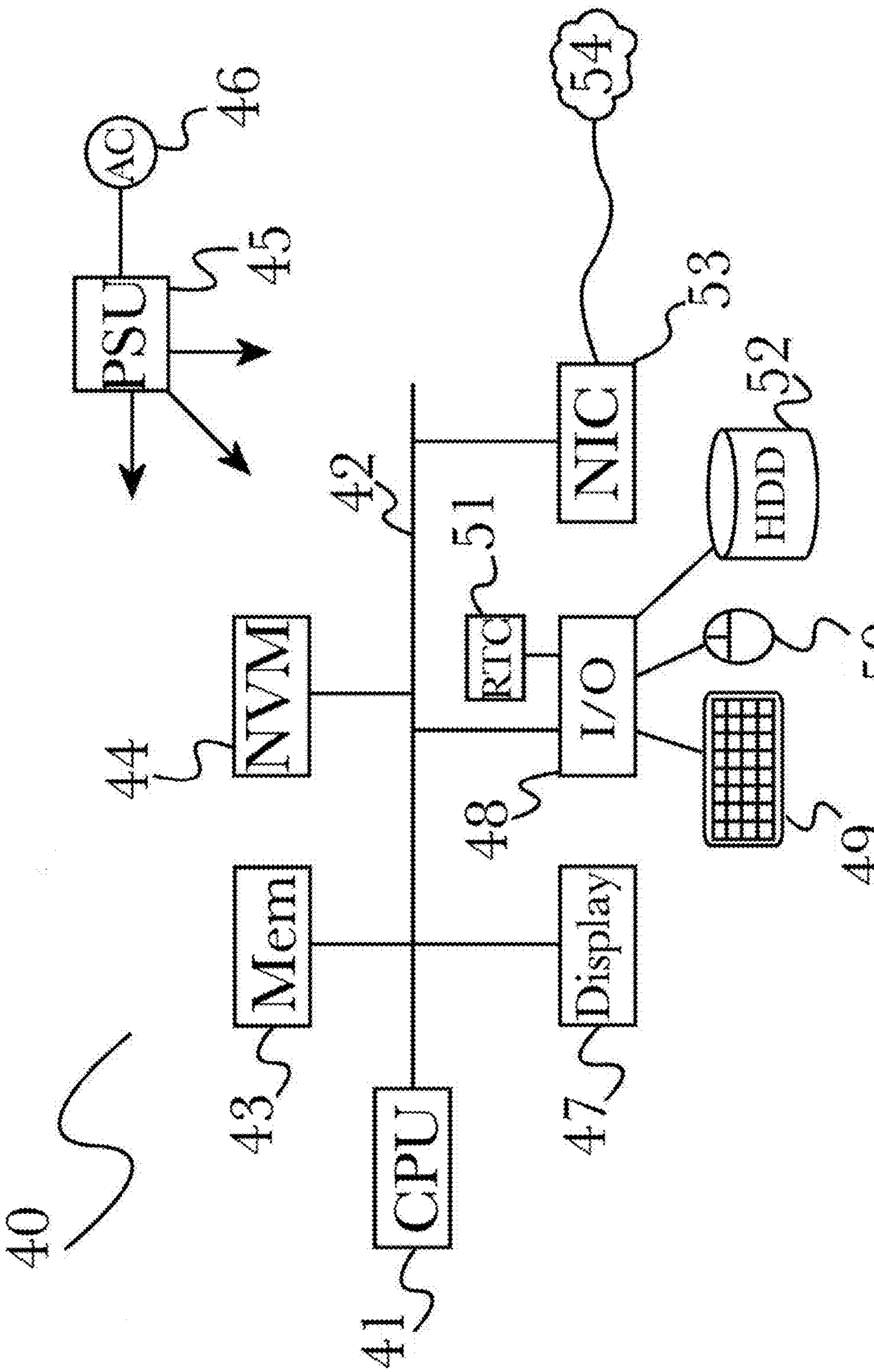

40
CPU
41
42
Mem
43
NVM
44
PSU
45
AC
46
Display
47
I/O
48
49
50
RTC
51
HDD
52
NIC
53
54

Fig. 16

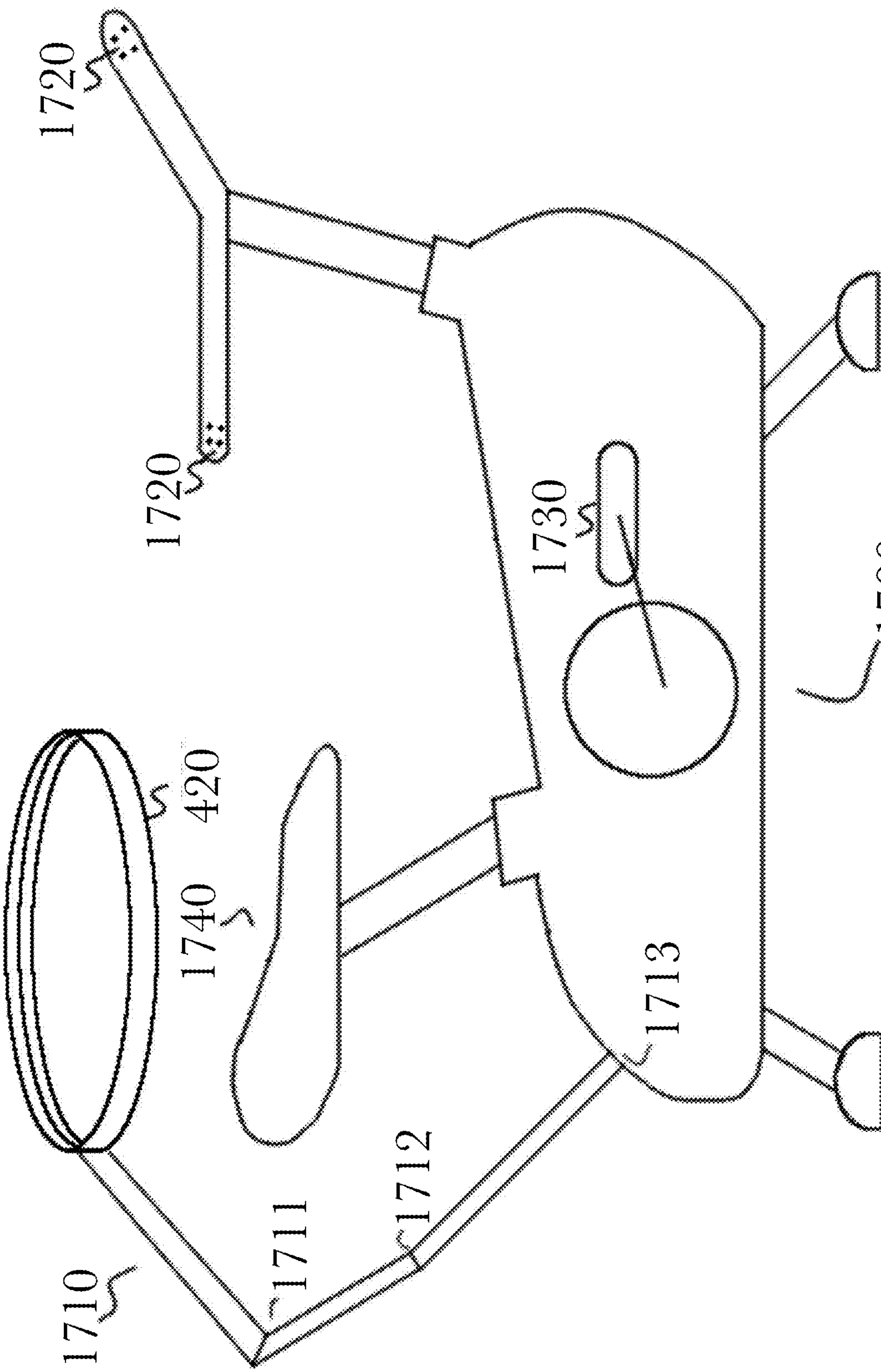


Fig. 17

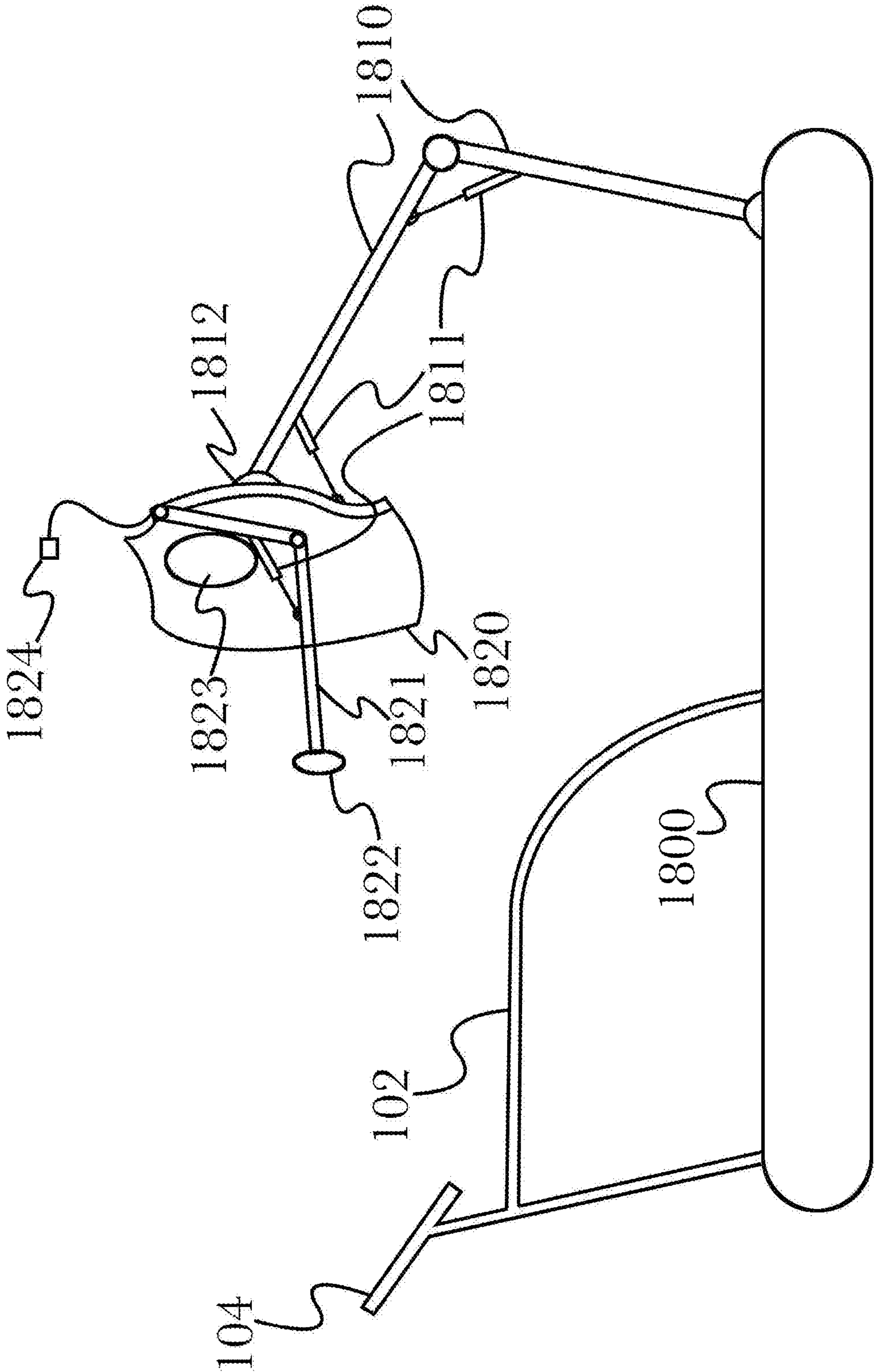


Fig. 18

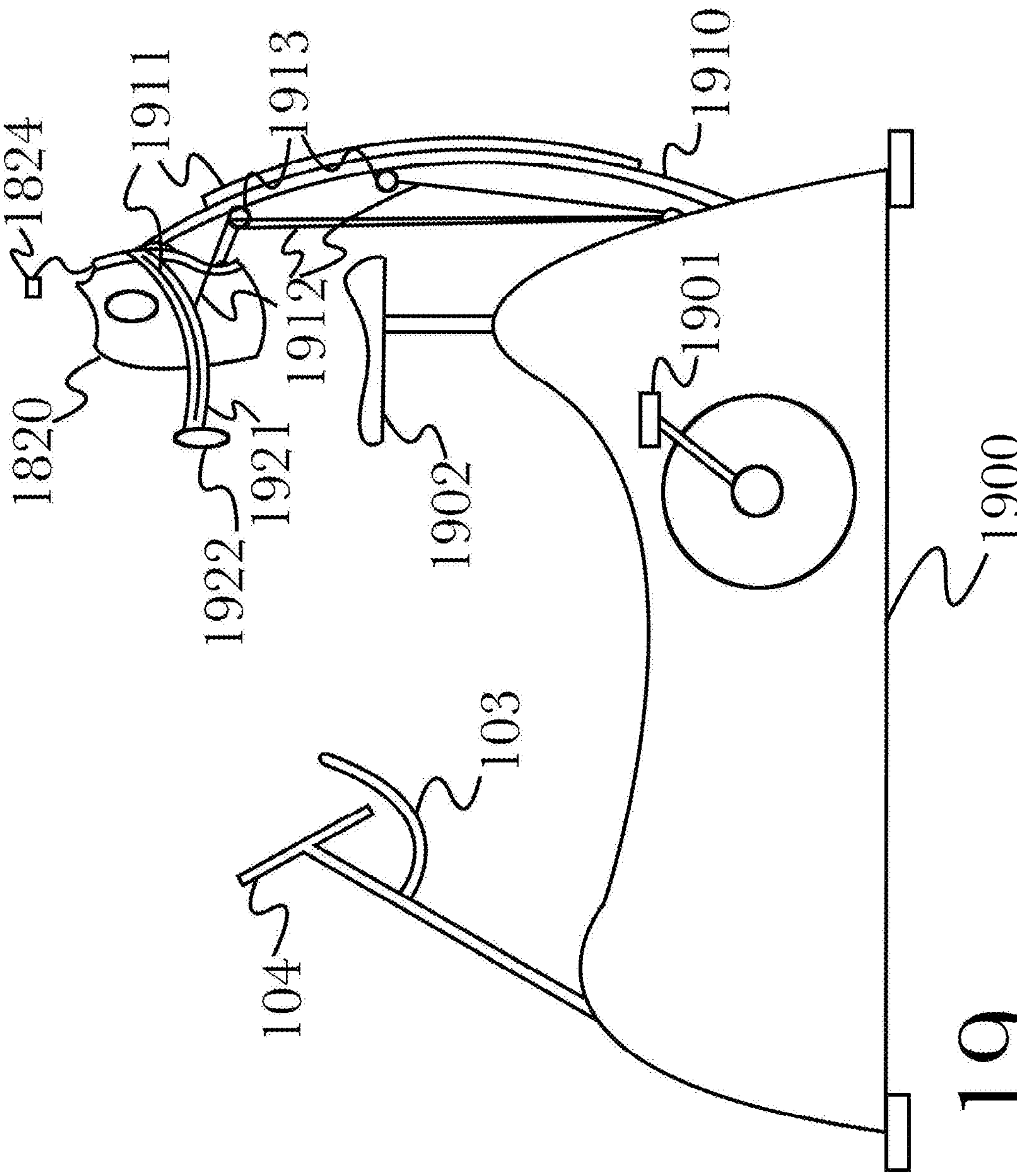

1824
1911
1913
1910
1820
1921
1922
1912
1902
1901
1900
103
104

Fig. 19

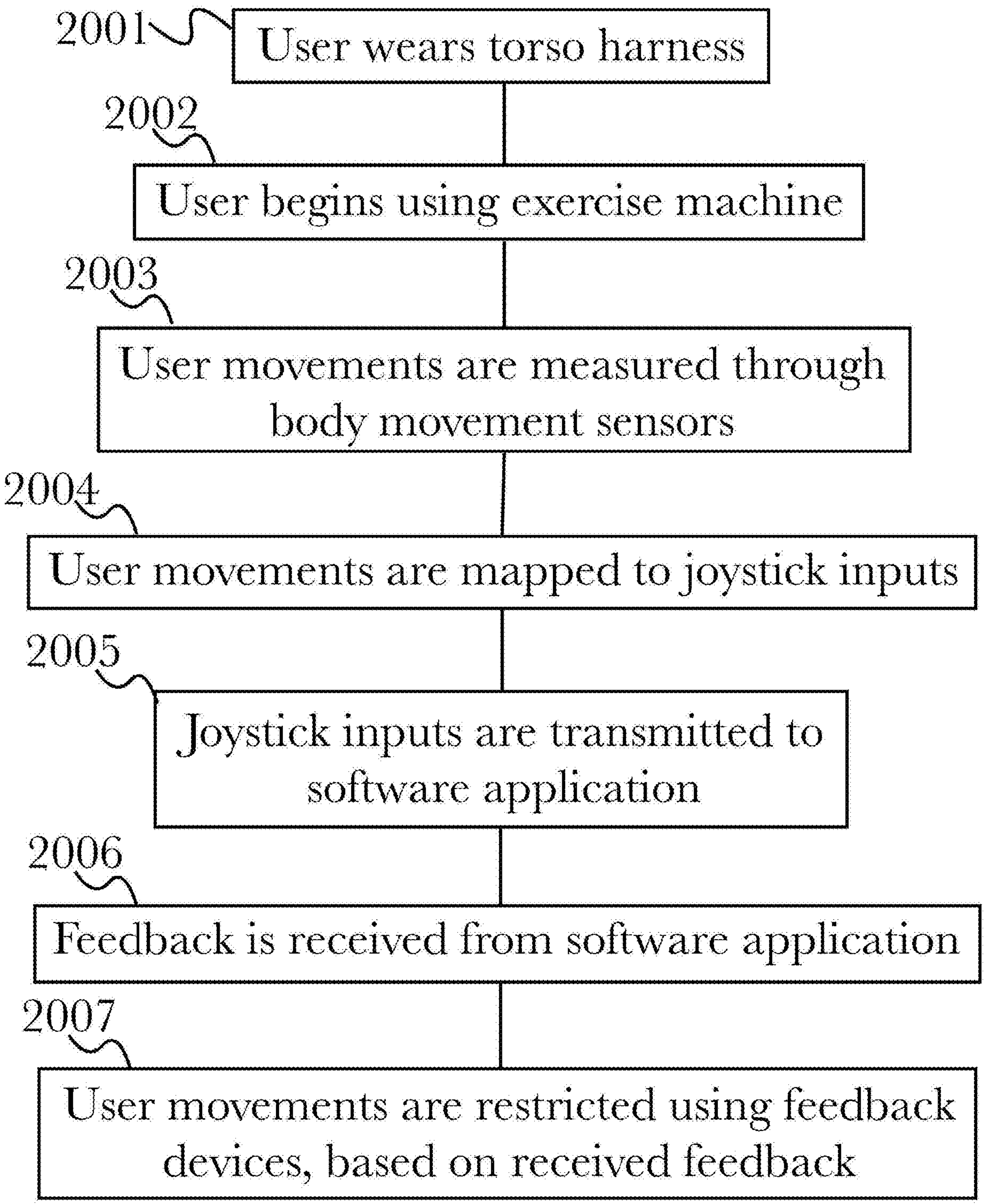

2001
User wears torso harness
2002
User begins using exercise machine
2003
User movements are measured through body movement sensors
2004
User movements are mapped to joystick inputs
2005
Joystick inputs are transmitted to software application
2006
Feedback is received from software application
2007
User movements are restricted using feedback devices, based on received feedback
2000

Fig. 20

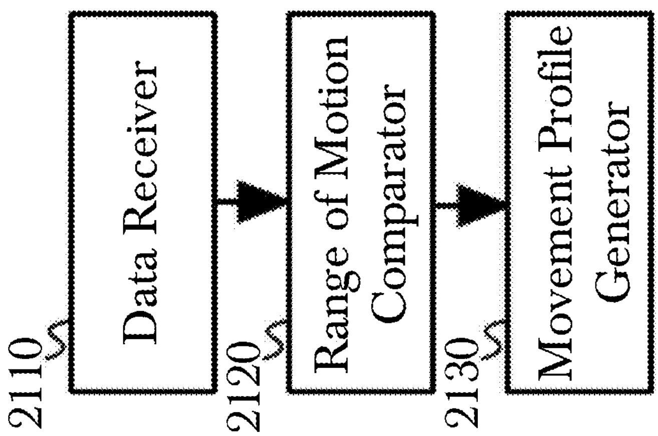
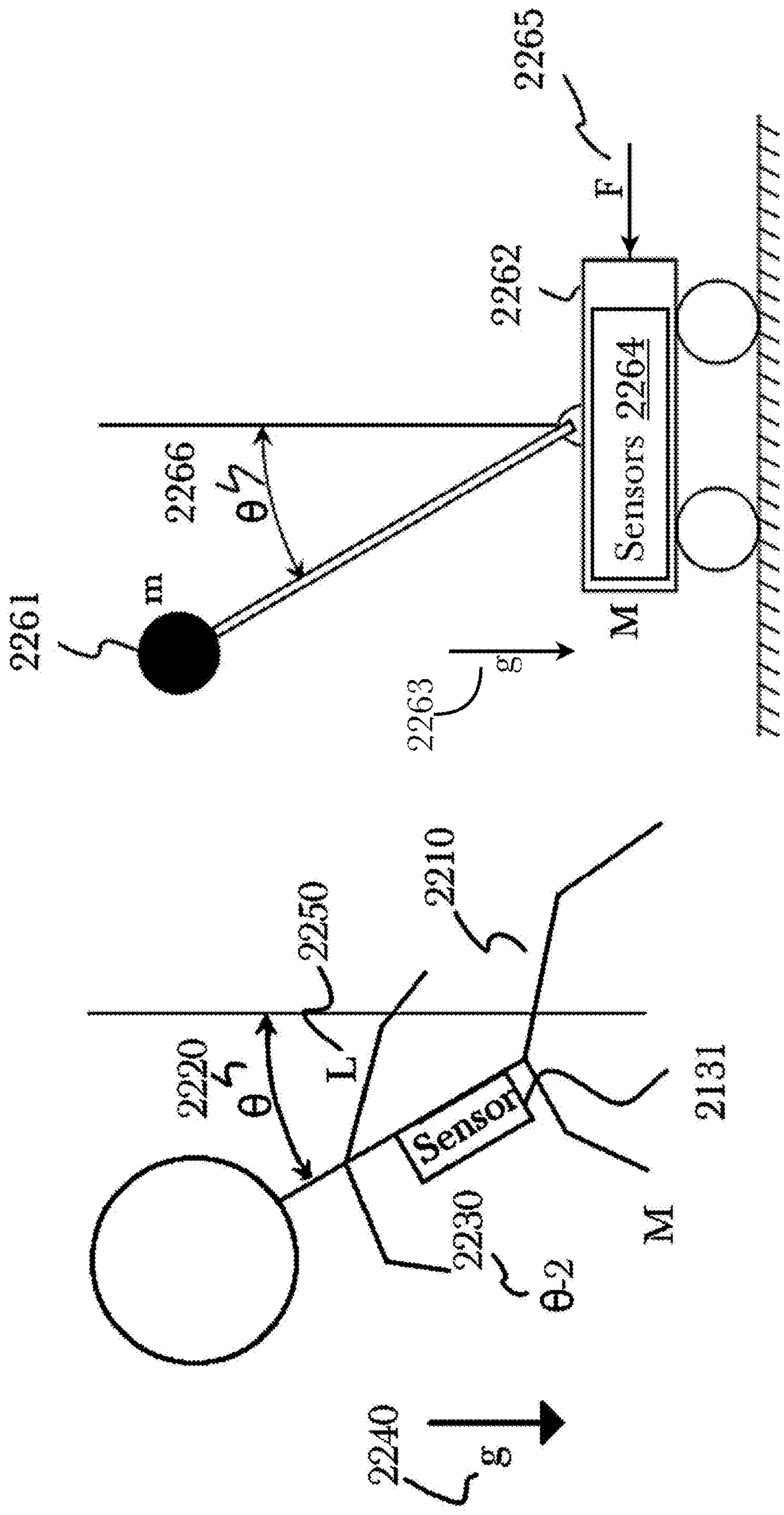

2110
Data Receiver
2120
Range of Motion Comparator
2130
Movement Profile Generator

Fig. 21



2265
F
2262
Sensors 2264
2266
θ
m
2261
M
g
2263
2210
2250
L
2220
θ
Sensor
2131
M
2230
θ-2
2240
g

Fig. 22

Movement Profile Generator

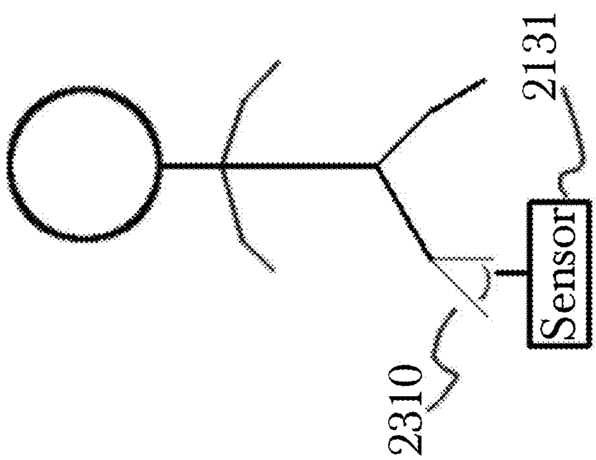


Fig. 23

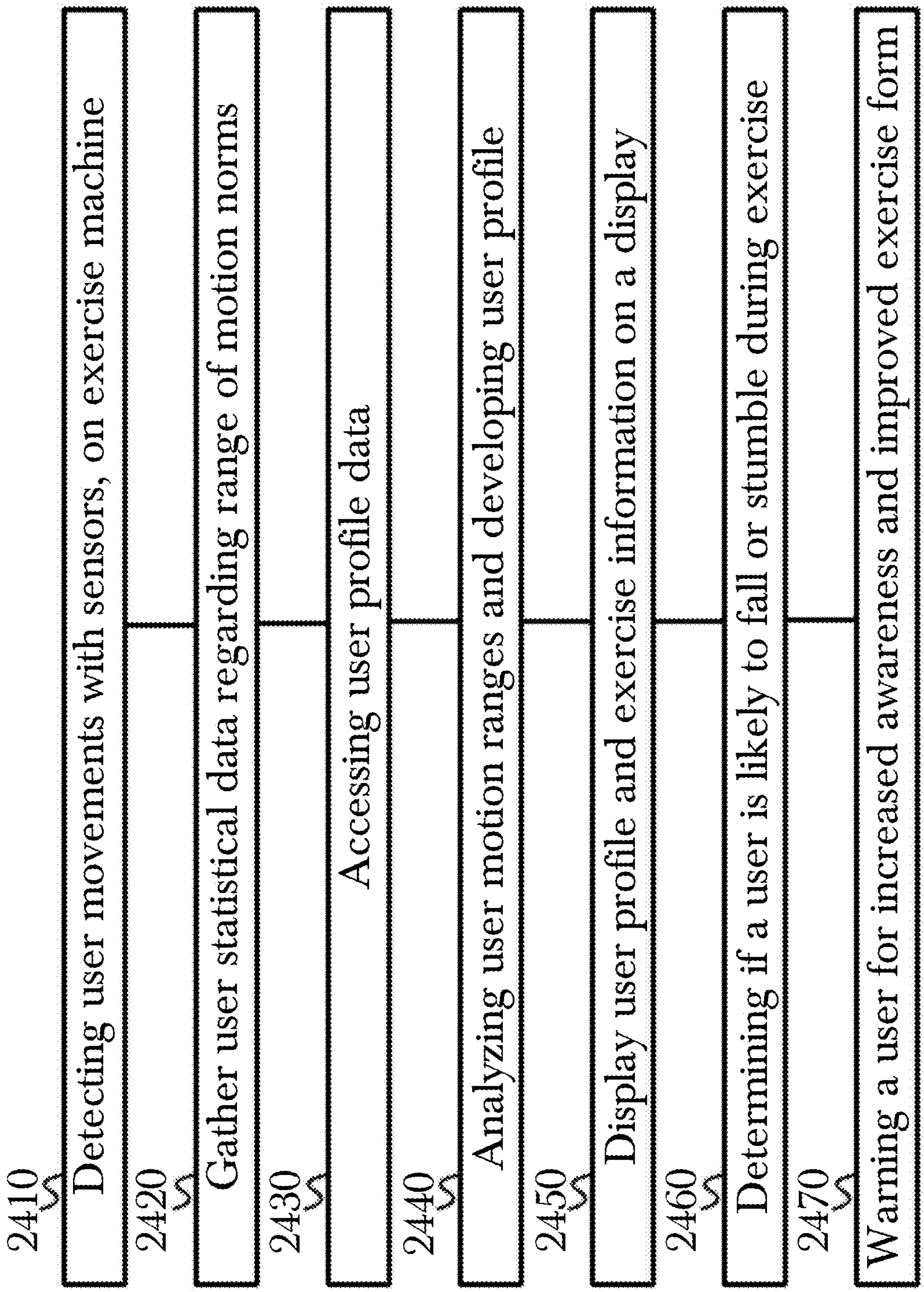
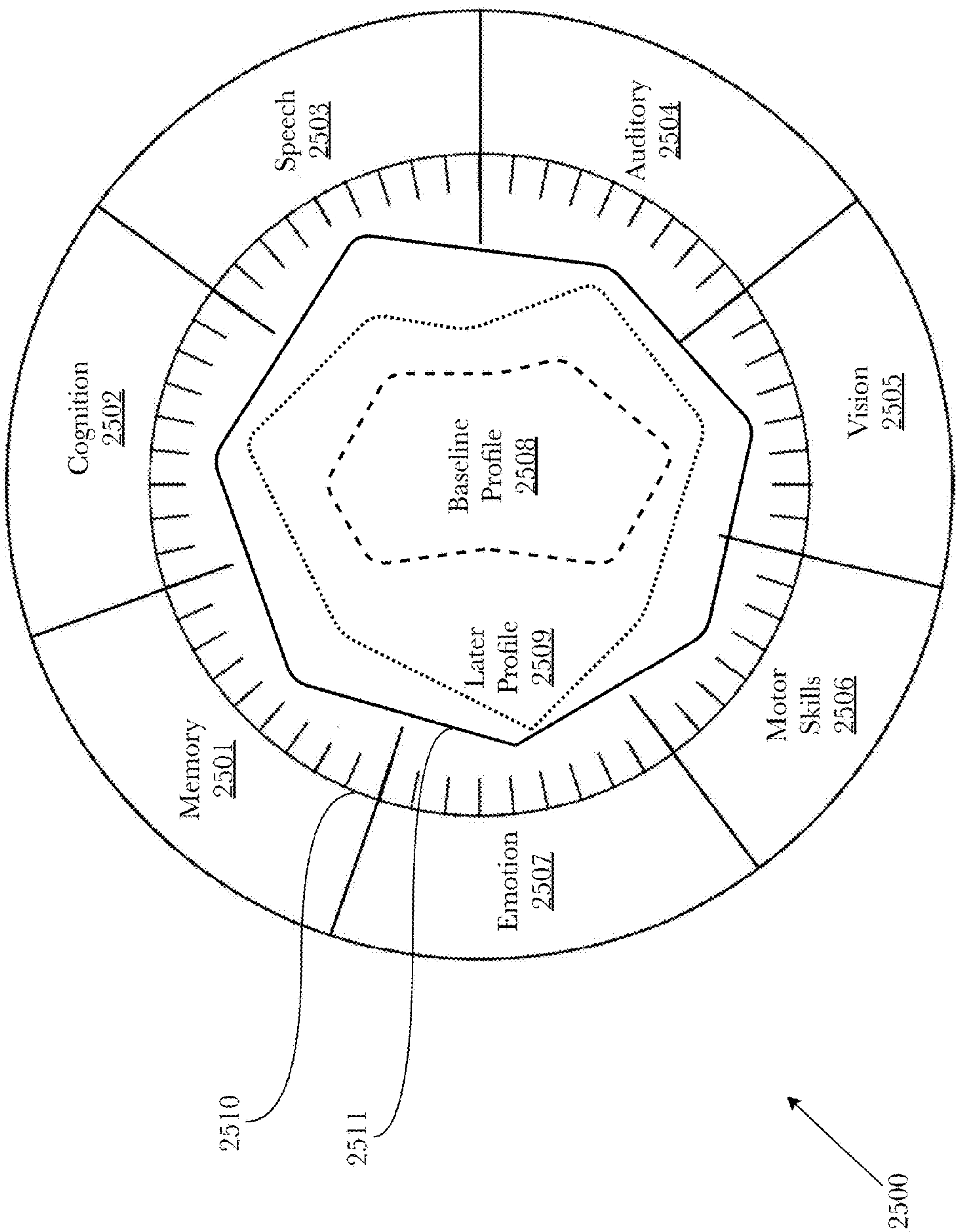

2410 Detecting user movements with sensors, on exercise machine
2420 Gather user statistical data regarding range of motion norms
2430 Accessing user profile data
2440 Analyzing user motion ranges and developing user profile
2450 Display user profile and exercise information on a display
2460 Determining if a user is likely to fall or stumble during exercise
2470 Warning a user for increased awareness and improved exercise form

Fig. 24



Speech 2503
Auditory 2504
Vision 2505
Motor Skills 2506
Emotion 2507
Memory 2501
Cognition 2502
Baseline Profile 2508
Later Profile 2509
2510
2511
2500

Fig. 25

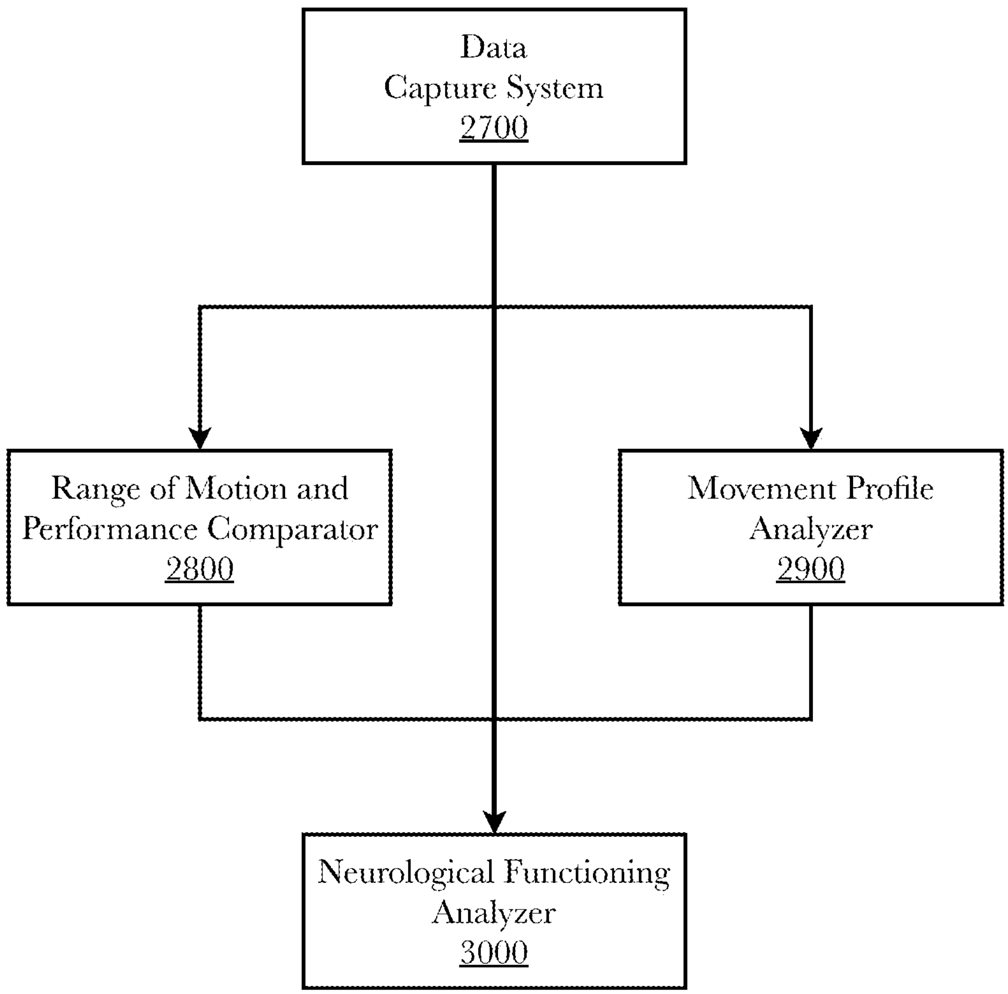

Data
Capture System
2700
Range of Motion and
Performance Comparator
2800
Movement Profile
Analyzer
2900
Neurological Functioning
Analyzer
3000

Fig. 26

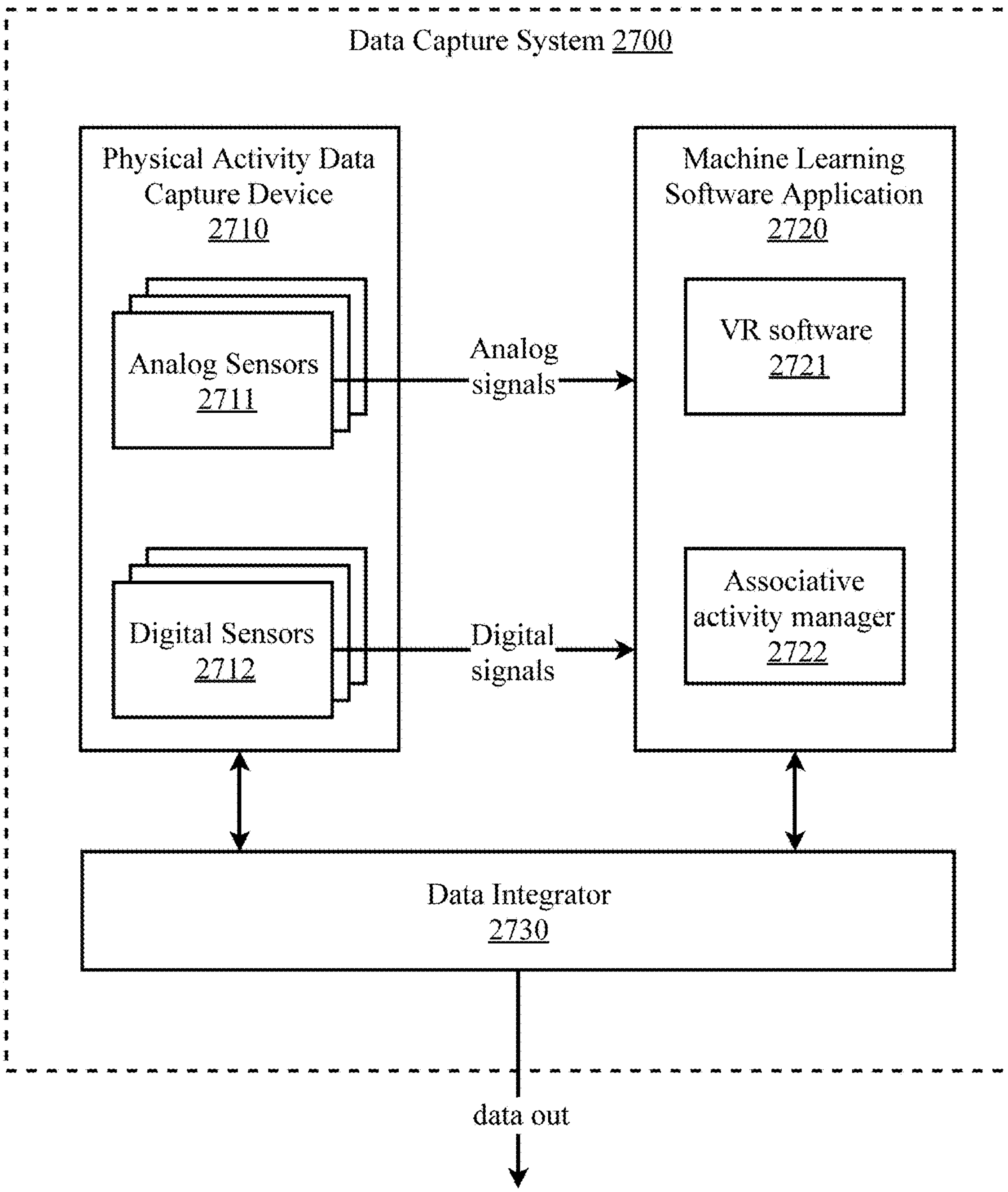

Data Capture System 2700
Physical Activity Data Capture Device 2710
Analog Sensors 2711
Digital Sensors 2712
Analog signals
Digital signals
Machine Learning Software Application 2720
VR software 2721
Associative activity manager 2722
Data Integrator 2730
data out

Fig. 27

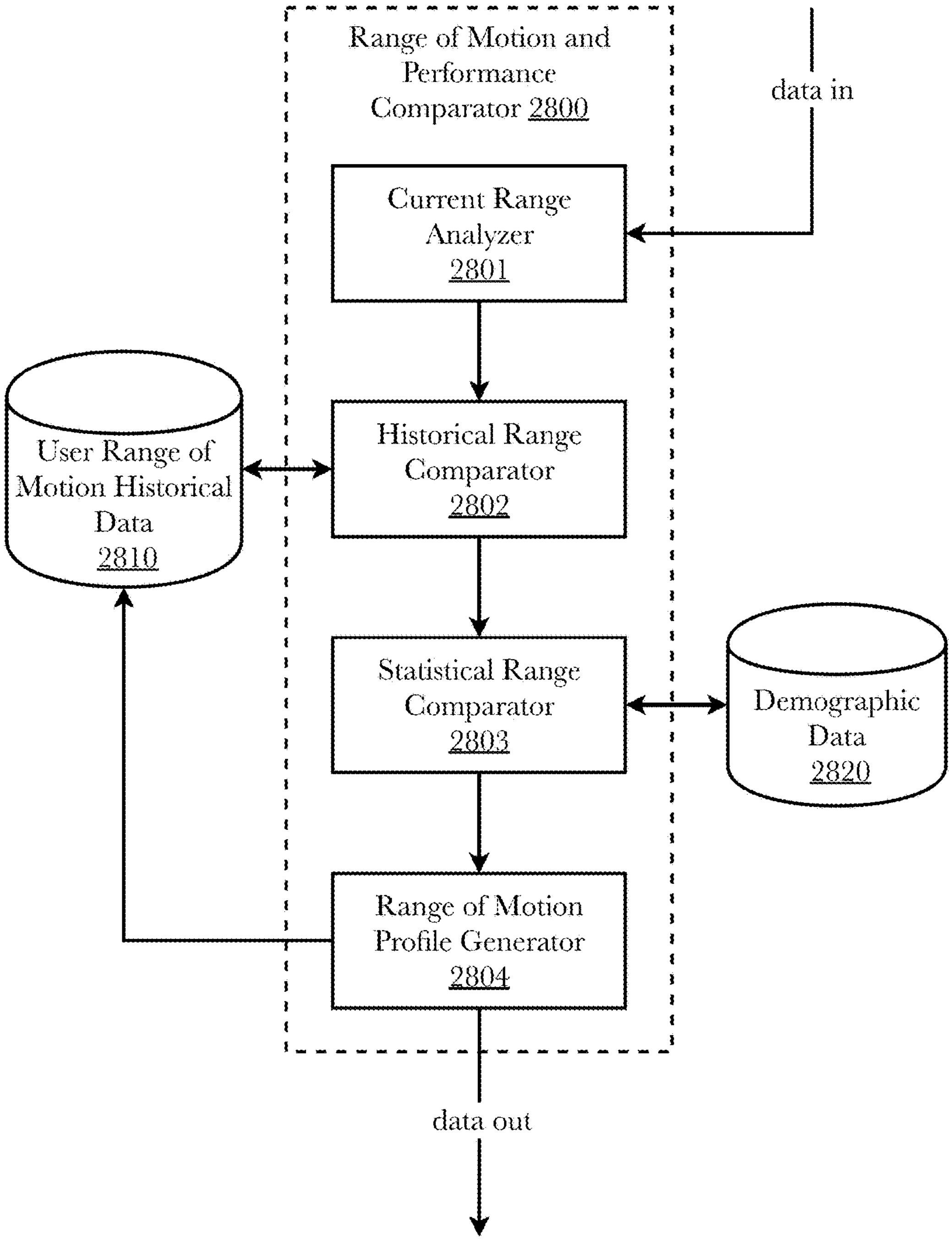

Range of Motion and Performance Comparator 2800
data in
Current Range Analyzer 2801
Historical Range Comparator 2802
User Range of Motion Historical Data 2810
Statistical Range Comparator 2803
Demographic Data 2820
Range of Motion Profile Generator 2804
data out

Fig. 28

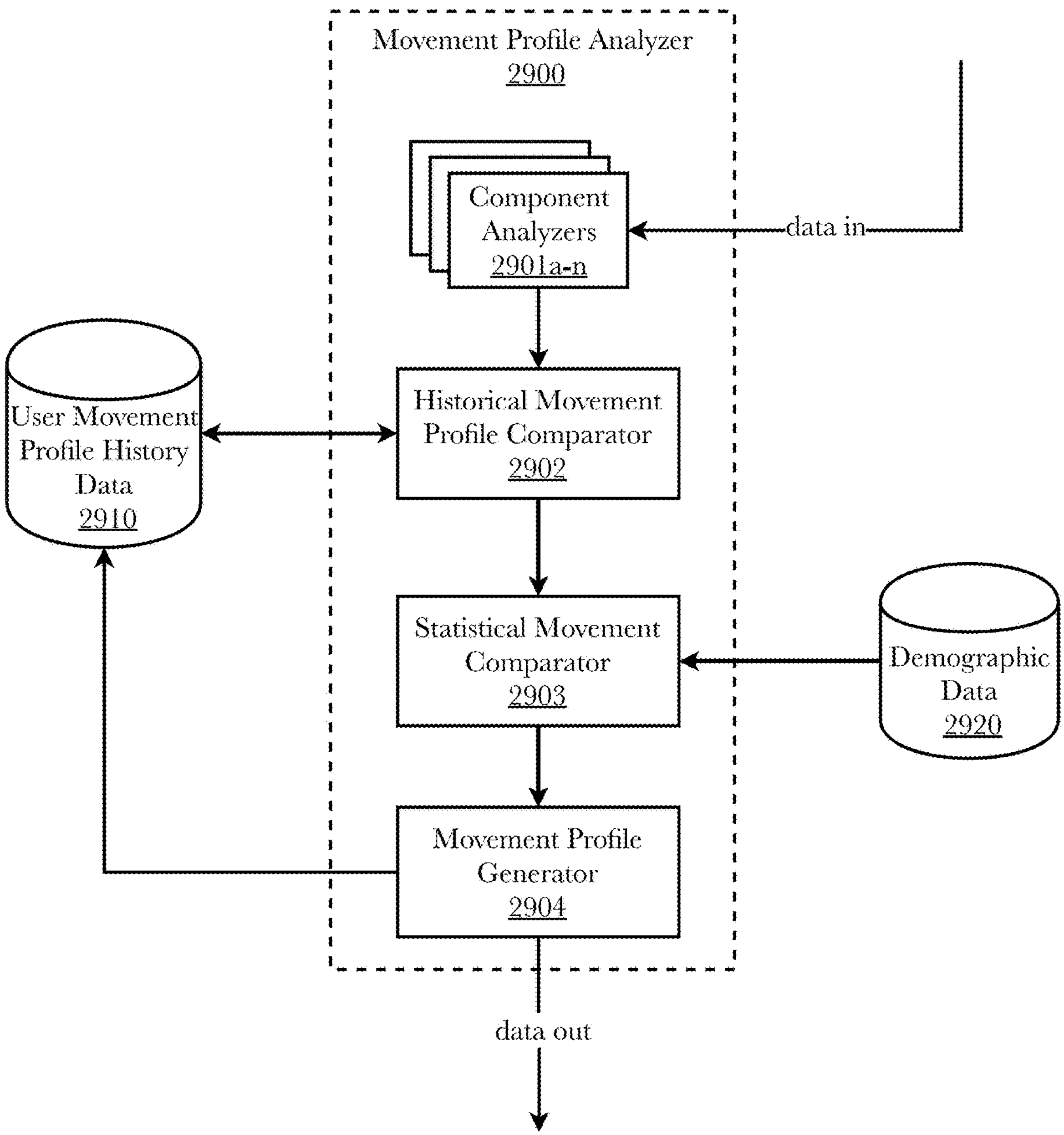

Movement Profile Analyzer
2900
Component Analyzers
2901a-n
data in
User Movement Profile History Data
2910
Historical Movement Profile Comparator
2902
Statistical Movement Comparator
2903
Demographic Data
2920
Movement Profile Generator
2904
data out

Fig. 29

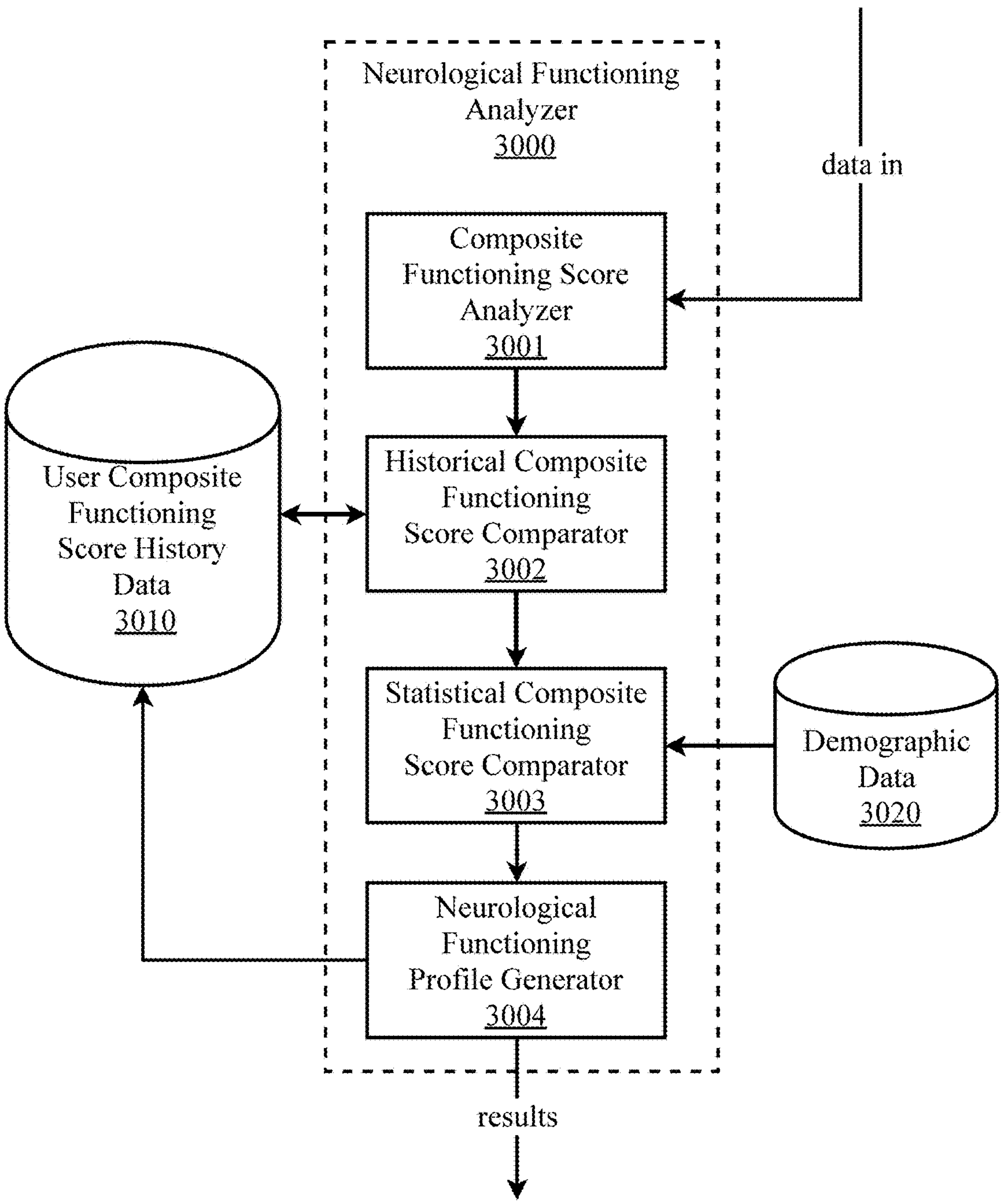

Neurological Functioning Analyzer 3000
data in
Composite Functioning Score Analyzer 3001
User Composite Functioning Score History Data 3010
Historical Composite Functioning Score Comparator 3002
Statistical Composite Functioning Score Comparator 3003
Demographic Data 3020
Neurological Functioning Profile Generator 3004
results

Fig. 30

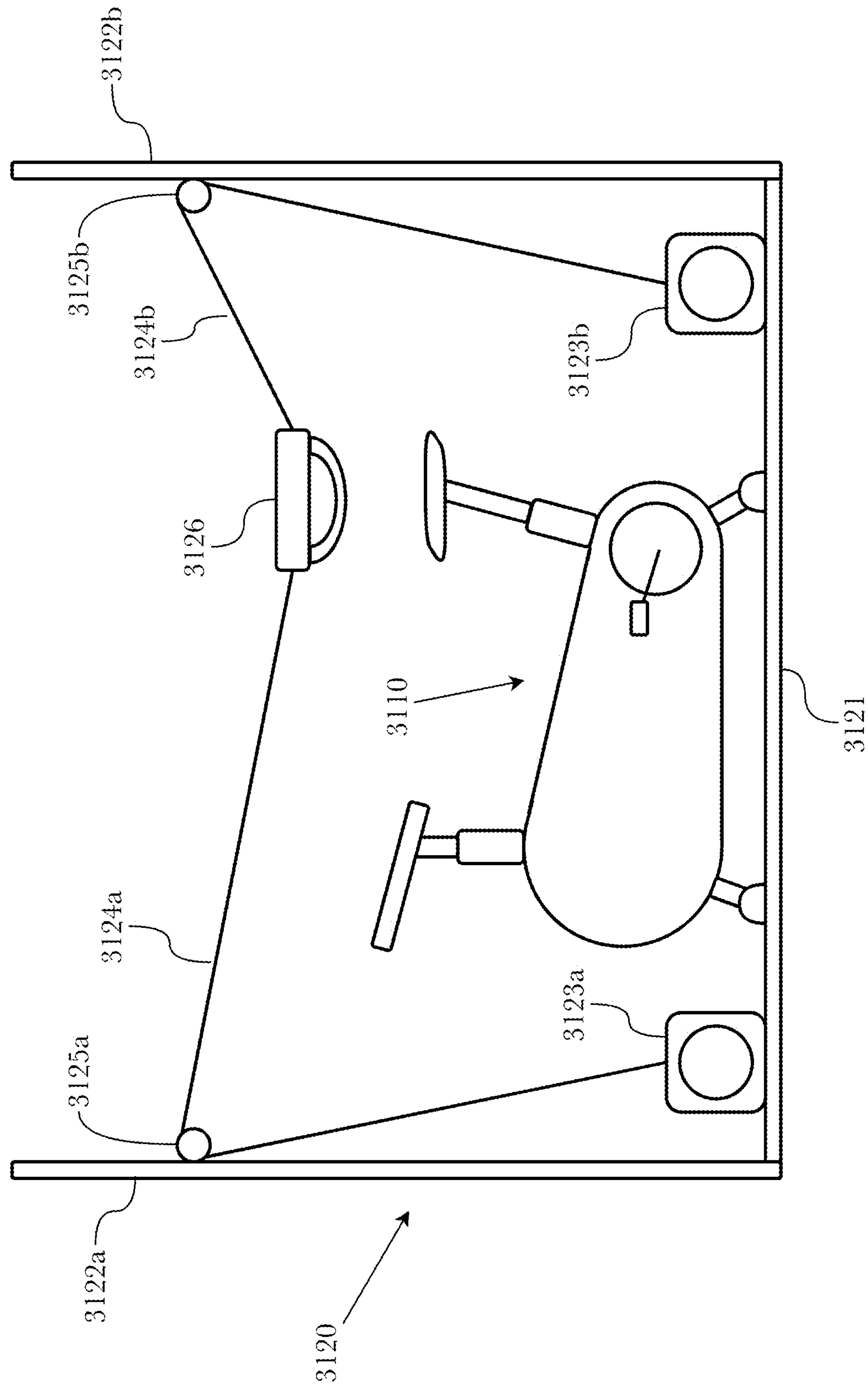

3122b
3125b
3124b
3123b
3126
3110
3121
3124a
3123a
3125a
3122a
3120

Fig. 31

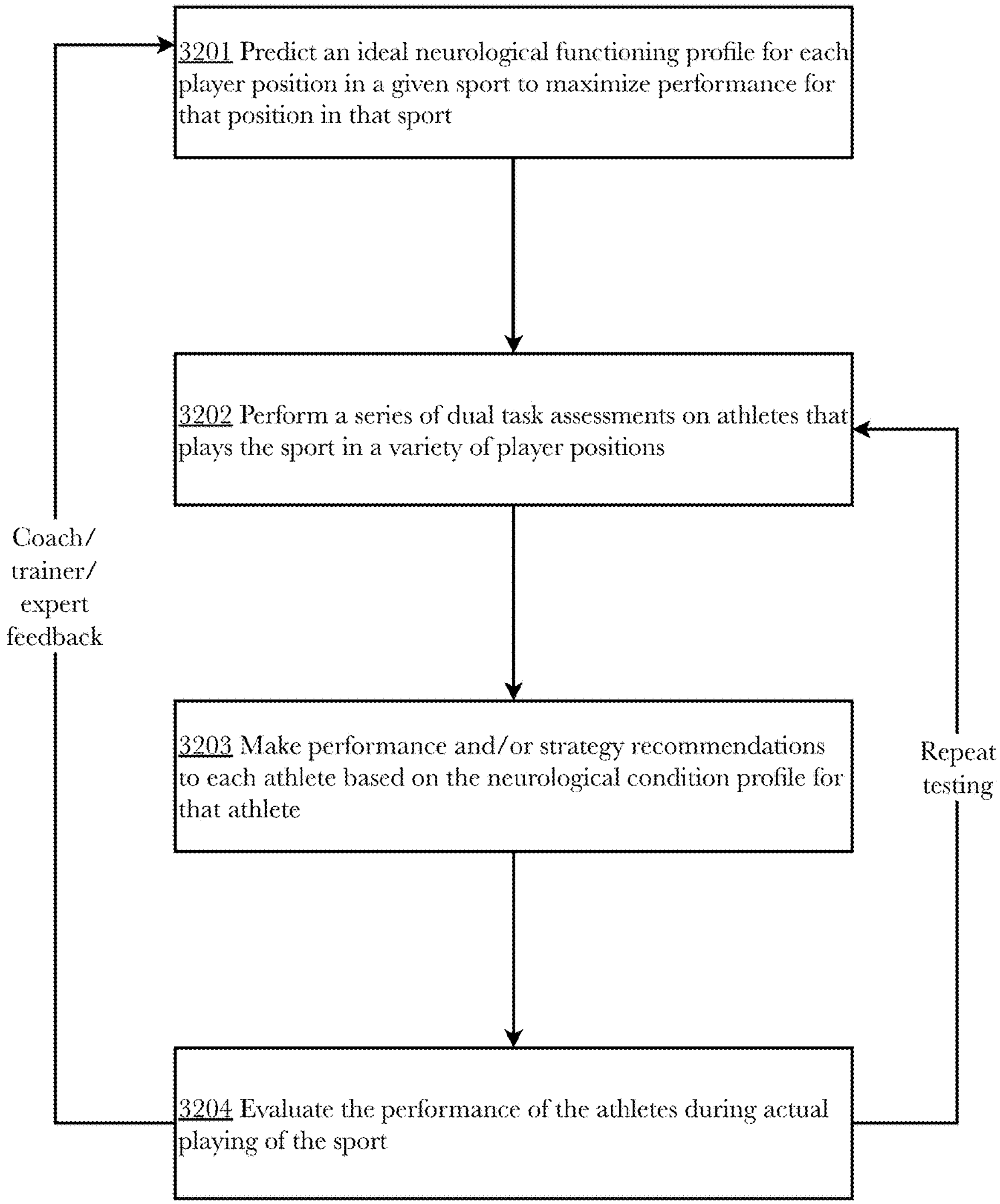

3201 Predict an ideal neurological functioning profile for each player position in a given sport to maximize performance for that position in that sport
3202 Perform a series of dual task assessments on athletes that plays the sport in a variety of player positions
3203 Make performance and/or strategy recommendations to each athlete based on the neurological condition profile for that athlete
3204 Evaluate the performance of the athletes during actual playing of the sport
Coach/ trainer/ expert feedback
Repeat testing

Fig. 32

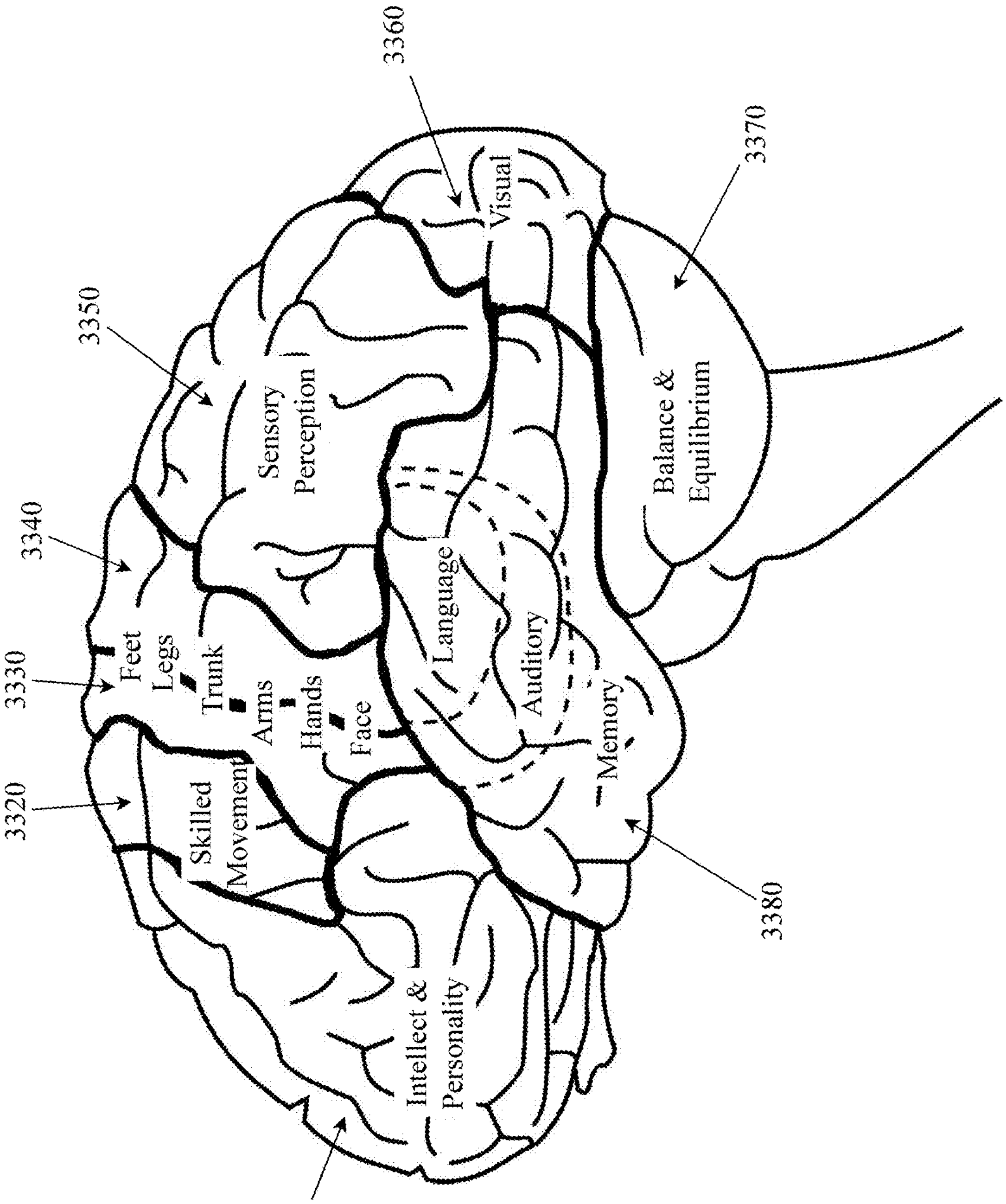


FIG. 33 (PRIOR ART)

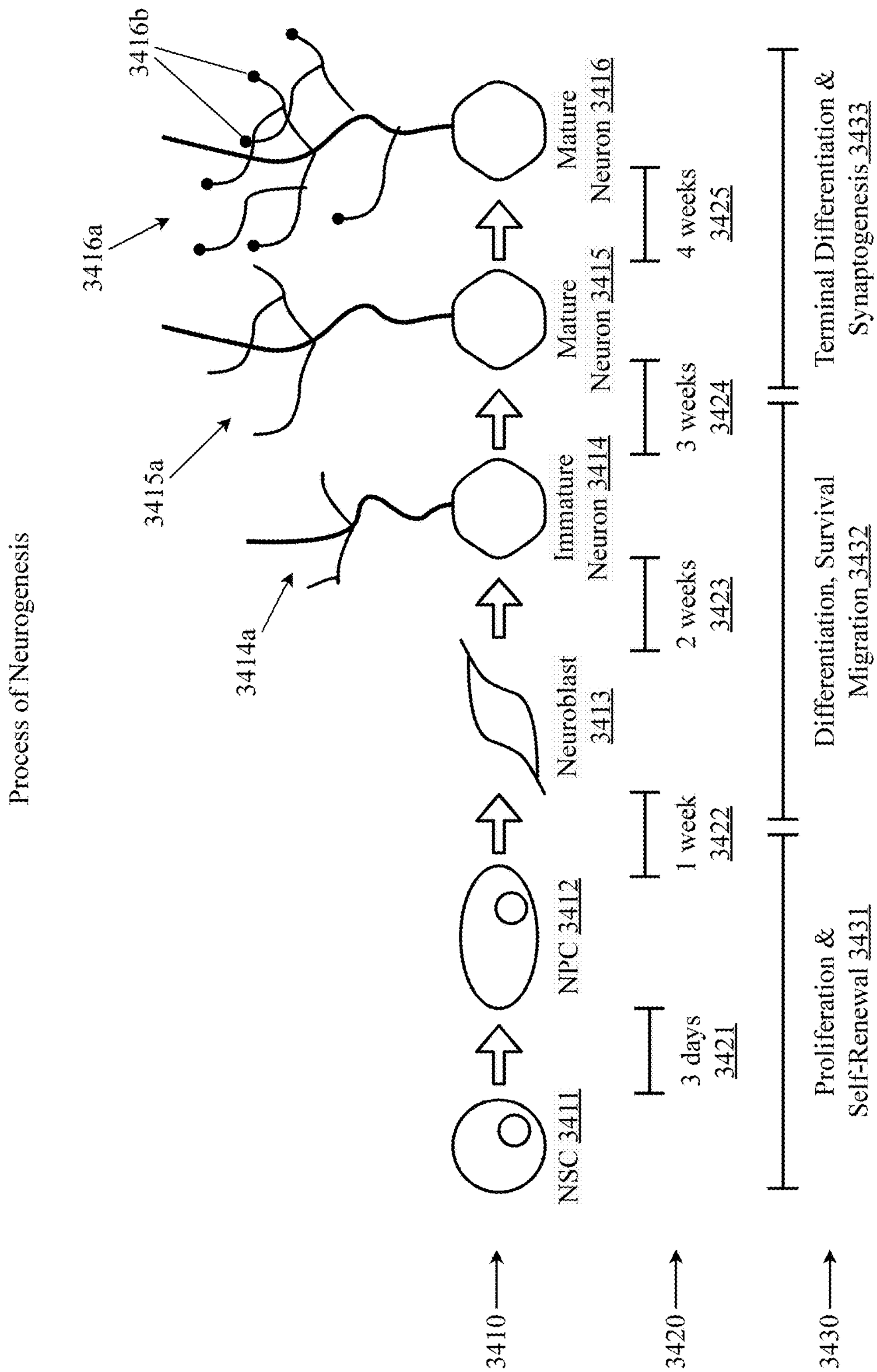


FIG. 34 (PRIOR ART)

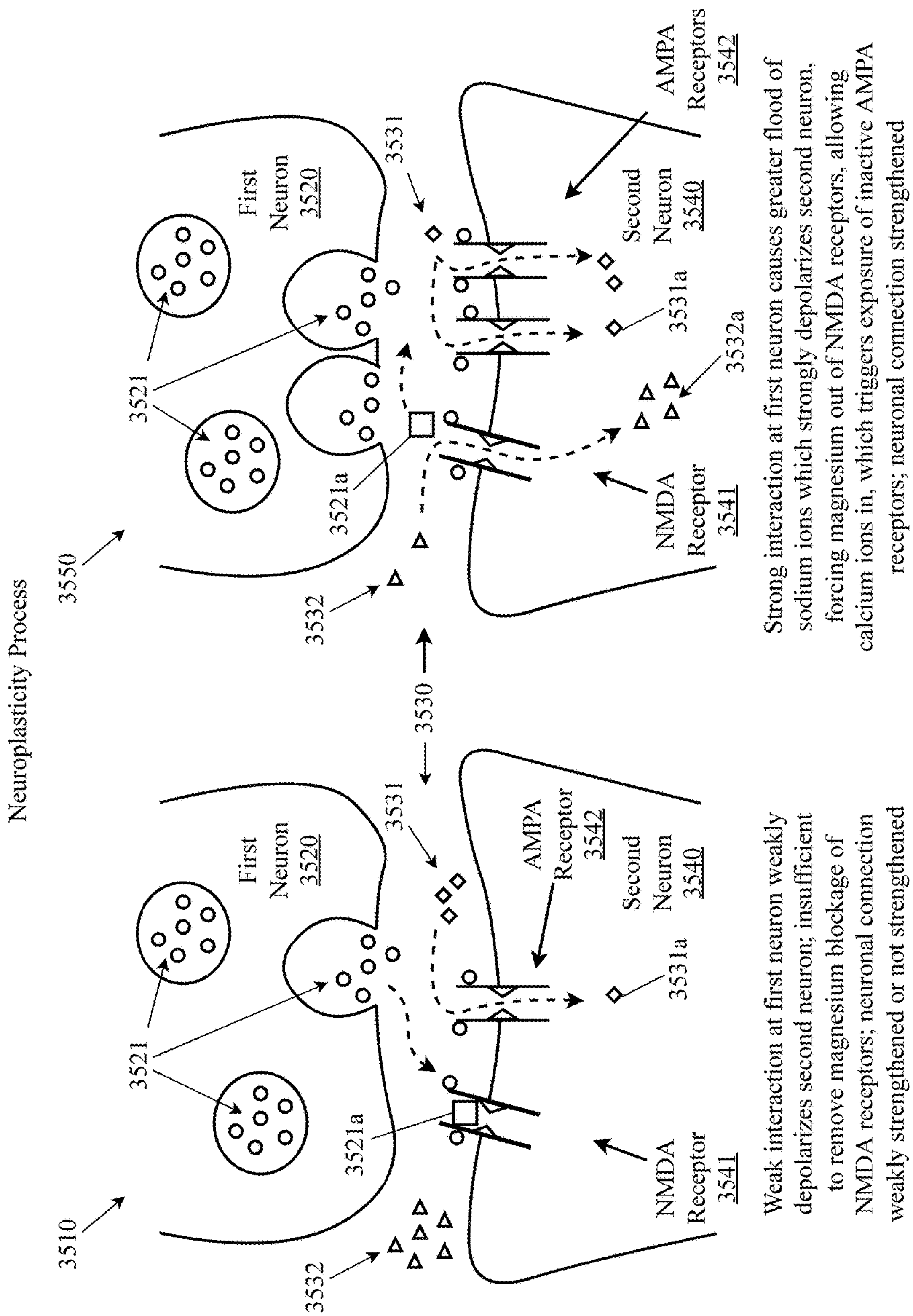


FIG. 35 (PRIOR ART)

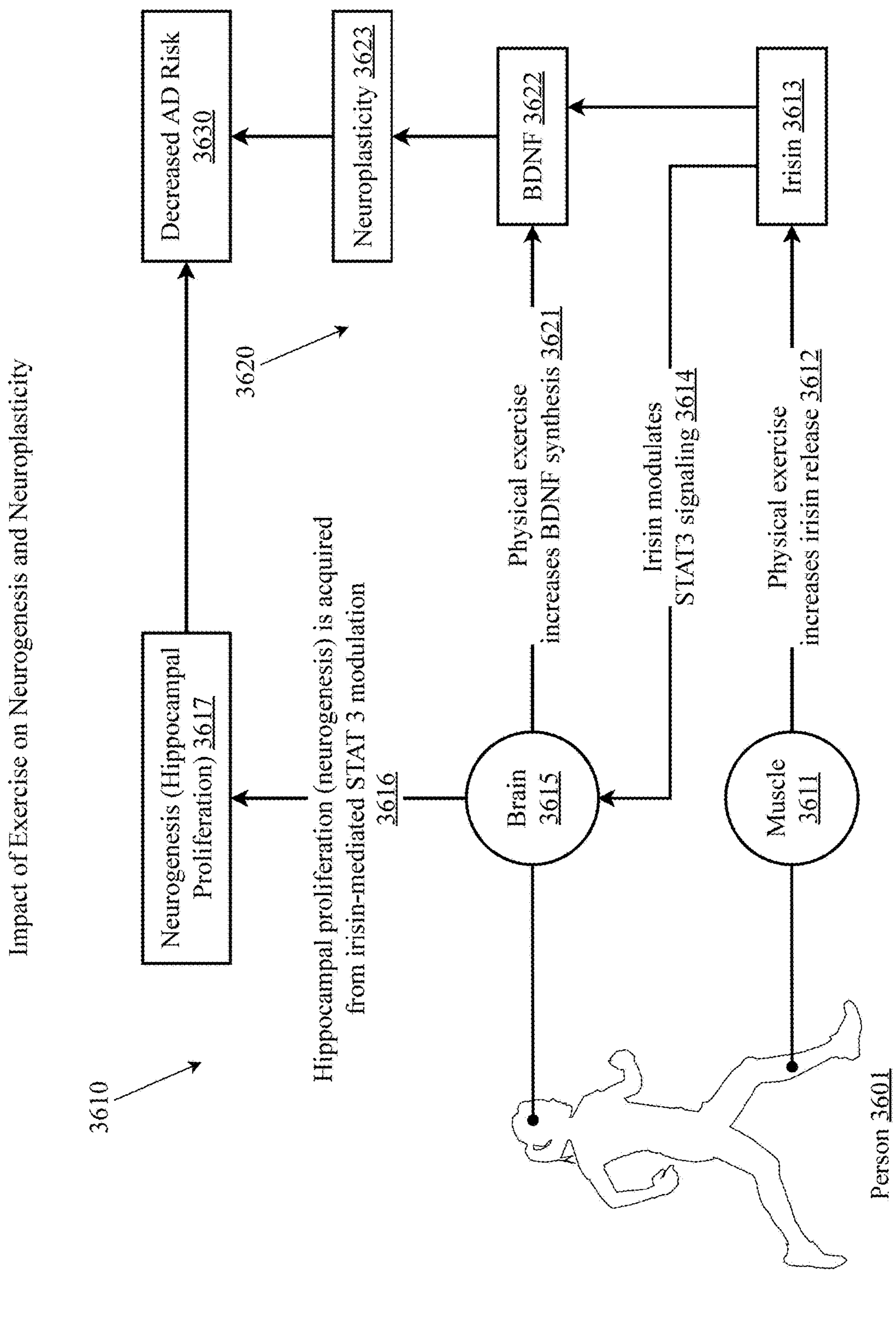


FIG. 36 (PRIOR ART)

Impact of Variable Exercise Intensity on Neurogenesis and Neuroplasticity

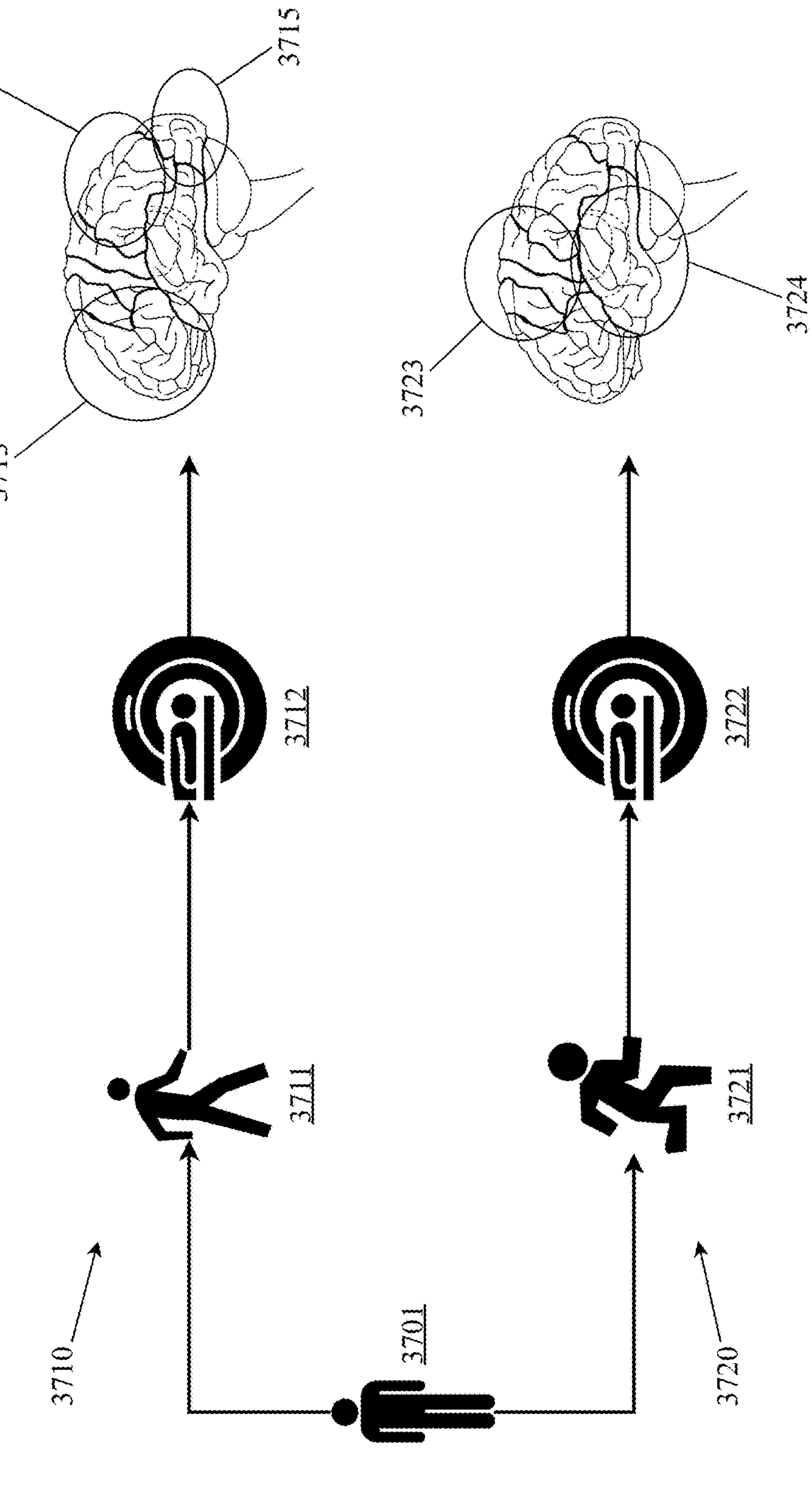
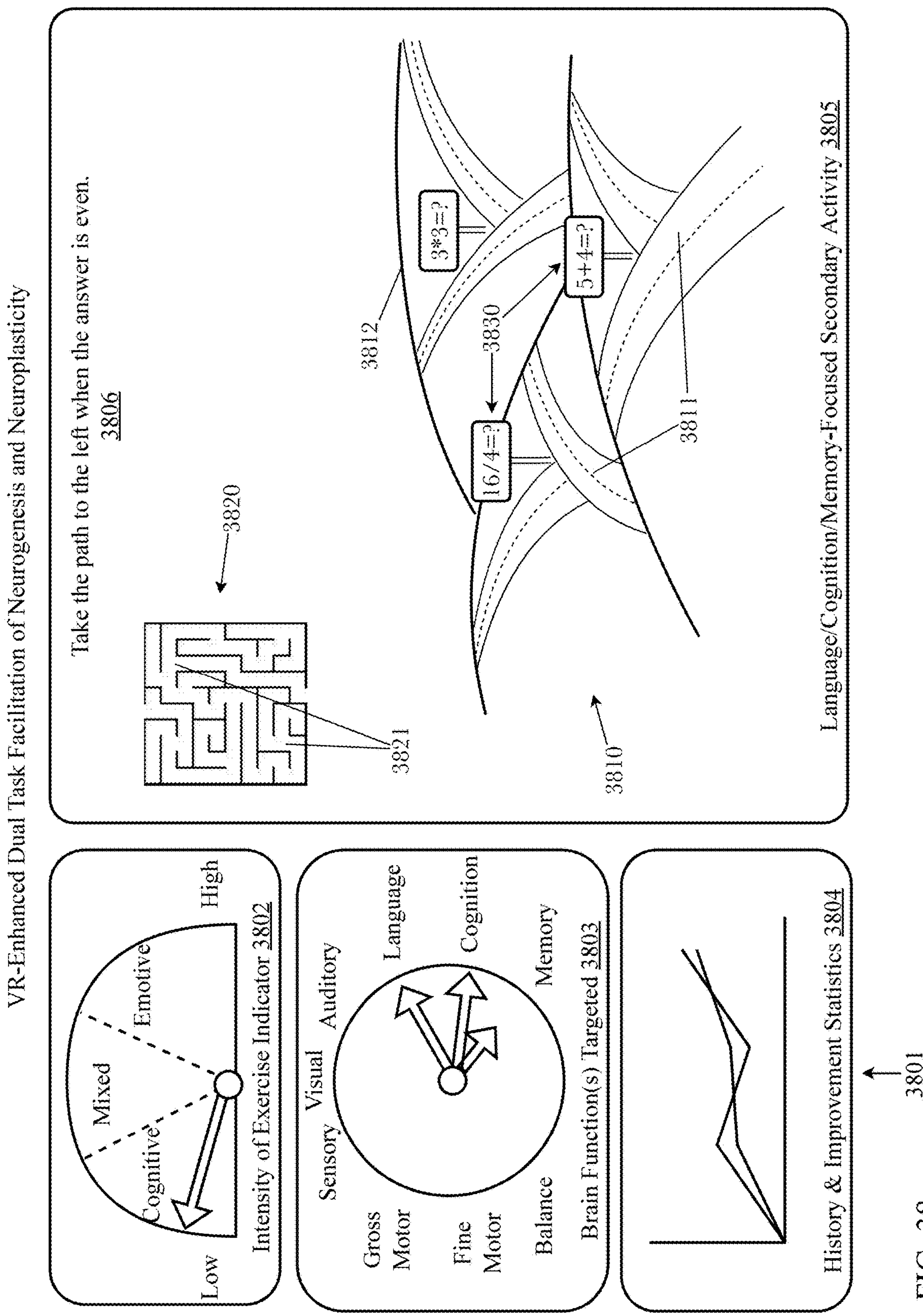


FIG. 37 (PRIOR ART)



VR-Enhanced Dual Task Facilitation of Neurogenesis and Neuroplasticity
Low
Cognitive
Mixed
Emotive
High
Intensity of Exercise Indicator 3802
Gross Motor
Sensory
Visual
Auditory
Language
Cognition
Memory
Balance
Fine Motor
Brain Function(s) Targeted 3803
History & Improvement Statistics 3804
3801
Take the path to the left when the answer is even.
3806
3820
3821
3810
3812
3830
3811
16/4=?
3*3=?
5+4=?
Language/Cognition/Memory-Focused Secondary Activity 3805

FIG. 38

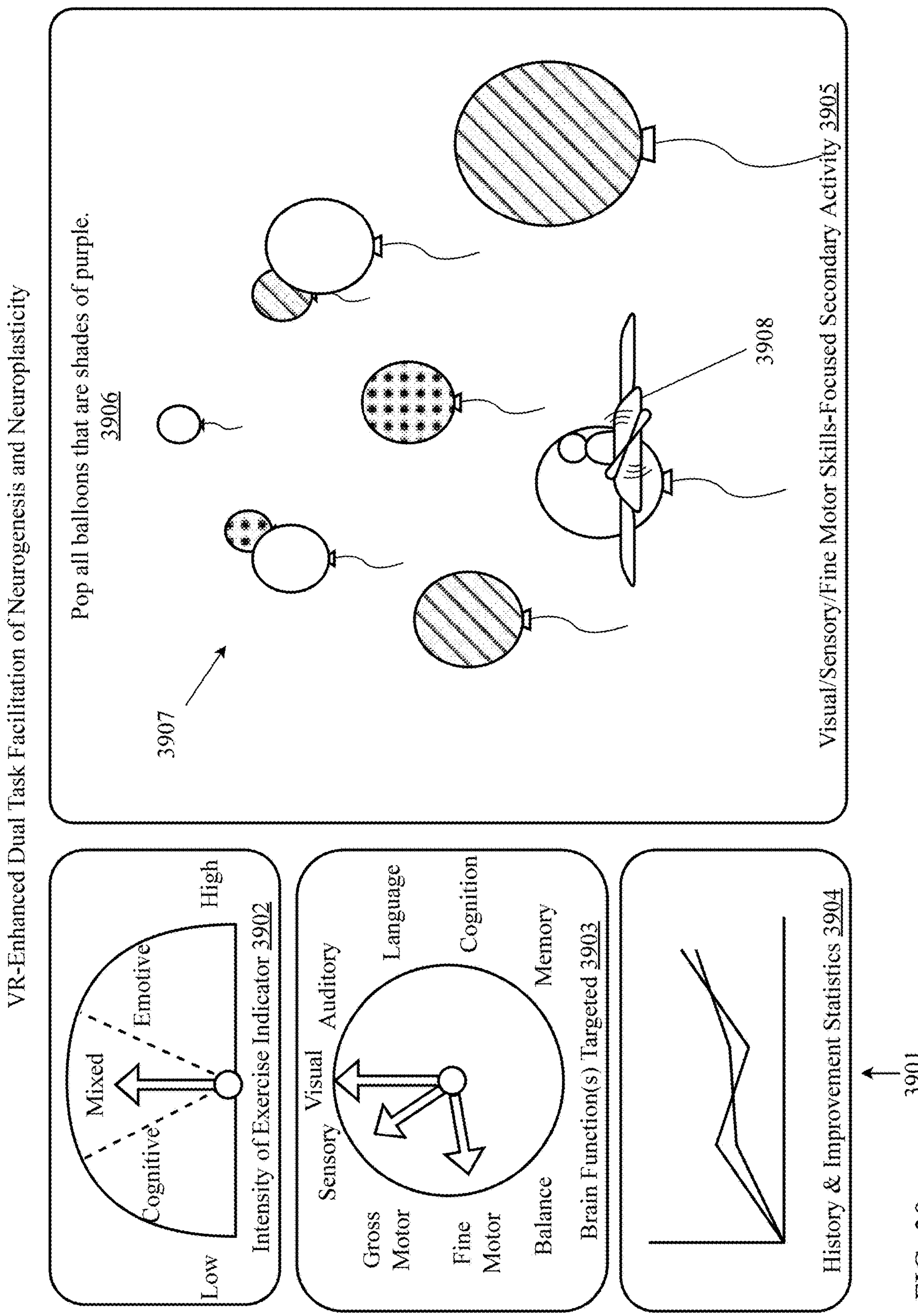

VR-Enhanced Dual Task Facilitation of Neurogenesis and Neuroplasticity
Pop all balloons that are shades of purple.
3906
3907
3908
Visual/Sensory/Fine Motor Skills-Focused Secondary Activity 3905
Low
Cognitive
Mixed
Emotive
High
Intensity of Exercise Indicator 3902
Gross Motor
Sensory
Visual
Auditory
Language
Cognition
Memory
Balance
Fine Motor
Brain Function(s) Targeted 3903
History & Improvement Statistics 3904
3901

FIG. 39

VR-Enhanced Dual Task Facilitation of Neurogenesis and Neuroplasticity

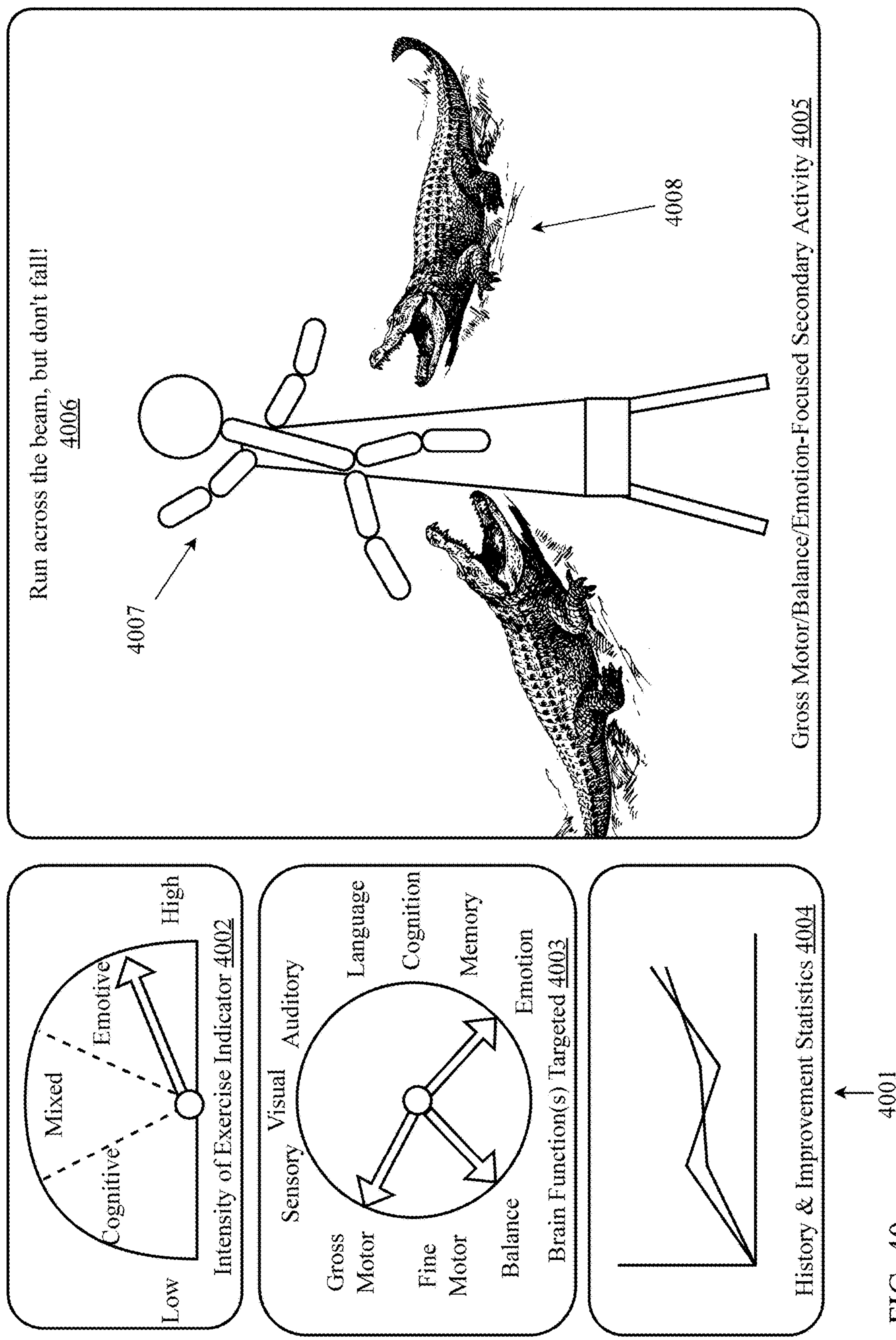


FIG. 40

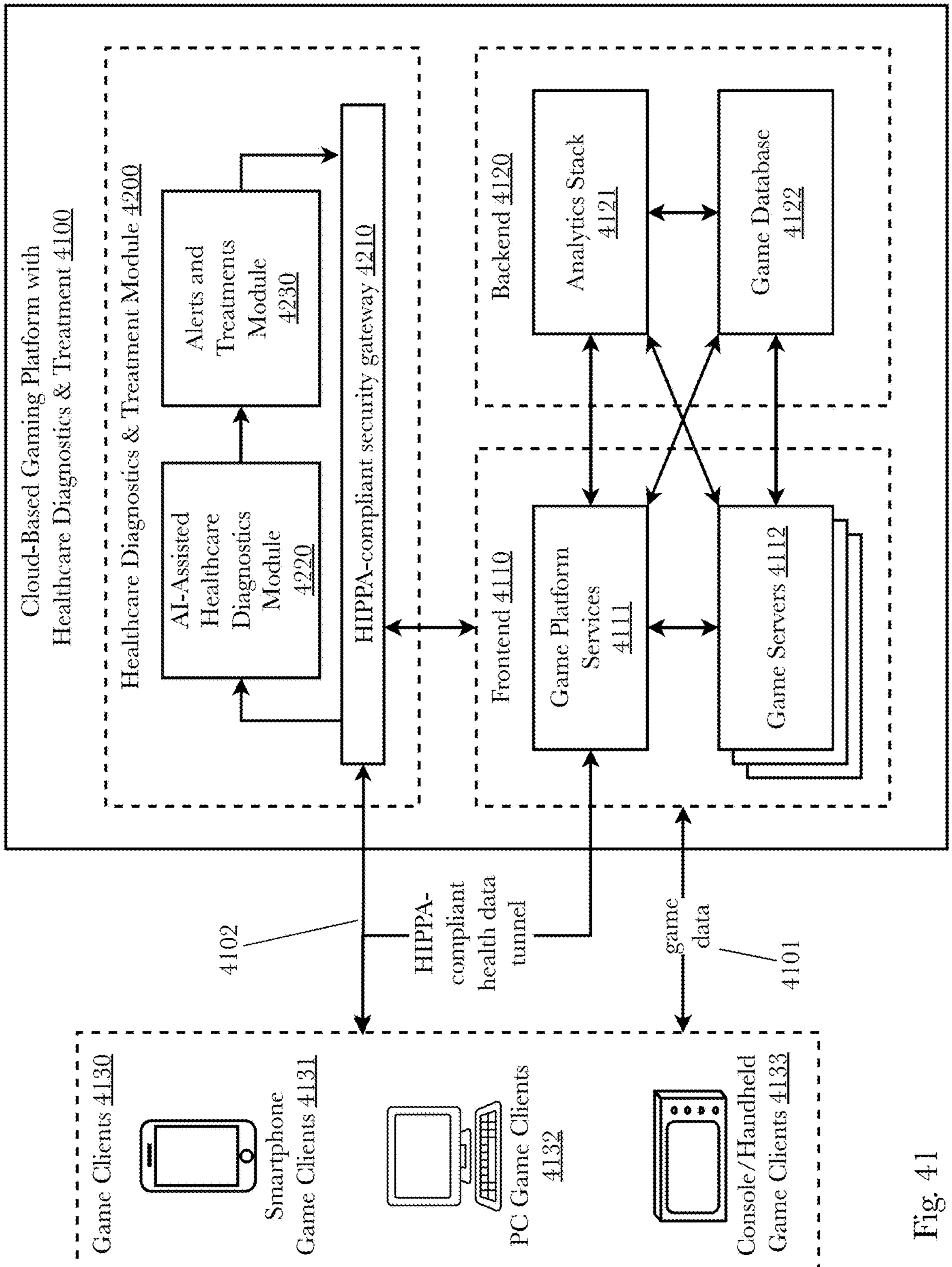

Cloud-Based Gaming Platform with Healthcare Diagnostics & Treatment 4100
Healthcare Diagnostics & Treatment Module 4200
AI-Assisted Healthcare Diagnostics Module 4220
Alerts and Treatments Module 4230
HIPPA-compliant security gateway 4210
Backend 4120
Analytics Stack 4121
Game Database 4122
Frontend 4110
Game Platform Services 4111
Game Servers 4112
4102
HIPPA-compliant health data tunnel
game data
4101
Game Clients 4130
Smartphone Game Clients 4131
PC Game Clients 4132
Console/Handheld Game Clients 4133

Fig. 41

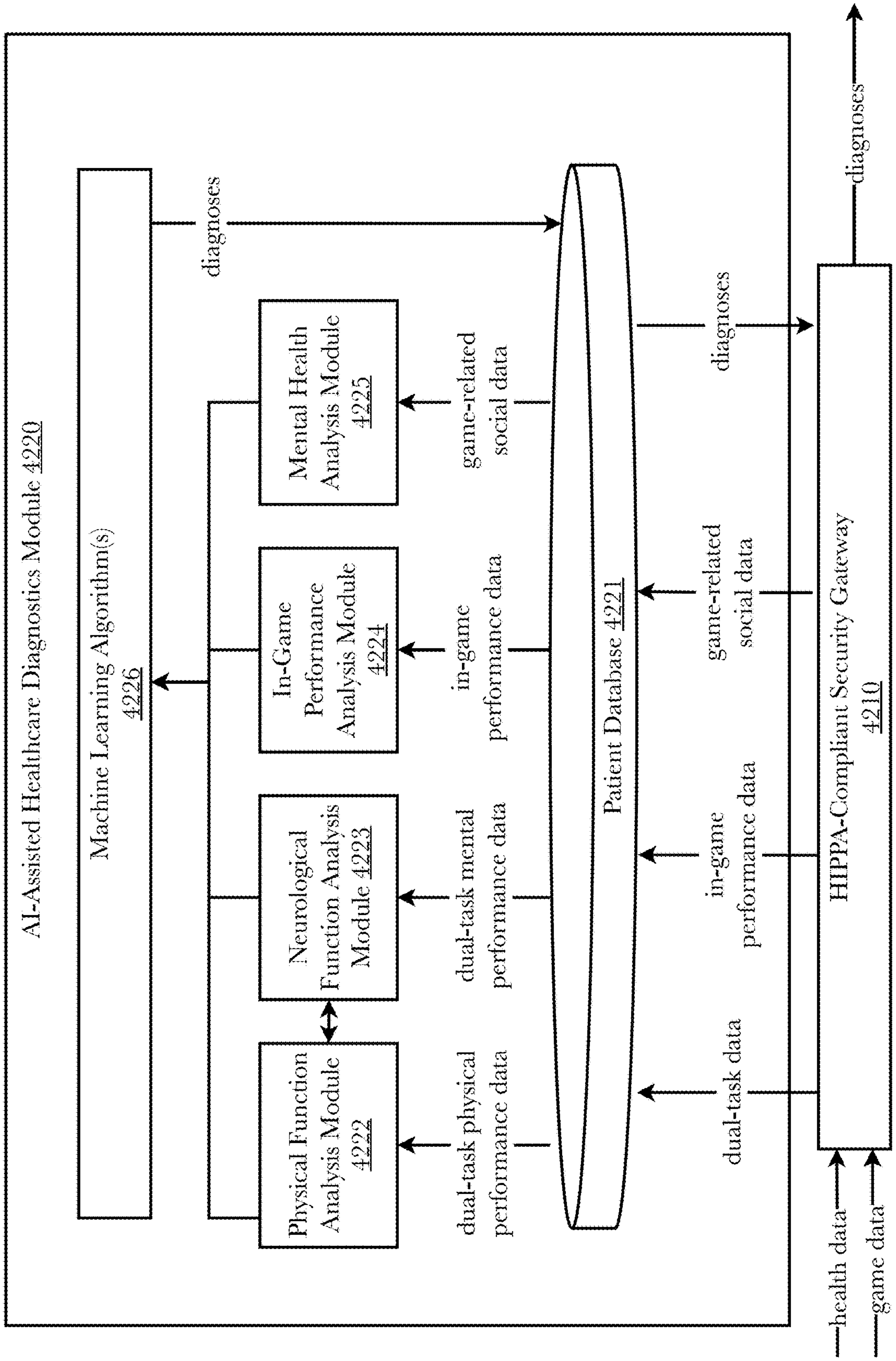

AI-Assisted Healthcare Diagnostics Module 4220
Machine Learning Algorithm(s) 4226
diagnoses
Physical Function Analysis Module 4222
Neurological Function Analysis Module 4223
In-Game Performance Analysis Module 4224
Mental Health Analysis Module 4225
dual-task physical performance data
dual-task mental performance data
in-game performance data
game-related social data
Patient Database 4221
diagnoses
dual-task data
in-game performance data
game-related social data
HIPPA-Compliant Security Gateway 4210
diagnoses
health data
game data

Fig. 42

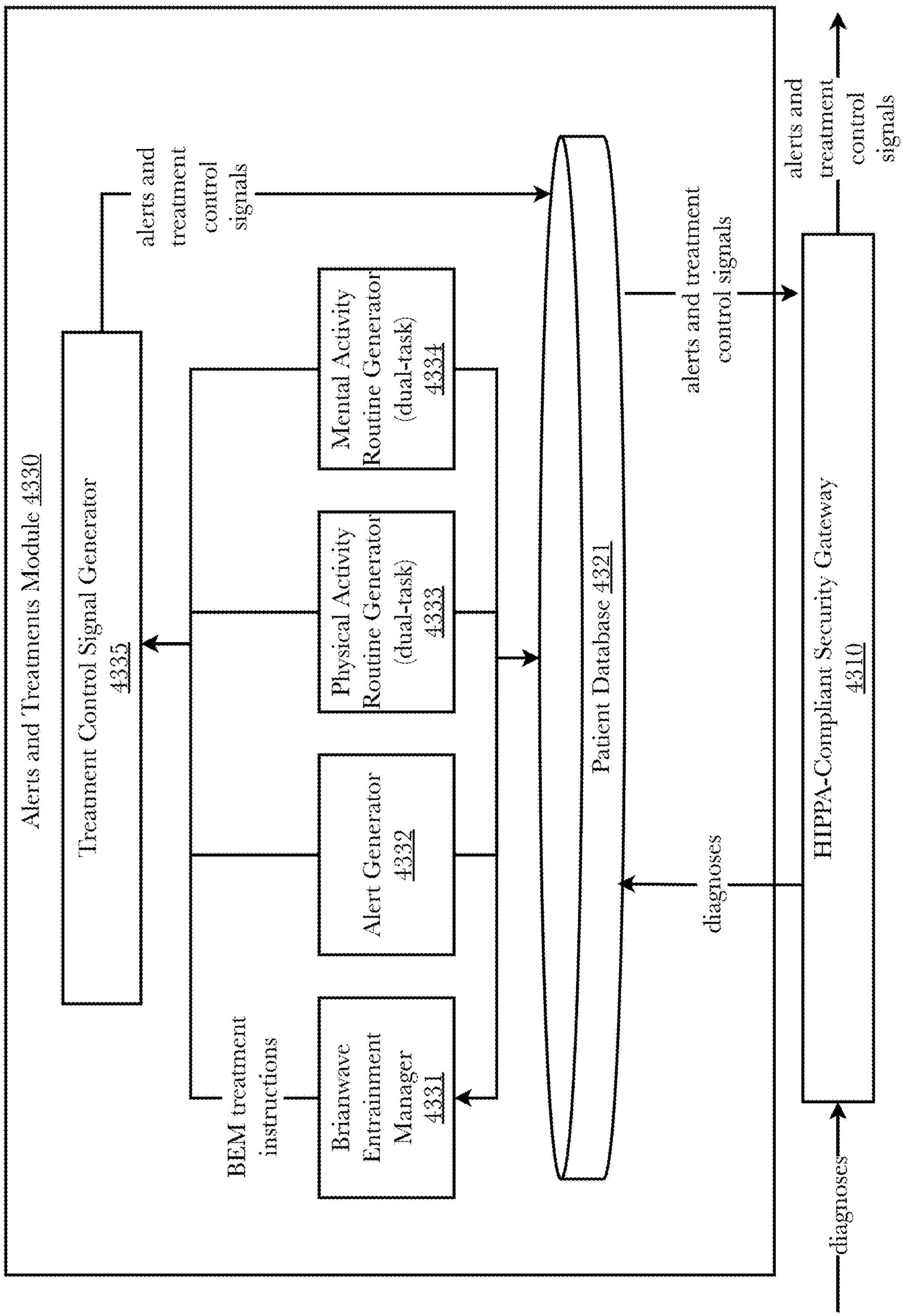

Alerts and Treatments Module 4330
Treatment Control Signal Generator 4335
alerts and treatment control signals
BEM treatment instructions
Brianwave Entrainment Manager 4331
Alert Generator 4332
Physical Activity Routine Generator (dual-task) 4333
Mental Activity Routine Generator (dual-task) 4334
Patient Database 4321
alerts and treatment control signals
diagnoses
HIPPA-Compliant Security Gateway 4310
alerts and treatment control signals
diagnoses

Fig. 43

TREATMENT OF NEUROLOGICAL FUNCTIONING AND CONDITIONS USING COMPUTER-ASSISTED DUAL-TASK METHODOLOGY

CROSS-REFERENCE TO RELATED APPLICATIONS

Priority is claimed in the application data sheet to the following patents or patent applications, each of which is expressly incorporated herein by reference in its entirety:

Ser. No. 18/171,330
Ser. No. 17/030,233
Ser. No. 17/030,195
Ser. No. 16/781,663
Ser. No. 16/354,374
Ser. No. 16/176,511
Ser. No. 16/011,394
Ser. No. 15/853,746
Ser. No. 15/219,115
Ser. No. 15/193,112
Ser. No. 15/187,787
Ser. No. 15/175,043
62/310,568
Ser. No. 14/846,966
Ser. No. 14/012,879
61/696,068
62/330,602
62/330,642
Ser. No. 16/927,704
Ser. No. 16/867,238
Ser. No. 16/793,915
Ser. No. 16/255,641
Ser. No. 16/223,034
62/697,973
Ser. No. 18/188,388
Ser. No. 18/080,343
Ser. No. 17/574,540
Ser. No. 17/888,449
Ser. No. 17/575,600
Ser. No. 16/951,281

BACKGROUND OF THE INVENTION

Field of the Art

The disclosure relates to the field of health devices, and more particularly to devices and methods for evaluation, detection, conditioning, and treatment of neurological functioning and conditions.

Discussion of the State of the Art

For many decades in the mid-20th century, the scientific consensus was that human adult brains stopped development upon reaching adulthood, and that neurogenesis and neuroplasticity did not occur in adults, and that the only changes that could occur were degenerative changes. Starting in the 1980s, some research countered that consensus, suggesting that neurogenesis and neuroplasticity do occur in adults. Today, it is generally accepted that neurogenesis and neuroplasticity can and do occur in the brains of human adults, that both physical activity and mental activity contribute to neurogenesis and neuroplasticity in adults, and that improvements in adult brain function can and do occur in specific regions of the brain. However, there is currently no methodology for treatment of neurological conditions of the brain, and particularly no methodology for treatment of targeted regions of the brain to cause neurogenesis and neuroplasticity in the targeted regions.

What is needed is a system and method for targeted treatment of human brain function using a combination of physical and mental activity to cause neurogenesis and neuroplasticity in targeted regions of the brain.

SUMMARY OF THE INVENTION

Accordingly, the inventor has conceived and reduced to practice, a system and method for targeted treatment of human brain function using a combination of physical and mental activity to cause neurogenesis and neuroplasticity in targeted regions of the brain using computer-enhanced dual-task analysis and treatment. The system and method involve having a subject engage in physical and mental activities at levels of intensity or stress associated with increased neurogenesis and neuroplasticity, the mental activities being associated with certain brain regions intended to be targeted by the treatment, wherein the physical and mental activities are assigned based on dual-task analyses of the subject's brain function and machine learning algorithms trained to optimize the stress levels associated with the physical and mental activities to maximize neurogenesis and neuroplasticity. The physical and mental activities may be continuously adjusted to ensure that the subject remains in appropriate ranges for neurogenesis and neuroplasticity for the brain region or regions being targeted for treatment.

According to a preferred embodiment, a system for treatment of neurological conditions is disclosed, comprising: a physical activity data capture device, configured to capture movement data associated with physical activity of persons exercising thereon; a computing device comprising a processor and a memory; a computer display configured to display outputs from a software application to persons undergoing treatment; the software application, comprising a first plurality of programming instructions stored in a memory which, when operating on the processor, causes the computing device to: receive a neurological condition of a person for treatment; select a primary activity for performance by the person, the primary activity comprising a physical activity associated with activation of portions of the human brain associated with the neurological condition and a recommended level of intensity of the physical activity also associated with activation of portions of the human brain associated with the neurological condition; select a secondary activity for performance by the person, the secondary activity being a mental activity to be engaged in by the person within a virtual reality environment controlled in whole or in part by the software application, the secondary activity also being associated with activation of portions of the human brain associated with the neurological condition; operate the secondary activity within the virtual reality environment; while the person engages in the primary task, record movement data associated with the physical activity using the physical activity data capture device; convert the movement data into movements in the virtual reality environment; display the virtual reality environment comprising the secondary activity and the movements to the person via the display; determine from the movement data an actual level of intensity of the physical activity, and either: provide feedback to the person via the display as to the actual level of intensity relative to the recommended level of intensity, such that the person can self-regulate the actual level of intensity to be similar to the recommended level of intensity; or adjust the operation of the physical activity data capture device to cause the actual level of intensity to more closely correspond with the recommended level of intensity; and record activity data associated with the person's engagement in the secondary activity within the virtual reality environment.

According to another preferred embodiment, a method for treatment of neurological conditions is disclosed, comprising the steps of: a computer display configured to display outputs from a software application to persons undergoing treatment; using a software application operating on a computing device comprising a processor and a memory to: receive a neurological condition of a person for treatment; select a primary activity for performance by the person, the primary activity comprising a physical activity associated with activation of portions of the human brain associated with the neurological condition and a recommended level of intensity of the physical activity also associated with activation of portions of the human brain associated with the neurological condition; select a secondary activity for performance by the person, the secondary activity being a mental activity to be engaged in by the person within a virtual reality environment controlled in whole or in part by the software application, the secondary activity also being associated with activation of portions of the human brain associated with the neurological condition; operate the secondary activity within the virtual reality environment; while the person engages in the primary task, record movement data associated with the physical activity using a physical activity data capture device, configured to capture movement data associated with physical activity of persons exercising thereon; convert the movement data into movements in the virtual reality environment; display the virtual reality environment comprising the secondary activity and the movements to the person via a computer display configured to display outputs from the software application to the person undergoing treatment; determine from the movement data an actual level of intensity of the physical activity, and either: provide feedback to the person via the display as to the actual level of intensity relative to the recommended level of intensity, such that the person can self-regulate the actual level of intensity to be similar to the recommended level of intensity; or adjust the operation of the physical activity data capture device to cause the actual level of intensity to more closely correspond with the recommended level of intensity; and record activity data associated with the person's engagement in the secondary activity within the virtual reality environment.

According to an aspect of an embodiment, a neurological functioning analyzer is used to: stop operation of the secondary activity; capture and analyze the movement data while the secondary task is stopped; stop operation of the primary activity and restart operation of the secondary activity; capture and analyze the activity data while the primary activity is stopped; restart operation of the primary activity; capture and analyze the movement data and the activity data while the primary activity and secondary activity are performed simultaneously; and calculate a composite functioning score for the person, the composite functioning score comprising an indication of the level of functionality of an aspect of the person's nervous system based on differences in performance between the separate performance of the primary activity and secondary activity relative to the combined performance of the primary activity and secondary activity; and generate a neurological condition profile of the person comprising an indication of relative functioning of the aspect of the person's nervous system.

According to an aspect of an embodiment, one or more primary tasks and one or more secondary activities are assigned to the person, a plurality of composite functioning scores are calculated comprising indications of the level of functionality of a plurality of aspects of the person's nervous system, and the neurological functioning profile comprises indications of relative functioning of the plurality of aspects of the person's nervous system.

According to an aspect of an embodiment, the neurological functioning analyzer creates a composite functioning score spatial map based on the neurological functioning profile.

According to an aspect of an embodiment, the software application changes the primary activity or secondary activity or selects a different primary activity or secondary activity based on changes to the person's composite function score while undergoing treatment.

According to an aspect of an embodiment, the neurological functioning analyzer determines an effectiveness of the treatment of the neurological condition based on changes to the person's neurological condition profile over time.

According to an aspect of an embodiment, the composite functioning score is compared to statistical composite functioning score data for a larger population to determine the person's relative position in the larger population with respect to the composite functioning score.

According to an aspect of an embodiment, the primary activity is exercise on a treadmill, elliptical machine, or stationary bicycle.

According to an aspect of an embodiment, the secondary activity involves overcoming a computer or video game-like challenge or a computer simulation of a life-like scenario or challenge.

According to an aspect of an embodiment, the system selects the primary activity and secondary activity based on changes to the person's neurological condition profile over time that suggest improvements in the person's performance.

BRIEF DESCRIPTION OF THE DRAWING FIGURES

The accompanying drawings illustrate several embodiments of the invention and, together with the description, serve to explain the principles of the invention according to the embodiments. It will be appreciated by one skilled in the art that the particular embodiments illustrated in the drawings are merely exemplary, and are not to be considered as limiting of the scope of the invention or the claims herein in any way.

FIG. **1** is a side view of an exemplary variable-resistance exercise machine with an embedded or a wireless computing device controlling the interactive software applications of the invention.

FIG. **2** is a top-down view of an exemplary variable-resistance exercise machine with an embedded or a wireless computing device controlling the interactive software applications of the invention.

FIG. **3** is a diagram illustrating an exemplary system for a virtual reality or mixed reality enhanced exercise machine, illustrating the use of a plurality of connected smart devices and tethers, and showing interaction via the user's body as a control stick.

FIG. **4** is a diagram of an exemplary apparatus for natural torso tracking and feedback for electronic interaction, illustrating the use of multiple tethers and a movable torso harness.

FIG. **5** is a diagram illustrating a variety of alternate tether arrangements.

FIG. **6** is a diagram of an additional exemplary apparatus for natural torso tracking and feedback for electronic interaction, illustrating the use of angle sensors to detect angled movement of tethers.

FIG. **7** is a diagram illustrating an exemplary apparatus for natural torso tracking and feedback for electronic interaction, illustrating the use of multiple tethers and a movable torso harness comprising a plurality of angle sensors positioned within the movable torso harness.

FIG. **8** is a block diagram of an exemplary system architecture for natural body interaction for mixed or virtual reality applications.

FIG. **9** is a block diagram of an exemplary system architecture for a stationary exercise bicycle being connected over local connections to a smartphone, an output device other than a phone, and a server over a network, according to an aspect.

FIG. **10** is a diagram of an exemplary hardware arrangement of a smart phone or computing device running a user identification component and communicating over a network, according to an aspect.

FIG. **11** is a block diagram of a method of mixed or virtual reality software operating to receive input through different sources, and send output to devices, according to an aspect.

FIG. **12** is a diagram illustrating an exemplary virtual reality or mixed reality enhanced exercise machine, illustrating the use of a plurality of optical sensors to detect body movement of a user during use of an exercise machine.

FIG. **13** is a block diagram illustrating an exemplary hardware architecture of a computing device.

FIG. **14** is a block diagram illustrating an exemplary logical architecture for a client device.

FIG. **15** is a block diagram showing an exemplary architectural arrangement of clients, servers, and external services.

FIG. **16** is another block diagram illustrating an exemplary hardware architecture of a computing device.

FIG. **17** is a block diagram of an exemplary virtual reality or mixed reality enhanced exercise machine, illustrating the use of a stationary bicycle with hand controls on the handles, and a belt-like harness attachment.

FIG. **18** is a diagram of another exemplary virtual reality or mixed reality enhanced exercise machine, illustrating the use of a treadmill exercise machine with a vest-type harness with a plurality of pistons to provide a hardware-based torso joystick with full-body tracking.

FIG. **19** is a diagram of another exemplary virtual reality or mixed reality enhanced exercise machine, illustrating the use of a stationary bicycle with a vest-type harness with a plurality of strain sensors and tethers.

FIG. **20** is a flow diagram illustrating an exemplary method for operating a virtual and mixed-reality enhanced exercise machine.

FIG. **21** is a system diagram of a key components in the analysis of a user's range of motion and balance training.

FIG. **22** is a diagram showing a system for balance measurement and fall detection.

FIG. **23** is a system diagram of a sensor measuring the range of motion of a user during a specific exercise.

FIG. **24** is a method diagram illustrating behavior and performance of key components for range of motion analysis and balance training.

FIG. **25** is a composite functioning score spatial map showing the relative ability of an individual in several physical and mental functional measurement areas.

FIG. **26** is an overall system architecture diagram for a neurological functioning analyzer.

FIG. **27** is a system architecture diagram for the data capture system aspect of a neurological functioning analyzer.

FIG. **28** is a system architecture diagram for the range of motion comparator aspect of a neurological functioning analyzer.

FIG. **29** is a system architecture diagram for the movement profile analyzer aspect of a neurological functioning analyzer.

FIG. **30** is a system architecture diagram for the neurological functioning analyzer aspect of a neurological condition evaluator.

FIG. **31** is an exemplary human/machine interface and support system for using body movements to interface with computers while engaging in exercise.

FIG. **32** is an exemplary method for application of the system to improve the performance of a sports team.

FIG. **33** (PRIOR ART) is a side-view diagram of the human brain showing the major areas of brain function.

FIG. **34** (PRIOR ART) is a diagram showing the process of neurogenesis in the human brain.

FIG. **35** (PRIOR ART) is a diagram showing the process of neuroplasticity in the human brain.

FIG. **36** (PRIOR ART) is a diagram showing the influence of exercise on neurogenesis and neuroplasticity.

FIG. **37** (PRIOR ART) is a diagram showing the influence of variable exercise intensity on neurogenesis and neuroplasticity.

FIG. **38** is a diagram showing an exemplary application of dual-task facilitation of neurogenesis and neuroplasticity for treatment of language, cognition, and memory brain functions.

FIG. **39** is a diagram showing an exemplary application of dual-task facilitation of neurogenesis and neuroplasticity for treatment of visual, sensory, and fine motor skill brain functions.

FIG. **40** is a diagram showing an exemplary application of dual-task facilitation of neurogenesis and neuroplasticity for treatment of gross motor skill, balance, and emotion brain functions.

FIG. **41** is a block diagram showing an exemplary architecture for a cloud-based gaming platform with integrated healthcare diagnostics and treatment.

FIG. **42** is a block diagram showing an exemplary architecture for an AI-assisted healthcare diagnostics module aspect of a cloud-based gaming platform with integrated healthcare diagnostics and treatment.

FIG. **43** is a block diagram showing an exemplary architecture for an alerts and treatments module aspect of a cloud-based gaming platform with integrated healthcare diagnostics and treatment.

DETAILED DESCRIPTION OF THE INVENTION

The inventor has conceived, and reduced to practice, a system and method for targeted treatment of human brain function using a combination of physical and mental activity to cause neurogenesis and neuroplasticity in targeted regions of the brain using computer-enhanced dual-task analysis and treatment. The system and method involve having a subject engage in physical and mental activities at levels of intensity or stress associated with increased neurogenesis and neuroplasticity, the mental activities being associated with certain brain regions intended to be targeted by the treatment, wherein the physical and mental activities are assigned based on dual-task analyses of the subject's brain function and machine learning algorithms trained to optimize the stress levels associated with the physical and mental activities to maximize neurogenesis and neuroplasticity. The physical and mental activities may be continuously adjusted to ensure that the subject remains in appropriate ranges for neurogenesis and neuroplasticity for the brain region or regions being targeted for treatment.

Exercise has been shown to have numerous positive effects on the brain, including enhancing neurogenesis (the growth of new neurons) and neuroplasticity (the brain's ability to change and adapt in response to experience). Studies have shown that regular exercise increases the production of new neurons in the hippocampus, a region of the brain that is involved in learning and memory. Exercise promotes the release of growth factors such as brain-derived neurotrophic factor (BDNF), which stimulate the production of new neurons and support their survival. Exercise also increases blood flow to the brain, which can enhance the delivery of oxygen and nutrients necessary for neurogenesis. Additionally, exercise has been shown to reduce stress and inflammation, which can impair neurogenesis. Exercise has also been shown to enhance neuroplasticity, which is the brain's ability to change and adapt in response to experience. Regular exercise can increase the strength and number of connections between neurons, known as synapses. This can lead to improvements in cognitive function and the ability to learn and remember new information. Exercise has also been shown to increase the production of neurotransmitters, such as dopamine and serotonin, which play important roles in regulating mood, motivation, and attention. These changes in neurotransmitter levels can lead to improved mental well-being and cognitive function.

Other studies have shown that time spent engaging in mental activities has similar effects. Intellectual activities such as working on complicated problems, solving puzzles, and similar cognitive activities also enhance neurogenesis and neuroplasticity.

However, while these effects of exercise and mental activities have been demonstrated in a general sense, there are no known methodologies for targeted neurogenesis and neuroplasticity which seek to improve particular areas of the brain or particular functions of the brain based on particular forms of exercise or particular types of mental activities. Further, there are no known methodologies for targeted neurogenesis and neuroplasticity which take advantage of interactions between the effects of exercise on the brain and the effects of mental activities on the brain. The combination of targeted neurogenesis and neuroplasticity from particular forms of exercise with targeted neurogenesis and neuroplasticity from particular types of mental activities has the potential to dramatically improve brain function. Methodologies as described herein for targeted neurogenesis and neuroplasticity are of benefit to all persons, but could have a dramatic impact on the lives of those suffering from neurological deficiencies such as elderly persons suffering from dementia, victims of brain trauma from accidents who suffer reduced cognition, and stroke victims who suffer reduced ability to move or communicate.

As lifespans have improved in the past few decades, particularly in more developed countries, the mean and median age of populations have increased. The greatest risk factor for neurodegenerative diseases is aging, so older persons are more likely to suffer from degenerative diseases and conditions affecting the nervous system such as amyotrophic lateral sclerosis, Parkinson's disease, Alzheimer's disease, fatal familial insomnia, Huntington's disease, Friedreich's ataxia, Lewy body disease, and spinal muscular atrophy. It has been estimated that some 20-40% of healthy people between 60 and 78 years old experience discernable decrements in cognitive performance in one or more areas including working, spatial, and episodic memory, and cognitive speed. Early stages of neurodegenerative diseases are difficult to detect, the causes of such diseases are not well understood, and treatments for such diseases are non-existent.

Without using one of the costly brain scan technologies, it remains difficult to detect, assess, and treat poor functioning of the nervous system, whether such poor functioning is due to injury to the brain, neurodegenerative disease, psychological or physical trauma, or changes in brain chemistry, diet, stress, substance abuse, or other factors. For certain neurological conditions, such as Chronic Traumatic Encephalopathy (CTE), none of the current brain scan technologies are able to reliably capture diagnostic data. Other neurological deficits and conditions can be evaluated or diagnosed using assessments using readily available equipment and observational analysis, such as the Cognitive Performance Test (CPT) and Timed Up and Go Test (TUG) but lack the sensitivity suitable for nuanced or early deficit detection. Each of these types of poor nervous system function can impact different parts of the brain and/or nervous system in different ways. Due to the complexity of interactions in the nervous system and the brain's ability to adapt its function in many areas, it remains difficult to detect poor functioning and to identify which neurological functions and anatomical aspects and regions are impacted early enough to implement an effective treatment protocol.

However, recent research studies have demonstrated that physical activity, especially aerobic exercise, can improve neurogenesis and other neurological functions, whether related to physical brain and nervous system impairments or mental health/emotional issues. In addition, evolutionary biologists have hypothesized that early humans began their cognitive revolution when they ventured into the African savannah and started walking upright. In fact, more recent research studies on the cerebellum, an ancient part of the brain that coordinates the motor control, have discovered unexpected connections between the cerebellum and other parts of the brain. Specifically, according to a team of researchers from the University of Washington, only 20 percent of the cerebellum connections was dedicated to areas involved in physical motion, while 80 percent was connected to areas involved in functions such as abstract thinking, planning, emotion, memory and language. The cerebellum doesn't actually execute tasks like thinking, just as it doesn't directly control movement. Instead, it monitors and coordinates the brain areas that are doing the work and makes them perform better.

Therefore, simultaneous testing of primary physical tasks such as walking or running and the secondary activities that include various mental, other physical activities as well as emotional experiences (commonly known as a dual task assessment), and the correlation of results therefrom can be used to evaluate and treat specific neurological functional areas to create a profile of relative neurological functioning and see where deficiencies may be present. Therefore, changes in a person's walking gait while the person is engaged in other secondary activities like solving a logic puzzle could be analyzed and compared against the normal or average dual-tasking costs of the same population group for relative functioning as well as anomalies. Such anomalies for the given brain functions or regions could be indicative of abnormal central nervous system functions. Further, the combination of the dual-tasked physical and secondary activities can help identify the abnormally-performing neurological functions or even help isolate affected neurological regions. For example, a walking gait/logic puzzle dual-task activity may indicate normal functioning in a given individual, indicating that autonomous physical activity and cognition are not affected. However, in the same individual another dual task of walking and listening within a virtual reality (VR) environment may result in gait changes or a complete stop of the walk as the neurological functions required for these tasks are different from walking and logic. In this case, it may indicate that there may be injury to or degeneration of the auditory cortex of the temporal lobe, potentially informing further diagnostic procedures. These same dual-task activities that allow for evaluation of brain function can be used as a form of targeted treatment for those brain functions. Neurogenesis and neuroplasticity occur in regions of the brain that are stimulated by mental activity, and this effect is enhanced by the type and amount of exercise engaged in by a person during the mental activity. As a result, a system combining numerous combinations of various dual-tasking activities, covering all neurological functions or regions, may be able to evaluate, detect, and treat neurological deficits and conditions even before they become noticeably symptomatic. For individuals for whom symptoms are already present, such a system can evaluate and track changes over time, and potentially slow down or reverse the progression of such deficits and conditions.

In various embodiments described herein, the system and method involve having a subject engage in a primary physical task, wherein movement data is gathered concerning indicators of physical function such as posture, balance, gait symmetry and stability, and consistency and strength of repetitive motion (e.g., walking or running pace and consistency, cycling cadence and consistency, etc.) along with other biometric data (e.g., heart rate, heart rate variability, galvanic skin response, pupil dilation, facial expression, electroencephalogram, etc.). Simultaneously, the person is asked to engage in a range of secondary activities that will each stimulate a specific neurological function or region and collectively cover all aspects of the nervous system. These secondary activities include mental activities, other physical activities, as well as emotional experiences, such as listening, reading, speaking, fine and gross motor movements, mathematics, logic puzzles, executive decisions, navigation, short- and longer-term memory challenges, empathic and traumatic scenarios, etc. The biometric and performance data from the primary physical task and the secondary activities are combined to generate a composite functioning score visualization indicating the relative functioning of primary physical tasks and the secondary neurological functions and, which can then be analyzed and compared against the population averages (from a larger population dataset) and benchmarks. In addition to seeing the calculated composite function score, in some cases experts and users may be given discretionary access to all or aspects of the underlying data used in computing the score.

Using this same dual-tasking analysis, it is also possible to evaluate, detect, and treat neurological conditions and changes involving mental health and emotional issues. For example, elevated heart rate, elevated blood pressure, or chest pain during exercise that are higher than an individual's normal history for these indicators can indicate emotional stress. The addition of story-telling or emotional experiences through computer games and/or simulations (and especially when such experiences are virtual-reality experiences) can help to elicit emotional and physiological responses or lack thereof. For example, a veteran suffering from PTSD (Post-Traumatic Stress Disorder) could be trained inside such a dual-tasking VR environment so that s/he can gradually regain her/his agency by overcoming progressively challenging physical and emotional scenarios—reactivating her/his dorsolateral prefrontal cortex and lateral nucleus of thalamus with the help of these combined physical and emotional activities (likely using parallel but not war-based scenarios). As a result, the veteran could potentially extricate herself or himself from such traumatic experiences by developing her/his closure stories.

The integration of a primary physical task with a secondary activity is also especially well-suited for the evaluation and conditioning of specific aspects of neurological functioning in individuals training for physical, mental, or combined forms of competition. After an initial array of primary physical challenges and associated tasks designed to evaluate specific neurological functioning areas to create a profile of relative functioning a more thorough understanding of the competitor's strengths and weaknesses in their specific mode of competition can be achieved. With the help of a conditioning recommendation algorithm, expert input, and competitor input a regimen of physical and secondary tasks specifically suited to improve performance of that competitor and mode of competition can be administered at prescribed or chosen frequency. Digital challenges can further be customized for competition and competitor specificity as the conditioning recommendation algorithm analyzes the efficacy of conditioning regimens for users aiming to improve in similar neurological functions, the specific user's response to conditioning inputs over time, and expert recommendations for users with similar neurological functioning profiles and objectives.

One or more different inventions may be described in the present application. Further, for one or more of the inventions described herein, numerous alternative embodiments may be described; it should be appreciated that these are presented for illustrative purposes only and are not limiting of the inventions contained herein or the claims presented herein in any way. One or more of the inventions may be widely applicable to numerous embodiments, as may be readily apparent from the disclosure. In general, embodiments are described in sufficient detail to enable those skilled in the art to practice one or more of the inventions, and it should be appreciated that other embodiments may be utilized and that structural, logical, software, electrical and other changes may be made without departing from the scope of the particular inventions. Accordingly, one skilled in the art will recognize that one or more of the inventions may be practiced with various modifications and alterations. Particular features of one or more of the inventions described herein may be described with reference to one or more particular embodiments or figures that form a part of the present disclosure, and in which are shown, by way of illustration, specific embodiments of one or more of the inventions. It should be appreciated, however, that such features are not limited to usage in the one or more particular embodiments or figures with reference to which they are described. The present disclosure is neither a literal description of all embodiments of one or more of the inventions nor a listing of features of one or more of the inventions that must be present in all embodiments.

Headings of sections provided in this patent application and the title of this patent application are for convenience only, and are not to be taken as limiting the disclosure in any way.

Devices that are in communication with each other need not be in continuous communication with each other, unless expressly specified otherwise. In addition, devices that are in communication with each other may communicate directly or indirectly through one or more communication means or intermediaries, logical or physical.

A description of an embodiment with several components in communication with each other does not imply that all such components are required. To the contrary, a variety of optional components may be described to illustrate a wide variety of possible embodiments of one or more of the inventions and in order to more fully illustrate one or more aspects of the inventions. Similarly, although process steps, method steps, algorithms or the like may be described in a sequential order, such processes, methods and algorithms may generally be configured to work in alternate orders, unless specifically stated to the contrary. In other words, any sequence or order of steps that may be described in this patent application does not, in and of itself, indicate a requirement that the steps be performed in that order. The steps of described processes may be performed in any order practical. Further, some steps may be performed simultaneously despite being described or implied as occurring non-simultaneously (e.g., because one step is described after the other step). Moreover, the illustration of a process by its depiction in a drawing does not imply that the illustrated process is exclusive of other variations and modifications thereto, does not imply that the illustrated process or any of its steps are necessary to one or more of the invention(s), and does not imply that the illustrated process is preferred. Also, steps are generally described once per embodiment, but this does not mean they must occur once, or that they may only occur once each time a process, method, or algorithm is carried out or executed. Some steps may be omitted in some embodiments or some occurrences, or some steps may be executed more than once in a given embodiment or occurrence.

When a single device or article is described herein, it will be readily apparent that more than one device or article may be used in place of a single device or article. Similarly, where more than one device or article is described herein, it will be readily apparent that a single device or article may be used in place of the more than one device or article.

The functionality or the features of a device may be alternatively embodied by one or more other devices that are not explicitly described as having such functionality or features. Thus, other embodiments of one or more of the inventions need not include the device itself.

Techniques and mechanisms described or referenced herein will sometimes be described in singular form for clarity. However, it should be appreciated that particular embodiments may include multiple iterations of a technique or multiple instantiations of a mechanism unless noted otherwise. Process descriptions or blocks in figures should be understood as representing modules, segments, or portions of code which include one or more executable instructions for implementing specific logical functions or steps in the process. Alternate implementations are included within the scope of embodiments of the present invention in which, for example, functions may be executed out of order from that shown or discussed, including substantially concurrently or in reverse order, depending on the functionality involved, as would be understood by those having ordinary skill in the art.

Definitions

The phrases "neurological functioning" and "neurological function" as used herein mean any and all aspects of neuroscience and neurology where input, output, processing, or combination thereof involve aspects of the nervous system. These include but are not limited to functional as well as anatomical aspects of cognitive, sensory, motor, emotional, and behavioral functions and experiences.

The term "expert" as used herein means an individual with specialization in an area via formal training, credentials, or advanced proficiency in a modality of interest to the user or with regard to neurological functioning. This includes but is not limited to physicians, psychiatrists, physical therapists, coaches, fitness trainers, high level athletes or competitors, and teachers.

The term "conditioning" as used herein means all aspects of the system that can be used for the improvement, training, treatment of or exposure to aspects of neurological functioning. This could be in the form of a prescribed regimen from an expert, recommendation algorithm, self-selected experiences, or combination thereof.

The phrase "composite function score" as used herein means a indicative of a relative level of neurological functioning comprised of weighted input of combined movement, biometric, and performance data sources collected by a given embodiment of the system, input by the user or an expert, historical performance and life history data from various sources, etc.

The phrase "dual task assessment" as used herein means measurement of baseline performance on a set of tasks and/or activities performed individually, as well as performance of the same set of tasks and/or activities simultaneously. While this is typically a single primary task (usually motor) combined with a single secondary activity (typically a neurological activity such as cognitive task), it should be taken herein to include other combinations of multiplexed tasks in combinations including, but not limited to, combinations in excess of two tasks and combinations that target a single or multiple aspects of neurological functioning.

The phrase "dual task cost" as used herein means any method for quantifying the difference in performance of a dual task assessment between the set of tasks performed individually and the same set of tasks performed simultaneously. Typically includes a comparison of each task performed in isolation to the performance on each of those tasks when performed simultaneously, either for a pair or larger combination of tasks.

The term "biometrics" as used herein mean data that can be input, directly measured, or computed using directly measured data from a user. This data includes but is not limited to physical and virtual movement, physiological, biological, behavioral, navigational, cognitive, alertness and attention, emotional, and brainwave measurements and patterns.

The phrase "primary task" as used herein means a first task or activity to be engaged in by an individual under assessment. The primary task will often, but not always, be a physical task or exercise such as walking on a treadmill.

The phrase "secondary activity" (or "associative activity") as used herein means a second task or activity to be engaged in by an individual under assessment. The secondary activity will often, but not always, be a mental or cognitive task such as performing arithmetic or identifying objects on a display.

Conceptual Architecture

FIG. **1** is a side view of a variable-resistance exercise machine with wireless communication for smart device control and interactive software applications **100** of the invention. According to the embodiment, an exercise machine **100** may have a stable base **101** to provide a platform for a user to safely stand or move about upon. Additional safety may be provided through the use of a plurality of integrally-formed or detachable side rails **102**, for example having safety rails on the left and right sides (with respect to a user's point of view) of exercise machine **100** to provide a stable surface for a user to grasp as needed. Additionally, side rails **102** may comprise a plurality of open regions **105***a-n* formed to provide additional locations for a user to grasp or for the attachment of additional equipment such as a user's smart device (not shown) through the use of a mountable or clamping case or mount. Formed or removable supports **106***a-n* may be used for additional grip or mounting locations, for example to affix a plurality of tethers (not shown) for use in interaction with software applications while a user is using exercise machine **100** (as described below, referring to FIG. **3**).

Exercise machine **100** may further comprise a rigid handlebar **103** affixed or integrally-formed on one end of exercise machine **100**, for a user to hold onto while facing forward during use. Handlebar **103** may further comprise a stand or mount **104** for a user's smart device such as (for example) a smartphone or tablet computer, so they may safely support and stow the device during use while keeping it readily accessible for interaction (for example, to configure or interact with a software application they are using, or to select different applications, or to control media playback during use, or other various uses). Handlebar **103** may be used to provide a stable handle for a user to hold onto during use for safety or stability, as well as providing a rigid point for the user to "push off" during use as needed, for example to begin using a moving treadmill surface (described below in FIG. **2**). During use, a user may also face away from handlebar **103**, using exercise machine **100** in the reverse without their view or range of motion being obscured or obstructed by handlebar **103** (for example, for use with a virtual reality game that requires a wide degree of movement from the user's hands for interaction).

As illustrated, the base **101** of exercise machine **100** may be formed with a mild, symmetrical curvature, to better approximate the natural range of movement of a user's body during use. Common exercise machines such as treadmills generally employ a flat surface, which can be uncomfortably during prolonged or vigorous use, and may cause complications with multi-directional movement or interaction while a user's view is obscured, as with a headset (described below in FIG. **3**). By incorporating a gradual curvature, a user's movements may feel more natural and require less reorientation or accommodation to become fluid and proficient, and stress to the body may be reduced.

FIG. **3** is a diagram illustrating an exemplary system for a virtual reality or mixed reality enhanced exercise machine **100** with wireless communication for smart device control and interactive software applications using a smart device, illustrating the use of a plurality of connected smart devices and tethers, and showing interaction via the user's body as a control stick. According to the embodiment, a user **301** may be standing, walking, or running on a variable-resistance exercise machine **100** with wireless communication for smart device control and virtual reality applications with a stable base **101** and two separate moveable surfaces **203***a*, **203***b* for separate movement of the user's legs. Exercise machine **100** may have fixed handlebars with affixed or integrally-formed controllers **305***a*, **305***b* for use as connected smart devices for interaction, and support rails **201***a*, **201***b* for a user to hold onto or affix tethers for safety or interaction when needed. User **401** may interact with software applications using a variety of means, including manual interaction via controller devices **305***a*, **305***b* that may be held in the hand for example to use as motion-input control devices or (as illustrated) may be affixed or integrally-formed into exercise machine **100**. This may provide a user with traditional means of interacting with software applications while using exercise machine **100**. Additionally, a user's body position or movement may be tracked and used as input, for example via a plurality of tethers **304***a-n* affixed to handlebars **201***a*, **201***b* and a belt, harness or saddle **303** worn by user **301**, or using a headset device **302** that may track the position or movement of a user's head as well as provide video (and optionally audio) output to the user, such as a virtual reality headset that displays images while blocking the user's view of the outside world, or an augmented reality or mixed reality headset that combines presented information with the user's view using transparent or semitransparent displays (for example, using transparent OLED displays, hologram displays, projected displays, or other various forms of overlaying a display within a user's normal field of vision without obstructing the user's view). Body tracking may be used to recognize additional input data from user **301** (in addition to manual input via controllers **305***a*, **305***b*), by tracking the position and movement of user **301** during use. For example, motion tracking within a headset device **302** may be used to recognize a variety of translational **310** or rotational **320** movement of user's **301** head, such as leaning to the side, or looking over the shoulder. Tethers **304***a-n* may recognize a variety of movement of user's **301** torso, such as leaning, crouching, side-stepping, or other body movement. This body tracking may then be utilized either as feedback to rehab programs (for example, to track a user's posture for physical therapy coaching or exercises such as holding yoga poses) or input similar to a control stick or joystick in manual controller arrangements, for example by interpreting the user's entire body as the "stick" and processing their body movements as if they were stick movements done manually (such as to control in-game character posture or movement, or to direct movement in certain applications such as vehicle simulations that may turn or accelerate in response to stick movements).

For example, a user **301** on exercise machine **100** may be playing a virtual reality skiing game or rehab program wherein they are given audio and video output via a headset **302** to immerse them in a virtual ski resort. When user **301** is not skiing, they may be able to use manual controls **305***a*, **305***b* for such operations as selecting from an on-screen menu, or typing text input such as to input their name or to chat with other players using text. When they begin skiing within the game, user **301** may be instructed in proper ski posture or technique, and may then use their body to control various aspects of their virtual skiing, such as leaning to the side **320** to alter their course and avoid trees or other skiers, or jumping **310** to clear rocks or gaps. Movement of their head may be detected by a headset **302** and used to control their view independently of their body as it is tracked by tethers **304***a-n*, allowing user **301** to look around freely without interfering with their other controls. In this manner, the user's entire body may serve as an input control device for the game, allowing and encouraging them to use natural body movements to control their gameplay in an immersive manner while still retaining the option to use more familiar manual control means as needed. Alternatively, specific body functions such as hip twisting are used as user feedback for rehabilitating programs, including rehab games.

FIG. **12** is a diagram illustrating an exemplary system **1200** for a virtual reality or mixed reality enhanced exercise machine **100**, illustrating the use of a plurality of optical sensors to detect body movement of a user during use of an exercise machine. As above (with reference to FIG. **3**), a user **301** may be standing, walking or running, sitting, or otherwise physically active during use of an exercise machine **100**. During use, the user's position, posture, movement, cadence, technique, or any other movement or position-related information may be detected, observed, or measured using a plurality of body movement sensors such as (for example, including but not limited to) tethers **304***a-n* that may optionally be affixed to handlebars **201***a-b* or other features of an exercise machine **100**, hardware sensors integrated into controllers **305***a-b* or a headset **302** the user may be using during exercise for virtual reality or mixed reality applications, or using a plurality of optical sensors **1201***a-n* that may be affixed to an exercise machine **100** or adjacent equipment, or that may be affixed to or positioned within an environment around exercise machine **100** to observe the user **301** during use. Optical sensors **1201***a-n* may be used in a variety of configurations or arrangements, such as using a single wide-angle sensor positioned to observe a user's movement or posture from a particular angle (which may be useful for coaching or physical therapy applications), or using more than one sensor placed about a user to observe their movement in three-dimensional space. A variety of hardware may be utilized in optical sensors **1201***a-n*, for example including (but not limited to) an infrared or other optical camera that may directly observe the user's movement, a structured-light emitter that projects a structured-light grid **1202** or other arrangement onto the user, exercise machine, or environment (and corresponding scanner or receiver that may observe the user's movement through detected changes in the structured-light projection), or a light-field sensor that detects or measures depth to observe a user's movement in three-dimensions. It should also be appreciated that various combination of optical sensors **1201***a-n* may be utilized to achieve a desired effect, for example using both structured light and a light-field sensor to observe a user's movement in precise detail in three dimensions. Additionally, some or all optical sensors **1201***a-n* utilized in some arrangements may be integrated into a user's headset **302** or an exercise machine **100** to provide "inside-out" tracking where tracking sensors are associated with the user rather than the environment, or they may be external devices as illustrated that may be introduced to enhance an existing exercise machine or environment.

Utilizing an exercise machine **100** in this manner allows for a variety of novel forms of user interaction within virtual reality or mixed reality applications. For example, a user's body movement during exercise may be tracked in three dimensions and along or around various axes to record movement with six degrees of freedom (6DOF) comprising both translation along, and rotation about, each of three spatial axes. This may be used with torso tracking as described above (referring to FIGS. **3-7**) to produce a 6DOF "torso joystick" virtual device that directs movement or other inputs within a software application. This may be used in a number of ways, for example including but not limited to aiding exercise through interactive coaching (either with a human coach or using software to simulate a coach by providing feedback to detected user movements), providing physical therapy, interacting with games or other applications during exercise, or using exercise combined with software interaction for an immersive virtual reality or mixed reality experience. For example, a user may control movement or expression of a virtual avatar or other user representation within a software application, such as using their own body movements to direct movement of a virtual character. Physical therapy or fitness coaching may utilize detected movements to assist a user with improving their abilities or technique, or to measure progress. Social interaction applications may utilize body movements during exercise, for example a chat or voice call application may utilize body movement as a form of nonverbal expression similar to emoji or other icons. Safety may also be enhanced by controlling the operation of software in response to detected user movements, for example displaying caution information or pausing an application if a user is detected to move outside a configured safety parameter (such as stepping off a running treadmill, for example).

FIG. **8** is a block diagram of an exemplary system architecture **800** for natural body interaction for mixed or virtual reality applications of the invention. According to the embodiment, a composition server **801** comprising programming instructions stored in a memory **11** and operating on a processor **12** of a computing device **10** (as described below, with reference to FIG. **13**), may be configured to receive a plurality of input data from various connected devices. Such input devices may include (but are not limited to) a variety of hardware controller devices **804** (such as a gaming controller [such as GOJI PLAY™ controllers], motion tracking controller, or traditional computer input devices such as a keyboard or mouse), a headset device **803** such as an augmented reality or mixed or virtual reality headset (for example, OCULUS RIFT™, HTC VIVE™, SAMSUNG GEAR VR™, MICROSOFT MIXED REALITY™, or other headset devices), a variety of fitness devices **805** (for example, fitness tracking wearable devices such as FITBIT™, MICROSOFT BAND™, APPLE WATCH™, or other wearable devices), or a variety of body input **802** tracking devices or arrangements, such as using a plurality of tethers attached to the environment and a harness worn by a user, configured to track movement and position of the user's body.

Various input devices may be connected to composition server **801** interchangeably as desired for a particular arrangement or use case, for example a user may wish to use a controller **804** in each hand and a headset **803**, but omit the use of fitness devices **805** altogether. During operation, composition server **801** may identify connected devices and load any stored configuration corresponding to a particular device or device type, for example using preconfigured parameters for use as a default configuration for a new controller, or using historical configuration for a headset based on previous configuration or use. For example, a user may be prompted (or may volunteer) to provide configuration data for a particular device, such as by selecting from a list of options (for example, "choose which type of device this is", or "where are you wearing/holding this device", or other multiple-choice type selection), or composition server **801** may employ machine learning to automatically determine or update device configuration as needed. For example, during use, input values may be received that are determined to be "out of bounds", for example an erroneous sensor reading that might indicate that a user has dramatically shifted position in a way that should be impossible (for example, an erroneous reading that appears to indicate the user has moved across the room and back again within a fraction of a second, or has fallen through the floor, or other data anomalies). These data values may be discarded, and configuration updated to reduce the frequency of such errors in the future, increasing the reliability of input data through use.

Composition server **801** may receive a wide variety of input data from various connected devices, and by comparing against configuration data may discard undesirable or erroneous readings as well as analyze received input data to determine more complex or fine-grained measurements. For example, combining input from motion-sensing controllers **804** with a motion-sensing headset **803** may reveal information about how a user is moving their arms relative to their head or face, such as covering their face to shield against a bright light or an attack (within a game, for example), which might otherwise be impossible to determine with any reliability using only the controllers themselves (as it may be observed that a user is raising their hands easily enough, but there is no reference for the position or movement of their head). These derived input values may then be combined into a single composite input data stream for use by various software applications, such as augmented reality or mixed or virtual reality productivity applications (for example, applications that assist a user in performing manual tasks by presenting virtual information overlays onto their field of vision, or by playing audio directions to instruct them while observing their behavior through input devices, or other such applications), or mixed or virtual reality applications or games, such as simulation games that translate a user's movement or position into in-game interaction, for example by moving a user's in-game character or avatar based on their physical movements as received from input devices. In some arrangements, composition server **801** may operate such software applications in a standalone manner, functioning as a computer or gaming console as needed. In other arrangements, composition server **801** may provide the composite data for use by an external computer **810**, such as a connected gaming console, mixed or virtual reality device, personal computer, or a server operating via a network in the cloud (such as for online gaming arrangements, for example). In this manner, the composite data functions of the embodiment may be utilized with existing hardware if desired, or may be provided in a standalone package such as for demonstrations or public use, or for convenient setup using a single device to provide the full interaction experience (in a manner similar to a household gaming console, wherein all the functions of computer components may be prepackaged and setup to minimize difficulty for a new user).

It should be appreciated that while reference is made to virtual reality applications, a wide variety of use cases may be possible according to the embodiment. For example, torso tracking may be used for fitness and health applications, to monitor a user's posture or gait while walking, without the use of additional virtual reality equipment or software. In some arrangements, some or all interaction between a user and a software application may be nonvisual, and in some arrangements no display device may be present. In such an arrangement, a user may interact with software entirely using feedback and movement of a worn harness **420** or tethers **304***a-n*, using resistance or software-guided actuation of tethers **304***a-n* (as described below, with reference to FIGS. **4-7**) or other elements. In other arrangements, various combinations of display devices and other electronic devices may be used for a mixed-reality setup, for example where a user's movement and interaction may be used by software to incorporate elements of the physical world into a digital representation of the user or environment. For example, a user may interact with games or fitness applications, participate in social media such as chat, calls, online discussion boards, social network postings, or other social content, or they may use body tracking to navigate user interface elements of software such as a web browser or media player. Software used in this manner may not need to be specially-configured to utilize body tracking, for example to navigate a web browser a user's body movements or reactions to feedback may be processed by a composition server **801** and mapped to generic inputs such as keystrokes or mouse clicks, for use in any standard software application without the need for special configuration.

It should be further appreciated that while reference is made to a treadmill-type exercise machine **100**, such an exercise machine is exemplary and any of a number of exercise machines may be utilized according to the aspects disclosed herein, for example including (but not limited to) a treadmill, a stationary bicycle, an elliptical machine, a rowing machine, or even non-electronic exercise equipment such as a pull-up bar or weight machine. Traditional exercise equipment may be outfitted with additional components to facilitate virtual reality or mixed reality interaction according to the aspects disclosed herein, for example by affixing a plurality of tethers **304***a-n* to a weight machine so that a user's movement during exercise may be used as interaction as described below (with reference to FIGS. **3-7**).

FIG. **25** is a composite functioning score spatial map **2500** showing the relative ability of a user in several physical and mental functional measurement areas (also referred to herein as "composite functioning scores" or "composite functioning score groups") **2501-2507**. The composite functioning score spatial map is a visual representation of a person's ability in several functional measurement areas **2501-2507**. The center of the composite functioning score spatial map **2500** represents zero ability, while the inner circle **2510** of the composite functioning score spatial map **2500** represents full ability (i.e., maximum functionality of a healthy individual while not dual-tasking). Greater functionality in a given composite functioning score **2501-2507** is represented by a greater profile coverage area in the direction of that functional measurement area. The average profile area of a representative population of individuals (e.g., of the same age as the individual being tested) is shown as the solid line profile average **2511** of the composite functioning score spatial map **2500**. The composite functioning score spatial map **2500** is a visual representation of data obtained from other components of the system and placed into a composite functioning score matrix or other data structure (not shown) which organizes the data relative to the various composite functioning scores.

In this example, there are seven groups of composite functioning scores, each representing either a physical ability, a mental ability, or a combined ability, and all of which together represent a picture of an individual's nervous system function. The memory **2501** and cognition **2502** composite functioning score groups represent purely mental activities, and present a picture of the individual's ability to think clearly. The speech **2503**, auditory **2504**, and vision **2505** composite functioning score groups represent combined physical/mental activities, as each represents some physical/mental interaction on the part of the individual. For example, speech requires the individual not only to mentally generate words and phrases on a mental level, but also to produce those words and phrases physically using the mouth and vocal cords. It is quite possible, for example, that the individual is able to think of the words, but not produce them, which represents one type of neurological condition. The speech **2503** composite functioning score group represents that combined ability, and the auditory **2504** and vision **2505** composite functioning score groups represent a similar combined ability. The motor skills **2506** composite functioning score group represents a mostly-physical ability to move, balance, touch, hold objects, or engage in other non-cognitive activities (recognizing, of course, that the nervous system controls those movements, but is not engaged in higher-level thinking). The emotional biomarker **2507** group represents the individual's emotional responses to certain stimuli during testing, as would be indicated by lack of empathetic responses to virtual reality characters in a story, responses indicating sadness or depression, etc.

From the data obtained from other components of the system, a profile of an individual's functional ability may be created and displayed on the composite functioning score spatial map. For example, a baseline profile **2508** may be established for an individual during the initial use or uses of the system (e.g., pre-treatment evaluation(s)), showing a certain level of ability for certain composite functioning scores. In the baseline profile **2508** example, all composite functioning scores indicate significant impairment relative to the population average **2511**, but the composite functioning scores for cognition **2502** and auditory **2504** ability are relatively stronger than the composite functioning scores for memory **2501**, speech **2503**, vision **2505**, and motor skills **2506**, and the emotional biomarker group **2507** indicates substantial impairment relative to the population average **2511**. Importantly, changes in the profile can show improvements or regressions in functionality, and changes over time in the profile can be tracked to show trends in improvement or regression. For example, a later profile **2509** for the same individual shows improvement in all biomarker groups, with substantial improvement in the cognition **2502**, auditory **2504**, motor skill **2506** biomarker groups, and dramatic improvement in the emotion **2507** composite functioning score groups, relative to the baseline profile **2508**. The biomarker group for emotion **2507** in the later profile **2509** shows performance matching or nearly matching that of the population average **2511**.

FIG. **26** is an overall system architecture diagram for a system for analyzing neurological functioning. In this example, the system comprises a data capture system **2700**, a range of motion comparator **2800**, a movement profile analyzer **2900**, and a neurological functioning analyzer **3000**. The data capture system **2700** captures data from sensors on the system such as motor speed sensors, angle sensors, accelerometers, gyroscopes, cameras, and other sensors which provide data about an individual's movement, balance, and strength, as well as information from software systems about tasks being performed by the individual while engaging in exercise. The range of motion comparator **2800** evaluates data from the data capture system **2700** to determine an individual's range of motion relative to the individual's personal history and relative to statistical norms, and to population averages. The movement profile analyzer **2900** evaluates data from the data capture system **2700** to generate a profile of the individual's physical function such as posture, balance, gait symmetry and stability, and consistency and strength of repetitive motion (e.g., walking or running pace and consistency, cycling cadence and consistency, etc.). The neurological functioning analyzer evaluates data from the data capture system **2700**, the range of motion comparator **2800**, and the movement profile analyzer **2900** to generate a profile of the user's nervous system function as indicated by composite functioning scores which indicate relative ability of an individual in one or more physical and mental functional measurement areas (also referred to herein as "composite functioning scores").

FIG. **27** is a system architecture diagram for the data capture system aspect of a neurological functioning analyzer. In this embodiment, the data capture system **2700** comprises a physical activity data capture device **2710** designed to capture information about an individual's movements while the individual is engaged in a primary physical activity and a software application **2720** designed to assign physical tasks and secondary activities, to engage the user in the physical tasks and secondary activities, and track and store responses to tasks and activities, as well as a data integrator **2730** configured to convert, calibrate, and integrate data streams from the physical activity data capture device **2710** and software application **2720**. The data capture system **2700** captures data from sensors **2711**, **2712** on the physical activity data capture device **2710** such as motor speed sensors, angle sensors, accelerometers, gyroscopes, cameras, and other sensors which provide data about the speed, operation, direction and angle of motion of the equipment, and about an individual's movement, balance, and strength.

The physical activity data capture device **2710** may be any type of device that captures data regarding the physical movements and activity of a user. In some embodiments, the physical activity data capture device **2710** may be a standalone device not associated with the activity being performed (e.g. a camera, ultrasonic distance sensor, heat sensor, pedometer, or other device not integrated into exercise equipment). In other embodiments, the physical activity data capture device **2710** may be exercise equipment or peripherals that captures motion and activity information of a user engaged in physical activity while using the device. For example, the physical activity data capture device **2710** may be in the form of exercise equipment such as stand-on or ride-on exercise machines like treadmills, stair stepping machines, stationary bicycles, rowing machines, and weight-lifting or resistance devices, or may be other equipment wherein the user stands separately from the equipment and pulls or pushes on ropes, chains, resistance bands, bars, and levers. The physical activity data capture device **2710** may be in the form of computer peripherals (e.g., game controllers, virtual reality headsets, etc.) that capture data while the user is performing physical movements related to a game or virtual reality environment, or exercise equipment that engage the user in physical activity, such as barbells, free weights, etc., which are configured to provide location and/or motion information such an integrated motion sensors or external cameras configured to detect the peripheral. The physical activity data capture device **2710** may be in the form of exercise equipment or peripherals and may be referred to as an exercise device. Sensors in the physical activity data capture device **2710** may be either analog **2711** or digital **2712**. Non-limiting examples of analog sensors **2711** are motor voltages and currents, resistors, potentiometers, thermistors, light sensors, and other devices that produce an analog voltages or currents. Most digital sensors are analog sensors **2711** with integrated analog-to-digital converters which output a digital signal, although some sensors are digital in the sense that they measure only discrete steps (e.g., an on/off switch). In most cases, signals from analog sensors **2711** will be converted to digital signals using an analog to digital converter **2701**. For signals from digital sensors **2712**, conversion is not necessary. In some cases, signals may need to be calibrated by a sensor calibrator, which corrects for sensor drift, out of range errors, etc., by comparing signals to known good values or to other devices.

The software application **2720** is any software designed to assign physical tasks and secondary activities, to engage the user in the physical tasks and secondary activities, and track and store data from physical tasks and responses to secondary activities. The software application **2720** may have, or may use or access, a number of different software components such as a virtual reality game or environment generator **2721**, a secondary activity manager **2722** which designs, selects, and/or implements testing protocols based on the user's profile. Many different configurations of the software are possible. The software application **2720** may be configured to present tasks to the user independent of inputs from the physical activity data capture device **2710**, such as performing playing games, performing math computations, remembering where certain symbols are located, visually following an object on a screen, or reading and speaking a given text. Alternatively, the software application **2720** may be configured to engage the user in mental or combined activities that correspond in some way to the inputs from the physical activity data capture device **2710**. For example, the user might be running on a treadmill, and the speed of the treadmill might be used as an input to a virtual reality environment which shows the user virtually running at a rate corresponding to the rate of the real world treadmill speed. The software application **2720** is configured to record data regarding, or evaluate and assign scores or values to, the user's responses and reactions to the tasks presented by the software application **2720**. For example, if the user is assigned the task of performing a mathematical calculation, the correctness of the user's response may be evaluated, scored, and recorded as data. As another example, the user may be presented with the task of speeding up or slowing down a running motion in response to a visual cue, and the speed of the user's reaction may be recorded as data. In such cases, a data integrator **2730** may be used to integrate the data from the physical activity data capture device **2710** with the data from the software application **2720**. In some embodiments, the data from the physical activity data capture device **2710** may be used to change the operation of the software application **2720**, and vice versa (i.e., the software application **2720** may also be used change the operation of the exercise equipment, for example, providing additional resistance or speeding up the operation of a treadmill). In some embodiments, the data integrator may not be a separate component, and its functionality may be incorporated into other components, such as the software application **2720**.

In some embodiments, the software application **2720**, another machine-learning based software application such as a task assignment software application (not shown), may be configured to assign physical tasks to the user to be performed in conjunction with the secondary activities assigned. Rather than simply performing continuously performing physical activity and recording the impact on the physical activity of performance of the secondary activities, the user may be assigned discrete physical tasks to perform while a mental activity is being performed. For example, the user may be assigned the physical task of pointing to a fixed spot on a display screen while reading aloud a text, and the steadiness of the user's pointing may be measured before, during, and after the reading, thus indicating an impact on the user's physical activity of the mental effort. Such dual-task testing may allow for more precise measurement and evaluation of relative functioning as different combinations of physical and secondary activities are evaluated together. In some embodiments, the secondary activity may be a second physical task or activity assigned to be performed simultaneously with a primary physical task or activity. Note that the terms "task" and "activity" as used herein are interchangeable, although the phrases "physical task" and "secondary activity" are often used for purposes of clarity and convenience.

FIG. **28** is a system architecture diagram for the range of motion comparator aspect of a neurological functioning analyzer. The range of motion and performance comparator **2800** evaluates data from the data capture system **2700** to determine an individual's range of motion and performance for the given secondary activity relative to the individual's personal history and relative to statistical norms. The range of motion and performance comparator **2800** comprises a current range analyzer **2801**, a historical range comparator **2802**, a statistical range comparator **2803**, and a range of motion and performance profile generator **2804**, as well as databases for user range of motion and performance historical data **2810** and demographic data **2820**. The current range analyzer **2801** ingests data related to an individual's movement and performance, and calculates a range of motion and performance of that individual while performing versus not performing the given secondary activity. For example, if an individual is given a primary physical task of standing in balance and an associative activity of popping a virtual balloon of a specific color as it appears randomly in the VR environment, the current range analyzer **2801** will start tracking the individual's balance while performing the associative activity and measure the accuracy and timing of balloon popping (for testing the individual's gross motor and executive functions). To conclude, the individual is instructed to start walking to warm up, and then repeat the same balloon popping activity while walking. The current range analyzer **2801** will finish capturing all the motion and performance data—the differences in the individual's accuracy and timing of balloon popping between standing and walking as well as the nuanced changes in the individual's walking movement during warmup and while balloon popping—and forwarding its analysis to the historical range comparator **2801**. The historical range comparator **2802** retrieves historical data for the individual (if such exists) from a user range of motion and performance historical data database **2810**, and compares the current data with historical data to determine trends in the individual's motion and performance over time. The statistical range comparator **2803** retrieves statistical range data for populations similar to the individual from a demographic data database **2820**, and determines a range of motion and performance of the individual relative to similar individuals by sex, age, height, weight, health conditions, etc. The range of motion and performance profile generator **2804** takes the data from the prior components, and generates and stores a range of motion profile for the individual which integrates these analyses into a comprehensive picture of the individual's range of motion functionality.

FIG. **29** is a system architecture diagram for the movement and performance profile analyzer aspect of a neurological functioning analyzer. The movement and performance profile analyzer **2900** evaluates data from the data capture system **2700** to generate a profile of the individual's physical function such as posture, balance, gait symmetry and stability, and consistency and strength of repetitive motion (e.g., walking or running pace and consistency, cycling cadence and consistency, etc.) and mental performance such as executive function, cognitive response, visual and auditory functions, emotional or empathetic reactions, etc. The movement and performance profile analyzer **2900** comprises a number of component analyzers **2901***a-n*, a historical movement and performance profile comparator **2902**, a statistical movement and performance comparator

2903, and a movement and performance profile generator 2904, as well as a user movement and performance profile history data database 2910 and a demographic data database 2920.

Many different aspects of movement and performance may be analyzed by the movement and performance profile analyzer 2900 through one or more of its many component analyzers 2901*a-n* such as the gait analyzer, balance analyzer, gross motor analyzer, fine motor analyzer, depth perception analyzer, executive function analyzer, visual function analyzer, auditory function analyzer, memory function analyzer, emotional response analyzer, etc. For example, the gait analyzer of the component analyzers 2901 ingests sensor data related to an individual's ambulatory movements (walking or running) while performing the given secondary activity, and calculates a step frequency, step symmetry, weight distribution, and other metrics related to an individual's gait. These calculations are then compared to expected calculations for an individual without performing the given the secondary activity. If an individual exhibits a limp while performing the given secondary activity (e.g., popping virtual balloons), the step frequency, step symmetry, and weight distribution will all be skewed with the impaired side showing a shorter step duration and less weight applied. The expected calculations may be determined from the full range of sensor values, per-exercise calibrations, statistical data, or other means appropriate to the specific application. The balance analyzer of the component analyzer 2901 performs a similar function with respect to an individual's balance. Wobbling, hesitation, or partial falls and recoveries while performing a range of secondary activities can be calculated from the data. The historical movement and performance comparator 2902 retrieves historical data for the individual (if such exists) from a user movement and performance historical data database 2910, and compares the current movement and performance data with historical data to determine trends in the movements and performances over time. The statistical movement and performance comparator 2903 retrieves statistical range of motion and performance data for populations similar to the individual from a demographic data database 2920, and compares movements and performances of the individual to similar individuals by sex, age, height, weight, health conditions, etc. The movement and performance profile generator 2905 takes the data from the prior components, and generates and stores a movement and performance profile for the individual which integrates these analyses into a comprehensive picture of the individual's movement and performance functionality.

FIG. 30 is a system architecture diagram for the neurological functioning analyzer aspect of a neurological condition evaluator. The neurological functioning analyzer evaluates data from the data capture system 2700, the range of motion and performance comparator 2800, and the movement and performance profile analyzer 2900 to generate a profile of the user's nervous system function as indicated by composite functioning scores which indicate relative ability of an individual in one or more physical and mental functional measurement areas (also referred to herein as "composite functioning scores"). The current composite functioning score analyzer 3001 ingests sensor data related to an individual's movement and performance, and calculates a set of current composite functioning scores for that individual based on the sensor data, the range of motion and performance profile, the movement and performance profile, and input from the software 2720 regarding secondary activities associated with physical movement data. The historical composite functioning score comparator 3002 retrieves historical data for the individual (if such exists) from a user composite functioning score historical data database 3010, and compares the current composite functioning score data with historical data to determine trends in the individual's bio-makers over time. The statistical composite functioning score comparator 3003 retrieves statistical composite functioning score data for populations similar to the individual from a demographic data database 3020, and determines a range of composite functioning score functionality of the individual relative to similar individuals by sex, age, height, weight, health conditions, etc. The neurological functioning profile generator 3004 takes the data from the prior components, and generates and stores a neurological functioning profile for the individual which integrates these analyses into a comprehensive picture of the individual's composite functioning score functionality. In some embodiments, one or more of the composite functioning scores may be determined from dual-task testing, in which a physical task and a mental task are performed simultaneously to detect areas of abnormal nervous system function, and/or identify which areas of the nervous system may be affected. For example, while performing mathematical tasks, an individual slows down significantly in his/her walk compared to the population data. It will indicate that the individual's composite functioning score for logical and mathematic functions is worse than his/her population cohort (by sex, age, height, weight, health conditions, etc.). The neurological functioning profile may include a composite functioning score spatial map as described above. In some embodiments, the neurological functioning analyzer may receive data directly from the data capture system 2700 and may perform independent neurological analyses without inputs from the range of motion and performance comparator 2800 or the movement and performance profile analyzer 2900, or may incorporate some or all of the functionality of those components.

Detailed Description of Exemplary Aspects

FIG. 2 is a top-down view of a variable-resistance exercise machine 100 with wireless communication for smart device control and interactive software applications of the invention. According to the embodiment, exercise machine 100 may comprise a stable base 101 to provide a platform for a user to safely stand or move about upon. Exercise machine 100 may further comprise right 201*a* and left 201*b* hand rails for a user to brace against or grip during use, to provide a stable support for safety as well as a mounting point for external devices such as a plurality of tethers, as described below with reference to FIG. 3. A plurality of steps 202*a-n* may be used to provide a user with a safe and easy means to approach or dismount exercise machine 100, as well as a nonmoving "staging area" where a user may stand while they configure operation or wait for exercise machine 100 to start operation. Unlike traditional treadmill machines common in the art, exercise machine 100 may be made with greater width to accommodate a wider range of free movement of a user's entire body (whereas traditional treadmills are designed to best accommodate only a jogging or running posture, with minimal lateral motion), and a plurality of separate moving surfaces 203*a-b* may be utilized to provide multiple separate surfaces that may move and be controlled independently of one another during use. For example, a user may move each of their legs independently without resistance applied, with separate moving surfaces 203*a-b* moving freely underfoot as a user applies pressure during their movement. This may provide the illusion of movement to a user while in reality they remain stationary with respect to their surroundings. Another use may be multiple separate moving surfaces **203***a-b*, with separate speeds of movement or degrees of resistance, so that as a user moves about during use they may experience physical feedback in the form of changing speed or resistance, indicating where they are standing or in what direction they are moving (for example, to orient a user wearing a virtual reality headset, as described below with reference to FIG. **3**). Moving surfaces **203***a-b* may be formed with a texture **204** to increase traction, which may improve user safety and stability during use as well as improve the operation of moving surfaces **203***a-b* for use in multidirectional movement (as the user's foot is less likely to slide across a surface rather than taking purchase and applying directional pressure to produce movement). Use of multiple, multidirectional moving surfaces **203***a-b* may also be used in various therapeutic or rehabilitation roles, for example to aid a user in developing balance or range of motion. For example, a user who is recovering from an injury or surgery (such as a joint repair or replacement surgery) may require regular physical therapy during recovery. Use of multidirectional moving surfaces **203***a-b* along with appropriate guidance from a rehabilitation specialist or physical therapist (or optionally a virtual or remote coach using a software application) may make regular therapy more convenient and accessible to the user, rather than requiring in-home care or regular visits to a clinic. For example, by enabling a therapist or coach to manually vary the movement and resistance of the moving surfaces **203***a-b*, they can examine a user's ability to overcome resistance to different movements such as at odd angles or across varying range of motion, to examine the user's physical health or ability. By further varying the resistance it becomes possible to assist the user with rehabilitation by providing targeted resistance training to specific movements, positions, or muscle groups to assist in recovery and development of the user's abilities.

Exercise machine **100** may be designed without a control interface commonly utilized by exercise machines in the art, instead being configured with any of a variety of wireless network interfaces such as Wi-Fi or BLUETOOTH™ for connection to a user's smart device, such as a smartphone or tablet computer. When connected, a user may use a software application on their device to configure or direct the operation of exercise machine **100**, for example by manually configuring a variety of operation settings such as speed or resistance, or by interacting with a software application that automatically directs the operation of exercise machine **100** without exposing the particular details of operation to a user. Additionally, communication may be bi-directional, with a smart device directing the operation of exercise machine **100** and with exercise machine **100** providing input to a smart device based at least in part on a user's activity or interaction. For example, a user may interact with a game on their smart device, which directs the operation of exercise machine **100** during play as a form of interaction with, and feedback to, the user. For example, in a racing game, exercise machine **100** may alter the resistance of moving surfaces **203***a-b* as a user's speed changes within the game. In another example, a user may be moving about on moving surfaces **203***a-b* while playing a simulation or roleplaying game, and their movement may be provided to the connected smart device for use in controlling an in-game character's movement. Another example may be two-way interactive media control, wherein a user may select media such as music for listening on their smart device, and then while using exercise machine **100** their level of exertion (for example, the speed at which they run or jog) may be used to provide input to their smart device for controlling the playback of media. For example, if the user slows down music may be played slowly, distorting the audio unless the user increases their pace. In this manner, exercise machine **100** may be used interchangeably as a control and feedback device or both simultaneously, providing an immersive environment for a wide variety of software applications such as virtual reality, video games, fitness and health applications, or interactive media consumption.

FIG. **4** is a diagram of an exemplary hardware arrangement **400** for natural torso tracking and feedback for electronic interaction according to a preferred embodiment of the invention, illustrating the use of multiple tethers **410***a-n* and a movable torso harness **420**. According to the embodiment, a plurality of tethers **410***a-n* may be affixed or integrally-formed as part of a handle or railing **430**, such as handlebars found on exercise equipment such as a treadmill, elliptical trainer, stair-climbing machine, or the like. In alternate arrangements, specifically-designed equipment with integral tethers **410***a-n* may be used, but it may be appreciated that a modular design with tethers **410***a-n* that may be affixed and removed freely may be desirable for facilitating use with a variety of fitness equipment or structural elements of a building, according to a user's particular use case or circumstance. Tethers **410***a-n* may then be affixed or integrally-formed to a torso harness **420**, as illustrated in the form of a belt, that may be worn by a user such that movement of their body affects tethers **410***a-n* and applies stress to them in a variety of manners. It should be appreciated that while a belt design for a torso harness **420** is shown for clarity, a variety of physical arrangements may be used such as including (but not limited to) a vest, a series of harness-like straps similar to climbing or rappelling equipment, a backpack, straps designed to be worn on a user's body underneath or in place of clothing (for example, for use in medical settings for collecting precise data) or a plurality of specially-formed clips or attachment points that may be readily affixed to a user's clothing. Additionally, a torso harness **420** may be constructed with movable parts, for example having an inner belt **421** that permits a user some degree of motion within the harness **420** without restricting their movement. Movement of inner belt **421** (or other movable portions) may be measured in a variety of ways, such as using accelerometers, gyroscopes, or optical sensors, and this data may be used as interaction with software applications in addition to data collected from tethers **410***a-n* as described below. In some embodiments, a saddle-like surface on which a user may sit may be used, with motion of the saddle-like surface measured as described generally herein.

As a user moves, his or her body naturally shifts position and orientation. These shifts may be detected and measured via tethers **410***a-n*, for example by detecting patterns of tension or strain on tethers **410***a-n* to indicate body orientation, or by measuring small changes in strain on tethers **410***a-n* to determine more precise movements such as body posture while a user is speaking, or specific characteristics of a user's stride or gait. Additionally, through varying the quantity and arrangement of tethers **410***a-n*, more precise or specialized forms of movement may be detected and measured (such as, for example, using a specific arrangement of multiple tethers connected to a particular area of a user's body to detect extremely small movements for medical diagnosis or fitness coaching). This data may be used as interaction with software applications, such as for virtual reality applications as input for a user to control a character in a game. In such an arrangement, when a user moves, this movement may be translated to an in-game character or avatar to convey a more natural sense of interaction and presence. For example, in a multiplayer roleplaying game, this may be used to facilitate nonverbal communication and recognition between players, as their distinct mannerisms and gestures may be conveyed in the game through detection of natural torso position and movement. In fitness or health applications, this data may be used to track and monitor a user's posture or ergonomic qualities, or to assist in coaching them for specific fitness activities such as holding a pose for yoga, stretching, or proper running form during use with a treadmill. In medical applications, this data may be used to assist in diagnosing injuries or deficiencies that may require attention, such as by detecting anomalies in movement or physiological adaptations to an unrecognized injury (such as when a user subconsciously shifts their weight off an injured foot or knee, without consciously realizing an issue is present).

Through various arrangements of tethers **410***a-n* and tether sensors (as described below, referring to FIGS. **5-7**), it may be possible to enable a variety of immersive ways for a user to interact with software applications, as well as to receive haptic feedback from applications. For example, by detecting rotation, tension, stress, or angle of tethers a user may interact with applications such as virtual reality games or simulations, by using natural body movements and positioning such as leaning, jumping, crouching, kneeling, turning, or shifting their weight in various directions to trigger actions within a software application configured to accept torso tracking input. By applying haptic feedback of varying form and intensity (as is described in greater detail below, referring to FIG. **5**), applications may provide physical indication to a user of software events, such as applying tension to resist movement, pulling or tugging on a tether to move or "jerk" a user in a direction, or varying feedback to multiple tethers such as tugging and releasing in varying order or sequence to simulate more complex effects such as (for example, in a gaming use case) explosions, riding in a vehicle, or walking through foliage.

FIG. **5** is a diagram illustrating a variety of alternate tether arrangements. According to various use cases and hardware arrangements, tethers **410***a-n* may utilize a variety of purpose-driven designs as illustrated. For example, a "stretchable" tether **510** may be used to measure strain during a user's movement, as the tether **510** is stretched or compressed (for example, using piezoelectric materials and measuring electrical changes). Such an arrangement may be suitable for precise measurements, but may lack the mechanical strength or durability for gross movement detection or prolonged use. An alternate construction may utilize a non-deforming tether **520** such as a steel cable or similar non-stretching material. Instead of measuring strain on the tether **520**, instead tether **520** may be permitted a degree of movement within an enclosure **522** (for example, an attachment point on a torso harness **420** or handlebar **430**), and the position or movement **521** of the tether **520** may be measured such as via optical sensors. In a third exemplary arrangement, a tether **530** may be wound about an axle or pulley **531**, and may be let out when force is applied during a user's movement. Rotation of the pulley **531** may be measured, or alternately a tension device such as a coil spring may be utilized (not shown) and the tension or strain on that device may be measured as tether **530** is extended or retracted. In this manner, it may be appreciated that a variety of mechanical means may be used to facilitate tethers and attachments for use in detecting and measuring natural torso position and movement, and it should be appreciated that a variety of additional or alternate hardware arrangements may be utilized according to the embodiments disclosed herein.

Additionally, through the use of various hardware construction it becomes possible to utilize both "passive" tethers that merely measure movement or strain, as well as "active" tethers that may apply resistance or movement to provide haptic feedback to a user. For example, in an arrangement utilizing a coiled spring or pulley **531**, the spring or pulley **531** may be wound to retract a tether and direct or impede a user's movement as desired. In this manner, various new forms of feedback-based interaction become possible, and in virtual reality use cases user engagement and immersion are increased through more natural physical feedback during their interaction.

By applying various forms and intensities of feedback using various tether arrangements, a variety of feedback types may be used to provide haptic output to a user in response to software events. For example, tension on a tether may be used to simulate restrained movement such as wading through water or dense foliage, walking up an inclined surface, magnetic or gravitational forces, or other forms of physical resistance or impedance that may be simulated through directional or non-directional tension. Tugging, retracting, or pulling on a tether may be used to simulate sudden forces such as recoil from gunfire, explosions, being grabbed or struck by a software entity such as an object or character, deploying a parachute, bungee jumping, sliding or falling, or other momentary forces or events that may be conveyed with a tugging or pulling sensation. By utilizing various patterns of haptic feedback, more complex events may be communicated to a user, such as riding on horseback or in a vehicle, standing on the deck of a ship at sea, turbulence in an aircraft, weather, or other virtual events that may be represented using haptic feedback. In this manner, virtual environments and events may be made more immersive and tangible for a user, both by enabling a user to interact using natural body movements and positioning, as well as by providing haptic feedback in a manner that feels natural and expected to the user. For example, if a user is controlling a character in a gaming application through a first-person viewpoint, it would seem natural that when their character is struck there would be a physical sensation corresponding to the event; however, this is not possible with traditional interaction devices, detracting from any sense of immersion or realism for the user. By providing this physical sensation alongside the virtual event, the experience becomes more engaging and users are encouraged to interact more naturally as their actions results in natural and believable feedback, meeting their subconscious expectations and avoiding excessive "immersion-breaking" moments, which in turn reduces the likelihood of users adopting unusual behaviors or unhealthy posture as a result of adapting to limited interaction schema.

Haptic feedback may be provided to notify a user of non-gaming events, such as for desktop notifications for email or application updates, or to provide feedback on their posture for use in fitness or health coaching. For example, a user may be encouraged to maintain a particular stance, pose, or posture while working or for a set length of time (for example, for a yoga exercise application), and if their posture deviates from an acceptable range, feedback is provided to remind them to adjust their posture. This may be used in sports, fitness, health, or ergonomic applications that need not utilize other aspects of virtual reality and may operate as traditional software applications on nonspecialized computing hardware. For example, a user at their desk may use an ergonomic training application that monitors their body posture throughout the work day and provides haptic reminders to correct poor posture as it is detected, helping the user to maintain a healthy working posture to reduce fatigue or injuries due to poor posture (for example, repetitive-stress injuries that may be linked to poor posture while working at a computer).

FIG. **6** is a diagram of an additional exemplary hardware arrangement **600** for natural torso tracking and feedback for electronic interaction according to a preferred embodiment of the invention, illustrating the use of angle sensors **612**, **621***a-n* to detect angled movement of a tether **620**. According to one exemplary arrangement, a tether **610** may be affixed to or passed through a rotating joint such as a ball bearing **611** or similar, to permit free angular movement. During movement, the angular movement or deflection **612** of a protruding bar, rod, or tether segment **613** may be measured (for example, using optical, magnetic, or other sensors) to determine the corresponding angle of tether **610**. In this manner, precise angle measurements may be collected without impeding range of motion or introducing unnecessary mechanical complexity.

In an alternate hardware arrangement, the use of angle sensors **621***a-n* enables tracking of a vertical angle of a tether **620**, to detect and optionally measure vertical movement or orientation of a user's torso. When tether **620** contacts a sensor **621***a-n*, this may be registered and used to detect a general vertical movement (that is, whether the tether is angled up or down). For more precise measurements, the specific hardware construction of a sensor **621***a-n* may be varied, for example using a pressure-sensing switch to detect how much force is applied and use this measurement to determine the corresponding angle (as may be possible given a tether **620** of known construction). It should be appreciated that various combinations of hardware may be used to provide a desired method or degree of angle detection or measurement, for example using a conductive tether **620** and a capacitive sensor **621***a-n* to detect contact, or using a mechanical or rubber-dome switch (as are commonly used in keyboard construction) to detect physical contact without a conductive tether **620**.

The use of angle detection or measurement may expand interaction possibilities to encompass more detailed and natural movements of a user's body. For example, if a user crouches, then all tethers **410***a-n* may detect a downward angle simultaneously. Additionally, data precision or availability may be enhanced by combining input from multiple available sensors when possible (for example, utilizing adaptive software to collect data from any sensors that it detects, without requiring specific sensor types for operation), for example by combining data from tethers **410***a-n* and hardware sensors such as an accelerometer or gyroscope, enabling multiple methods of achieving similar or varied types or precision levels of position or movement detection. Similarly, when a user jumps then all tethers may detect an upward angle simultaneously. However, if a user leans in one direction, it may be appreciated that not all tethers **410***a-n* will detect the same angle. For example, tethers **410***a-n* in the direction the user is leaning may detect a downward angle, while those on the opposite side would detect an upward angle (due to the orientation of the user's torso and thus a worn torso harness **420**). In this manner, more precise torso interaction may be facilitated through improved detection and recognition of orientation and movement. Additionally, it may be appreciated that sensors **621***a-n* may be utilized for other angle measurements, such as to detect horizontal angle. For example, if a user is wearing a non-rotating torso harness **420**, when they twist their body a similar stress may be applied to all attached tethers **410***a-n*. Without angle detection the precise nature of this movement will be vague, but with horizontal angle detection it becomes possible to recognize that all tethers **410***a-n* are being strained in a similar direction (for example, in a clockwise pattern when viewed from above, as a user might view tethers **410***a-n* during use), and therefore interpret the interaction as a twisting motion (rather than, for example, a user squatting or kneeling, which might apply a similar stress to the tethers **410***a-n* but would have different angle measurements).

FIG. **7** is a diagram illustrating an exemplary hardware arrangement of an apparatus for natural torso tracking and feedback for electronic interaction according to a preferred embodiment of the invention, illustrating the use of multiple tethers **410***a-n* and a movable torso harness **420** comprising a plurality of angle sensors **701***a-n* positioned within the movable torso harness **420**. According to the embodiment, a plurality of tethers **410***a-n* may be affixed or integrally-formed as part of a handle or railing **430**, such as handlebars found on exercise equipment such as a treadmill, elliptical trainer, stair-climbing machine, or the like. In alternate arrangements, specifically-designed equipment with affixed or integral tethers **410***a-n* may be used, but it may be appreciated that a modular design with tethers **410***a-n* that may be affixed and removed freely may be desirable for facilitating use with a variety of fitness equipment or structural elements of a building, according to a user's particular use case or circumstance as well as weight-holding strength of the tethers. Tethers **410***a-n* may then be affixed or integrally-formed to angle sensors **701***a-n* placed within or integrally-formed as a component of torso harness **420** (as illustrated in the form of a belt) that may be worn by a user such that movement of their body affects tethers **410***a-n* and applies detectable or measurable stress to tethers **410***a-n* and angular motion to angle sensors **701***a-n*. In this manner, it may be appreciated that angle sensors **701***a-n* may be utilized as integral or removable components of a torso harness **420**, as an alternative arrangement to utilizing angle sensors **701***a-n* placed or formed within railings **430** or other equipment components connected to distal ends of tethers **410***a-n* (with respect to the user's torso). According to various embodiments, sensors may be placed optionally on a belt, vest, harness, or saddle-like surface or at attachment points on safety railings, or indeed both.

FIG. **9** is a block diagram of an exemplary system architecture **900** of an exercise machine **100** being connected over local connections to a smartphone or computing device **930**, an output device other than a phone **910**, and a server over a network **940**. An exercise machine **100** may connect over a network **920**, which may be the Internet, a local area connection, or some other network used for digital communication between devices, to a server **940**. Such connection may allow for two-way communication between a server **940** and an exercise machine **800**. An exercise machine **100** may also be connected over a network **920** to a smartphone or computing device **930**, or may be connected directly to a smartphone or computing device **930** either physically or wirelessly such as with Bluetooth connections. An exercise machine **100** also may be connected to an output device **910** which may display graphical output from software executed on an exercise machine **100**, including Mixed or virtual reality software, and this device may be different from a smartphone or computing device **930** or in some implementations may in fact be a smartphone or computing device **930**. A remote server **940** may contain a data store **941**, and a user verification component **942**, which may contain typical components in the art used for verifying a user's identity from a phone connection or device connection, such as device ID from a smartphone or computing device or logging in with a user's social media account.

FIG. **10** is a diagram of an exemplary hardware arrangement of a smart phone or computing device **1030** executing software **1010** and communicating over a network **1020**. In an exemplary smart phone or computing device **1030**, key components include a wireless network interface **1031**, which may allow connection to one or a variety of wireless networks including Wi-Fi and Bluetooth; a processor **1032**, which is capable of communicating with other physical hardware components in the computing device **1030** and running instructions and software as needed; system memory **1033**, which stores temporary instructions or data in volatile physical memory for recall by the system processor **1032** during software execution; and a display device **1034**, such as a Liquid Crystal Display (LCD) screen or similar, with which a user may visually comprehend what the computing device **1030** is doing and how to interact with it. It may or may not be a touch enabled display, and there may be more components in a computing device **1030**, beyond what are crucially necessary to operate such a device at all. Software **1010** operating on a processor **1033** may include a mixed or virtual reality application, a user verification system, or other software which may communicate with a network-enabled server **1040** and exercise machine **100** software for the purposes of enhanced mixed or virtual reality.

FIG. **11** is a block diagram of a method of mixed or virtual reality software operating to receive input through different sources, and send output to devices. Mixed or virtual reality software which may be run on a phone or computing device **1030** or another device, outputs data to a visual device for the purpose of graphically showing a user what they are doing in the software **1110**. Such display may be a phone display **1034**, or a separate display device such as a screen built into an exercise machine **100** or connected some other way to the system, or both display devices. During software execution, user input may be received either through buttons **1130** on the exercise machine **100**, **1120**, or through input from a belt-like harness **420**, such as user orientation or movements. Such received data may be sent **1140** to either a mobile smart phone or computing device **1030**, or to a server **1040** over a network **1020**, or both, for processing, storage, or both. Data may be stored on a server with a data store device **1041** and may be processed for numerous uses including user verification with a user verification component **1042**. Data may be processed either by software running on an exercise machine **100**, a smart phone or computing device **1030**, or some other connected device which may be running mixed or virtual reality software, when input is received from a user using either buttons on an exercise machine **100**, a belt-like harness **420**, or both, and optionally using hardware features of an exercise machine **100** such as handlebars, pedals, or other features in mixed or virtual reality software for tasks such as representing movement in a simulation.

FIG. **17** is a block diagram of an exemplary virtual reality or mixed reality enhanced exercise machine, illustrating the use of a stationary bicycle **1700** with hand controls on the handles **1720**, and a belt-like harness attachment **420**. A stationary exercise bicycle device **1700**, which may be of any particular design including a reclining, sitting, or even unicycle-like design, possesses two pedals **1730** as is common for stationary exercise bicycles of all designs. On handlebars of a stationary exercise bicycle may exist buttons and controls **1720** for interacting with a virtual reality or mixed reality augmented piece of software, allowing a user to press buttons in addition to or instead of pedaling, to interact with the software. A belt-like harness attachment **420** is attached via a mechanical arm **1710** to a stationary exercise bicycle **1700**, which may monitor motion and movements from a user during the execution of virtual reality software. A mechanical arm **1710** may have an outer shell composed of any material, the composition of which is not claimed, but must have hinges **1711**, **1712**, **1713** which allow for dynamic movement in any position a user may find themselves in, and angular sensors inside of the arm at the hinge-points **1711**, **1712**, **1713** for measuring the movement in the joints and therefore movement of the user. A stationary bicycle device **1700** may also have a pressure sensor in a seat **1740**, the sensor itself being of no particularly novel design necessarily, to measure pressure from a user and placement of said pressure, to detect movements such as leaning or sitting lop-sided rather than sitting evenly on the seat.

FIG. **18** is a diagram of another exemplary virtual reality or mixed reality enhanced exercise machine, illustrating the use of a treadmill exercise machine **100**, **1800** a vest-type harness **1820** with a plurality of pistons **1811** to provide a hardware-based torso joystick with full-body tracking. According to this embodiment, a treadmill or other exercise machine **100**, **1800** may comprise a plurality of rigid side tails **102** for a user to grip for support as needed during use (for example, as a balance aid or to assist getting on the machine and setting up other equipment properly) as well as a rigid stand or mount **104** for a user's smartphone or other computing device, that may be used to operate a virtual reality or mixed reality software application. Exercise machine **100**, **1800** may further comprise a jointed arm **1810** or similar assembly that may be integrally-formed or removably affixed to or installed upon exercise machine **100**, **1800**. Arm **1810** may utilize a plurality of pistons **1811** to provide for movement during use in order to follow the movements of a user's body, as well as to provide tension or resistance to motion when appropriate (for example, to resist a user's movements or to provide feedback) and motion detection of a user's movement during use, according to various aspects described previously (referring to FIGS. **3**-**7**, for example) by measuring movement of a piston **1811** or arm **1810** and optionally applying tension or resistance to piston **1811** to retard movement of arm **1810** and constrain user movement or simulate specific forms of physical feedback. For example, if a user is moving an avatar in a virtual reality software application, when the avatar encounters an obstacle such as another avatar, object, or part of the environment, resistance may be applied to piston **1811** to prevent the user from moving further, so that their avatar is effectively prevented from moving through the obstacle and thereby facilitating the immersive experience of a solid object in a virtual environment. Additional arms may be used for a user's limbs **1921** and may incorporate straps **1922** to be affix about a user's arm, wrist, or other body part, to incorporate more detailed movement tracking of a user's arms and/or legs rather than just torso-based tracking. A vest-type harness **1920** may be used in place of a belt **420**, to allow for more natural movement or to provide greater area upon which to affix additional arms **1821**, pistons **1811**, or any of a variety of sensors, for example such as accelerometers **1822** or gyroscopes **1823** for detecting body orientation (not all optional sensors are shown for the sake of clarity). For example, a vest **1820** may have integrated feedback actuators **1812** for use in first-person software applications to simulate impacts or recoil, or it may incorporate heating or cooling elements to simulate different virtual environments while worn. Additionally, vest **1820** may incorporate electrical connectors **1824** for various peripheral devices such as controllers **305***a-b* or a headset **302**, reducing the risk of tangles or injury by keeping cables short and close to the user so they cannot cause issues during movement or exercise.

FIG. **19** is a diagram of another exemplary virtual reality or mixed reality enhanced exercise machine, illustrating the use of a stationary bicycle This present application is a continuation-in-part of Ser. No. 16/176,511, titled "VIRTUAL REALITY AND MIXED REALITY ENHANCED EXERCISE MACHINE", and filed on Oct. 31, 2018, which with a vest-type harness **1820** with a plurality of strain sensors **1911** and tethers **1912**, according to an aspect of the invention. According to this embodiment, rather than a jointed arm **1810** and pistons **1811**, a solid flexible arm **1910** may be used to detect user movement while positioned on a seat **1902** to use exercise machine **100**, for example while the user is seated to use pedals **1901** on a stationary bike or elliptical training machine. Through a plurality of strain gauges **1911** that detect the flexion or extension of the solid arm. Tethers **1912** may be used for either movement tracking or providing feedback to a user, or both, and may optionally be connected or routed through joints or interconnects **1913** to allow for a greater variety of attachment options as well more precise feedback (for example, by enabling multiple angles from which a tether **1912** may apply force, to precisely simulate different effects). Additional arms may be used for a user's limbs **1921** and may incorporate straps **1922** to be affix about a user's arm, wrist, or other body part, to incorporate more detailed movement tracking of a user's arms and/or legs rather than just torso-based tracking. Additional arms **1921** may also incorporate additional tethers **1912** and strain sensors **1911** to track movement and apply feedback to specific body parts during use, further increasing precision and user immersion. A vest-type harness **1820** may be used in place of a belt **420**, to allow for more natural movement or to provide greater area upon which to affix additional arms **1921**, tether **1912**, or any of a variety of sensors, for example such as accelerometers or gyroscopes for detecting body orientation (not all optional sensors are shown for the sake of clarity). For example, a vest **1820** may have integrated feedback actuators for use in first-person software applications to simulate impacts or recoil, or it may incorporate heating or cooling elements to simulate different virtual environments while worn. Additionally, vest **1820** may incorporate electrical connectors **1914** for various peripheral devices such as controllers **305***a-b* or a headset **302**, reducing the risk of tangles or injury by keeping cables short and close to the user so they cannot cause issues during movement or exercise.

FIG. **20** is a flow diagram illustrating an exemplary method **2000** for operating a virtual and mixed-reality enhanced exercise machine, according to one aspect. According to the aspect, a user may wear **2001** a torso harness such as a belt **420** or vest **1820** harness, while they engage in the use **2002** of an exercise machine **100**. While using the exercise machine **100**, the user's movements may be detected and measured **2003** through the use of a plurality of body movement sensors such as (for example, including but not limited to) strain sensors **1911**, tethers **410***a-c*, **1912**, pistons **1811**, or optical sensors **1201***a-n*. These measured user movements may then be mapped by a composition server **801** to correspond to a plurality of movement inputs of a virtual joystick device **2004**. These virtual joystick inputs may then be transmitted **2005** to a software application, for example a virtual reality or mixed reality application operating on a user device such as (for example, including but not limited to) a smartphone **930**, personal computing device, or headset **302**. Composition server **801** may then receive feedback from the software application **2006**, and may direct the operation of a plurality of feedback devices such as tethers **410***a-c*, **1912** or pistons **1811** to resist or direct the user's movement **2007** to provide physical feedback to the user based on the received software feedback.

FIG. **21** is a system diagram of a key components in the analysis of a user's range of motion and balance training. A datastore containing statistical data **2110** on a user's age category, gender, and other demographic data, as well as a datastore containing balancing algorithms **2120**, are connected to a collection of components integrated into an exercise system **2130**, including a plurality of sensors **2131**, a movement profile analyzer **2132**, a balance trainer **2133**, and a tuner **2134**. A plurality of sensors **2131** may be connected to varying parts of an exercise system, tethered to a user, or otherwise connected to or able to sense a user during exercise, and may inform a movement profile analyzer **2132** of the performance of a user's exercise during such exercise. A movement profile analyzer **2132** may use data from a datastore containing statistical data on a user **2110** to generate movement profile of how a user performs and moves during exercise, in comparison with how they may be expected to move, and pass this data on to a balance trainer **2133** which is further connected to a datastore containing balance algorithms **2120**. A balance trainer **2133** accesses and utilizes balance algorithms **2120** in conjunction with assembled movement profile data **2132** and determines if a user is in need of correcting their form or balance during exercise. A tuner **2134** is connected to a datastore containing user profile data **2150** and also connected to a balance tuner **2133**, enabling a user's individual preferences or specifications, or exercise needs, to inform adjustments for a balance trainer **2133**, for example if a user would initially be detected as stumbling by a balance trainer **2133** but the user were to specify that they are not falling, and continue to exercise in this fashion for whatever reason (such as physical limitations), a tuner **2134** may adjust the balance trainer **2133** in this instance. Such information is stored in a user's profile data **2150**. A display **2140** is connected to core components **2130** and may display the warnings generated by a balance trainer **2133** or offer a user the opportunity to offer adjustments or physical information to a tuner **2134** for adjusting a balance trainer **2133**.

FIG. **22** is a diagram showing a system for balance measurement and fall detection. A classic problem in control system theory is controlling an inverted pendulum such that it balances vertically without falling down. On the right side of the diagram is a drawing of the inverted pendulum problem in which a pendulum (a rod having some length, l, and some mass, m) **2261** is attached to a movable platform **2262**. Sensors **2264** on the platform **2262** detect at least the angle, θ, **2266** of the pendulum **2261** from vertical, and may also be configured to detect or calculate the rate of change of the angle **2266**, the acceleration of the platform **2262**, and other variables. As the pendulum **2261** falls away from vertical due to the force of gravity, g, **2263**, a control mechanism such as a proportional, integral, differential (PID) controller may calculate and apply a force F **2265** to the platform **2262** sufficient to swing the pendulum **2261** back to vertical against the force of gravity **2262**.

A similar system may be used to measure balance and detect and predict falls by a person with impaired balance abilities. A user **2210** may wear a sensor and electronics package **2131**, on the torso. The sensor and electronics package **2131** may be simply a collection of sensors (e.g. accelerometers, gyroscopes, etc.) configured to transmit data to an external computing device, or the sensor and electronics package **2131** may itself have a computing device. The user's body mass, m, can be entered manually or obtained from a wireless scale capable of communicating wirelessly with the sensor and electronics package **2131**. As the user's torso moves from the vertical position **2220**, the angle from vertical and rate of change of the angle, θ", **2230** from vertical can be measured, tracked, and used to make predictions about the likelihood of a fall. Angular momentum **2230** may be represented by θ" **2230**, a user's angle deviation from vertical being represented by θ **2220**, the force of gravity being represented by g **2240**, and the approximate height of a user's body-part acting similar to the bar of an inverted pendulum being represented by L **2250**. The data obtained from the sensors and electronics package **2131** may be used in conjunction with various algorithms (e.g. a PID controller) and the user's historical or manually-entered movement ability to determine when the rate of fall is likely to exceed the user's ability to accelerate toward the direction of fall fast enough to right the torso. It is therefore possible to analyze and characterize a user's motions that may lead to a stumble or fall.

FIG. **23** is a system diagram of a sensor measuring the range of motion of a user during a specific exercise. A user performing an exercise with their leg is shown, with a sensor **2131** and angular movement **2310**. A sensor **2131** may be used to characterize the angle of the user's motion, or be attached as an ankle weight for a more specific implementation (but by no means the only implementation of this process of using a sensor to measure an individual user's body parts during exercise), to achieve more information about user form in addition to or instead of using an inverted pendulum **2220** with a sensor **2131** inside.

FIG. **24** is a method diagram illustrating behavior and performance of key components for range of motion analysis and balance training. A user's movements may first be detected on or with an exercise machine, using a plurality of sensors **2131**, **2410**. Given a user's movements **2410**, statistical data on a user's demographics may be gathered **2420** using a datastore containing such information **2110**, to compare a user's movements with expected or anticipated norms based on acquired or default statistical data. A user's profile data **2150** may then be accessed **2430**, and using a user's profile data **2150** which may contain individual preferences or information beyond statistical norms **2110** or sensor-acquired exercise data **2131**, analyses of a user's range of motion may occur **2440**. Such analyses may include examining differences between a user's expected motion during an exercise, with their actual motion, measuring individual, anomalous movements during a user's exercise (such as a single motion that does not match with the rest of the user's movements), and other techniques to analyze anomalies in a user's displayed exercise ability. A user's profile is also generated from these analyses **2440**, allowing a history of a user's exercise performance to be recorded for future analysis and for comparison with future observed exercise patterns and performance. A user's profile and exercise performance, along with any other notes, may be displayed **2450** with a graphical or textual display **2140**, allowing a user to see for themselves their performance and deficiencies as determined by the system. A further step may be to detect if a user is detect to be likely to fall or stumble **2460**, such as if a leg movement is not proper for a running motion on a treadmill, and display or sound a warning to a user **2470** using a display **2140** or any other method that may be available to the physical activity data capture device for warning a user of possible injury or failure. These warnings may further be recorded in a user profile **2150** for access by a tuner **2134** and balance trainer **2133** to help the user be aware of patterns of exercise performance that may lead to similar incidents in the future, before they happen, thereby helping to ensure safety of physically at-risk exercise machine users.

FIG. **31** is an exemplary human/machine interface and support system for using body movements to interface with embedded or external computers while engaging in exercise. In this embodiment, an exercise machine **3110** is placed inside a frame **3120** which contains components for sensing the movement of an individual, providing haptic feedback, and providing support in case of a fall. In this embodiment, the exercise machine **3110** is depicted as a stationary bicycle, although any type of exercise machine **3110** (e.g., treadmill, stair-stepper, rowing machine, weight-lifting machines, etc.) may be used. The exercise machine **3110** may contain or be in communication with an embedded or external computer that communicates with other components of the system, although in some embodiments, the exercise machine **3110** is not communicatively coupled with other components. In some embodiments, no exercise machine **3110** at all is used, and the individual may freely engage in exercise or other physical movement such as running in place, jumping, dancing, lifting barbells or free weights, etc. The frame **3120** comprises a base **3121** and one or more vertical supports **3122***a,b*. Mounted to a point on the vertical supports are one or more pulleys or routing devices **3125***a,b*, which guide one or more tethers **3124***a,b* at a height above the waist level of the individual during exercise. The tethers **3124***a,b* are attached at one end to a belt, harness, vest, or other device **3126** attachable to the body of the individual, and at the other end to sensors/actuators **3123***a,b*. In this embodiment, the sensors/actuators **3126***a,b* are electric motors fitted with rotary encoders and the tethers **3124***a,b* are wound around a drum on the shaft of the motors. In this way, body movements of the individual may be sensed and recorded as rotational movements of the drum, and rotational movement data may be sent to a computing device which can perform calculations to determine position, distance of movement, speed of movement, acceleration, and other such calculations. For example, the linear distance of movement may be calculated from the number of rotations and the circumference of the drum. Linear speed may be calculated as the linear distance over time. The position of the individual may be calculated from speed and distance. The rotational movement, linear distance, linear speed, or other calculations may be used to control the computing device or the output from a computing device such as a game, virtual reality environment, etc. Further, the motors of the sensors/actuators **3123***a,b* may also act as actuators, and varying voltages and currents may be applied to the motors to provide haptic feedback to the individual, such as resistance to movement, jerking, or vibration. This haptic feedback may be provided in response to interactions with the computer, such as to indicate game events, interactions with the virtual reality environment, etc. In one aspect, the belt **3126**, tethers **3124***a,b*, and sensors/actuators **3123***a,b*, may be used to support the individual in case of a slip or fall. Such support may be provided passively (e.g., a fixed resistance provided by the motors), actively (e.g., by sensing an acceleration and applying a resistance to the tethers), or by mechanical means (e.g., seatbelt-type mechanical locking mechanism that locks the tether upon a sudden pull). Other embodiments may use additional vertical supports **3122***a,b*, tethers, **3124***a,b*, and sensors/actuators **3123***a,b*. For example, some embodiments may have vertical supports **3122***a,b* and associated equipment at the front and back, and at the left and right sides of the individual. Many other configurations are possible.

FIG. **32** is an exemplary method for application of the system to improve the performance of a sports team. In a first step, an ideal neurological functioning profile for each player position in a given sport is predicted by experts in the sport (e.g., coaches, trainers, athletes, sports bettors, etc.) for maximization of performance for that position in that sport **3201**. Then, dual task assessments are performed for athletes from a variety of positions that play the sport **3202**. Based on the neurological condition profile generated by the testing, performance and/or play strategy recommendations are made for performance improvements for that player for that position **3203**. Performance of the athletes during actual play is evaluated by the experts in the sport **3204**. The evaluation feedback from the experts is provided back to step **3201**, and athletes are retested at step **3202**. All of these steps may be performed repeatedly to continuously refine input and recommendations. In some embodiments, dual task assessments at step **3202** may be specifically selected to condition or train the aspects of neurological functioning determined to be ideal at step **3201** (i.e., step **3202** a can be both a conditioning/training step and an evaluation step).

FIG. **33** (PRIOR ART) is a side-view diagram of the human brain showing the major areas of brain function. The brain is composed of several functional areas that work together to regulate different aspects of our behavior, cognition, and emotions. The brain can be divided into eight major functional areas: the frontal lobe **3310**, the premotor cortex **3320**, the motor cortex **3330**, the sensory cortex **3340**, the parietal lobe **3350**, the occipital lobe **3360**, the cerebellum **3370**, and the temporal lobe **3380**. One or more of these major areas is activated during physical or mental activity, and the areas that are activated depend on the type of physical or mental activity. The mapping of these various areas has been demonstrated through magnetic resonance imaging (MRI) of the brain during various types of physical and mental activities.

The frontal lobe **3310** is the portion of the brain responsible for intellect and personality. It is this portion of the brain that is activated during mental activities such as planning, abstract thought, puzzle solving, coordinating complex cognitive behavior, expression of personality, social relationships, impulse control, and speech (in Broca's speech area; not shown).

The premotor cortex **3320** is the portion of the brain responsible for fine motor control, as well as attention, decision-making, and language processing. It is this portion of the brain that is activated during mental activities such planning and executing movements of the body. The premotor cortex is responsible for planning and coordinating complex movements, such as reaching, grasping, and manipulating objects. It receives input from various sensory regions of the brain and integrates this information to plan and execute movements that are appropriate for the situation. The premotor cortex is involved in a variety of other cognitive functions. For example, it has been implicated in attentional processes, such as the ability to shift attention from one task to another. It is also involved in decision-making, particularly when it comes to selecting between different movements or actions. Finally, the premotor cortex is involved in some aspects of language processing. For instance, some studies suggest that it plays a role in the comprehension of action-related words, such as "grasp" or "kick", and that it is involved in the production of speech gestures that accompany speech, such as lip movements and facial expressions.

The motor cortex **3330** is the portion of the brain responsible for planning, initiating, and controlling voluntary movements of the body. More specifically, the motor cortex plays a crucial role in the execution of both simple and complex movements, including fine motor skills such as writing or playing a musical instrument, as well as larger movements such as walking and running. The motor cortex is organized from top to bottom in a way that corresponds to the following parts of the body: feet, legs, torso/trunk, arms, hands, and face. The motor cortex **3330** receives input from other parts of the brain, such as the sensory cortex **3340**, which provides information about the current position and movement of the body. This information is used by the motor cortex to plan and execute appropriate movements. In addition to its role in voluntary movements, the motor cortex is also involved in reflexive movements. When a part of the body is stimulated, such as by touching a hot object, the motor cortex can initiate a reflexive movement to remove the body part from the source of the stimulation.

The sensory cortex **3340** is the portion of the brain responsible for processing and integrating sensory information from various parts of the body. The sensory cortex is part of the parietal lobe **3350** of the brain, and it is organized into different regions that correspond to the different senses, such as vision, hearing, touch, taste, and smell. The sensory cortex **3340** is organized from top to bottom in a way that corresponds to the following parts of the body: feet, legs, torso/trunk, arms, hands, and face, and each of these portions is associated with its corresponding portion in the motor cortex **3330**. When sensory information is received by the brain, it is first processed in the appropriate sensory receptor cells in the body. The information is then transmitted to the thalamus, which acts as a relay station, before being transmitted to the appropriate region of the sensory cortex for further processing. In the sensory cortex, neurons respond selectively to specific features of the sensory input, such as the orientation of visual stimuli, the frequency of auditory stimuli, or the location and intensity of tactile stimuli. This allows the brain to perceive and interpret sensory information in a meaningful way. The sensory cortex also plays an important role in integrating sensory information from different modalities, such as combining visual and auditory information to recognize a speaker's face and voice or combining tactile and visual information to identify an object by touch. The sensory cortex is a critical part of the brain that allows humans to perceive and interpret sensory information from the environment, which is essential for their ability to interact with the world around them.

The parietal lobe **3350** is the portion of the brain responsible for a variety of sensory processing, spatial processing, attention and alertness, and motor skill planning. The parietal lobe contains the somatosensory cortex, which is responsible for processing touch, temperature, and pain sensations from different parts of the body. The somatosensory cortex is organized into specific regions that correspond to different parts of the body. The parietal lobe is involved in spatial perception and processing, which includes understanding the location and movement of objects in space, as well as our own body position and movement. The parietal lobe is important for maintaining attention and alertness, as well as for directing attention to specific stimuli.

The occipital lobe **3360** is the portion of the brain responsible for intellect and personality. The occipital lobe contains the primary visual cortex, which is responsible for processing and interpreting visual information from the eyes. This includes recognizing shapes, colors, and patterns, as well as perceiving depth, movement, and other visual features. The occipital lobe is also involved in the storage and retrieval of visual memories, including memories of objects, faces, and scenes. The occipital lobe plays a role in spatial perception and processing, including our ability to navigate our environment and recognize the relative location of objects in space. The occipital lobe is likewise involved in recognizing objects, faces, and other visual stimuli, and it works in conjunction with other parts of the brain to allow us to identify and categorize different visual stimuli.

The cerebellum **3370** is the portion of the brain responsible for functions related to movement, coordination, balance, and posture. The cerebellum is involved in fine-tuning and coordinating voluntary movements, including posture, balance, and coordination. The cerebellum receives sensory information from the inner ear, eyes, and muscles, and uses this information to help maintain balance and coordination. The cerebellum plays an important role in motor learning and the development of new motor skills, including the ability to adapt to changing environmental conditions. The cerebellum has also been implicated in a variety of cognitive functions, including attention, working memory, and language processing. Lastly, the cerebellum has been found to be involved in the regulation of emotional responses, and damage to the cerebellum has been associated with mood disorders such as depression and anxiety.

The temporal lobe **3380** is the portion of the brain responsible for functions related to auditory processing, language comprehension, and memory. The temporal lobe contains the primary auditory cortex, which is responsible for processing and interpreting auditory information from the ears. This includes recognizing sounds, discriminating between different frequencies and pitches, and localizing the source of sounds. The temporal lobe is involved in the comprehension of spoken and written language, including the recognition of words and the understanding of grammar and syntax. The temporal lobe plays an important role in the storage and retrieval of long-term memories, particularly those related to visual and auditory information. The temporal lobe has some involvement in the recognition and categorization of objects, faces, and other visual stimuli, and has been implicated in the processing of emotions, particularly with respect to the recognition of facial expressions and the regulation of emotional responses.

FIG. **34** (PRIOR ART) is a diagram showing the process of neurogenesis in the human brain. Adult neurogenesis is a multistage process by which neurons are generated and integrated into existing neuronal circuits. Neurogenesis plays a fundamental role in postnatal brain, where it is required for neuroplasticity. Perturbation of adult neurogenesis contributes to several human central nervous system conditions, including cognitive impairment and neurodegenerative diseases.

Neurons start their development neural stem cells (NSCs) **3411**. A neural stem cell **3411** is a type of central nervous system cell that can give rise to various types of neural cells, including neurons, astrocytes, and oligodendrocytes. Neural stem cells are located in various regions of the brain, including the subventricular zone (SVZ) lining the lateral ventricles and the subgranular zone (SGZ) of the dentate gyrus in the hippocampus. Neural stem cells **3411** have the ability to divide and differentiate into different types of neural cells, allowing them to play a critical role in the development and maintenance of the nervous system. During development, neural stem cells give rise to the various types of neural cells that make up the brain and spinal cord. In adulthood, they are involved in the maintenance and repair of the nervous system, such as the replacement of damaged or dying neurons.

The process of neurogenesis can be divided into three major stages, proliferation **3431**, differentiation **3432**, and synaptogenesis **3433**. This process starts at the proliferation stage **3431** wherein neural stem cells **3411** develop into neuronal progenitor cells (NPCs) **3412**, which are capable of proliferation and multi-potential differentiation but are unable to self-renew. This process takes one to three days **3421**. Next, the NPCs enter the differentiation phase **3432**, wherein the NPCs exit from the cell cycle and are committed to the neuronal lineage, becoming neuroblasts **3413** which will eventually mature into neurons. This process takes about a week **3422**. After the commitment to the neuronal lineage, the neuroblasts **3413** begin to develop into neurons, starting as immature neurons **3414*a*** which begin to extend their axonal projections and dendritic growths **3414*a***. This process takes roughly two weeks **3423**. In the following three weeks **3424**, the immature neurons **3414** grow into mature neurons **3415** with further growth of their axonal projections and dendritic growths **3415*a***. In the synaptogenesis stage **3433**, newly generated neurons finalize their axonal projections and dendritic growths **3415*a*** and establish their synaptic contacts **3416*b*** into the pre-existing circuitry of the brain. This process takes about four weeks **3425**. The entire process of neurogenesis takes about 2-4 months from the birth of neural stem cells **3411** to full integration of mature neurons with surrounding cells and incorporation into the brain circuitry.

FIG. **35** (PRIOR ART) is a diagram showing the process of neuroplasticity in the human brain. Neuroplasticity is the ability of the brain to change and adapt in response to experiences and environmental stimuli. The process of neuroplasticity involves the rewiring of neural connections and the creation of new synaptic connections between neurons. The extent of neuroplasticity varies depending on factors such as age, genetics, and the nature of the environmental input. In general, the brain is most plastic during early development and retains some degree of plasticity throughout life, although the extent of plasticity may decline with age. However, various interventions such as physical exercise, cognitive training, and exposure to enriched environments can promote neuroplasticity and help the brain adapt and change throughout the lifespan.

The process of neuroplasticity can be described in general terms as follows. The brain receives input from the environment, such as sensory stimuli or learning experiences. The input triggers the activation of neural networks in the brain, leading to the release of neurotransmitters and the firing of action potentials in neurons. The activation of neural networks leads to changes in synaptic strength, such as the strengthening or weakening of existing synaptic connections. Over time, repeated neural activation can lead to the growth and formation of new synapses, as well as the pruning of existing synapses that are not frequently used. The changes in neural connectivity and synaptic strength can lead to changes in behavior, such as improvements in cognitive function, the development of new skills, or the recovery from brain injury. \

A major chemical process by which neuronal connections are strengthened is via glutamate, the most abundant neurotransmitter. Glutamate is an excitatory neurotransmitter, meaning that its release by a first neuron into a neuronal connection makes it more likely that a second neuron of the neuronal connection will fire. Two main types of receptors are involved in this chemical process, N-methyl-D-aspartate (NMDA) and α-amino-3-hydroxy-5-methyl-4-isoxazole-propionic acid (AMPA), both of which bind with and are activated by glutamate.

The diagram on the left **3510** shows a weak neuronal interaction at a first neuron **3520** resulting in insufficient chemical change to strengthen a neuronal connection with a second neuron **3540**. Weak interactions at a first neuron **3520**, such as those resulting from minimal exercise or mild mental activity, cause a small release of glutamate **3521** into the neuronal connection **3530** between the two neurons. The glutamate binds with both NMDA receptors **3541** and AMPA receptors **3542**, causing them to activate. However, magnesium **3521*a*** blocks the NDMA receptors **3541**, preventing calcium ions **3532** from entering the second neuron **3540**. Although a small number **3531*a*** of sodium ions **3531** flow through AMPA receptors **3542** into second neuron **3540**, the flow of sodium ions **3531** is insufficient to force magnesium **3521*a*** out of NDMA receptors **3541** to allow calcium ions **3532** to flow into second neuron **3540**. The result is that the second neuron **3540** does not fire, and the neuronal connection **3530** is not strengthened.

The diagram on the right **3550** shows a strong neuronal interaction at first neuron **3520** resulting in sufficient chemical change to strengthen the neuronal connection with second neuron **3540**. Strong interactions at a first neuron **3520**, such as those resulting from significant exercise or strong mental activity, cause a large release of glutamate **3521** into the neuronal connection **3530** between the two neurons. The glutamate binds with a larger number of both NMDA receptors **3541** and AMPA receptors **3542**, causing them to activate. The larger number of activated AMPA receptors **3531** causes a greater flood **3531*a*** of sodium ions **3531** into second neuron **3540** which strongly depolarizes second neuron **3540**. The depolarization is sufficient to force magnesium **3521*a*** out of NDMA receptors **3541** to allow calcium ions **3532** to flow into second neuron **3540**. The flood **3532*a*** of calcium ions **3532** into second neuron **3540** which triggers exposure of inactive AMPA receptors **3542**, allowing more glutamate **3521** to attach to them, which repeats the process until the released glutamate **3521** is exhausted. The result is that the second neuron **3540** does fires, and the neuronal connection **3530** is strengthened.

FIG. **36** (PRIOR ART) is a diagram showing the influence of exercise on neurogenesis and neuroplasticity. Exercise has been shown to have numerous positive effects on the brain, including enhancing neurogenesis (the growth of new neurons) and neuroplasticity (the brain's ability to change and adapt in response to experience). Studies have shown that regular exercise increases the production of new neurons in the hippocampus, a region of the brain that is involved in learning and memory. Two major chemical pathways by which exercise affects neurogenesis and neuroplasticity are the irisin **3613** production pathway **3610** and the brain-derived neurotropic factor (BDNF) **3622** production pathway **3620**.

Regarding the irisin production pathway **3610**, irisin **3612** is a hormone that is released by muscles **3611** during exercise **3612**. In addition to other beneficial effects like improving glucose metabolism and reducing inflammation, irisin **3612** modulates signal transducer and activator of transcription 3 (STAT3) signaling **3614**. STAT3 is a protein that plays a critical role in signal transduction and gene expression. It is involved in a variety of cellular processes, including cell proliferation, differentiation, and survival, as well as regeneration of various tissues including the liver, heart, and muscle. STAT3 signaling causes neurogenesis **3615**, **3616** in the brain **3614**.

Regarding the BDNF **3622** production pathway **3620**, BDNF is synthesized **3621** by the brain **3614** during exercise. BDNF supports the survival and growth of existing neurons and helps to create new connections between neurons, which is important for learning and memory. It is also involved in regulating mood, anxiety, and stress responses. Research has suggested that low levels of BDNF may be associated with various neurological and psychiatric disorders, including depression, schizophrenia, and Alzheimer's disease. Therefore, BDNF has become a target of interest for the development of new treatments for these conditions. Synthesis of BDNF **3622** increases neuroplasticity **3623**.

Studies have indicated that both pathways **3610**, **3620** can lead to a generalized decreased risk of Alzheimer's disease and dementia, although results are still preliminary and targeted treatments do not yet exist.

FIG. **37** (PRIOR ART) is a diagram showing the influence of variable exercise intensity on neurogenesis and neuroplasticity. As described above, exercise promotes neurogenesis and neuroplasticity at least through the irisin production pathway **3610** and the BDNF production pathway **3620**. However, additional evidence using magnetic resonance imaging (MRI) **3712**, **3722** before and after exercise suggests that differences in exercise intensity can lead to neurogenesis and neuroplasticity in different areas of the brain. Studies have shown that, for a given person **3701**, low intensity exercise **3711** such as walking appears to activate areas of the brain associated with cognitive control and attention processing as the frontal cortex **3713**, the parietal lobe **3714**, and the occipital lobe **3715**. Conversely, high intensity exercise **3721** such as running appears to activate areas of the brain associated with emotional processing (as well as fatigue), such as the motor cortex and sensory cortex **3723** and the temporal lobe **3724**.

The differential impact on neurogenesis and neuroplasticity of differences in exercise intensity suggests that targeted neurological treatments could be developed in part by varying the intensity of exercise during treatment, although the study results are still preliminary and targeted treatments do not yet exist.

FIG. **38** is a diagram showing an exemplary application of virtual-reality-enhanced, dual-task facilitation of neurogenesis and neuroplasticity for treatment of language, cognition, and memory brain functions. Here, a series of computer display information windows **3801** are shown containing information about intensity of exercise **3802**, brain function(s) targeted **3803**, and history and improvement statistics **3804** of an individual undergoing evaluation or treatment. Another computer display window shows an exemplary virtual reality (VR) environment window **3805** inside of a virtual reality headset. Depending on configuration, information windows **3801** may be visible inside the VR headset or may be displayed on a separate screen outside of the VR headset.

Brain functions targeted as shown in brain functions target window **3803** are a combination of an intensity of exercise of the individual and a secondary (mental) activity performed by the individual in the virtual reality environment as shown in virtual reality (VR) window **3805** inside of a virtual reality headset. In this example, intensity of exercise indicator **3802** indicates that the individual undergoing evaluation or treatment is engaged in a low intensity exercise such as walking as suggested or dictated by the virtual reality environment of VR environment window **3805**. The walking may be performed, for example, on a treadmill. This low intensity exercise has been chosen to facilitate activation areas of the brain associated with language, cognition, and memory, as shown in brain functions targeted window **3803**. The low intensity exercise may be reflected in the VR environment as, for example, movement along pathways in the VR environment.

Correspondingly, a cognitive walking game has been chosen for the VR environment of **3805** to further engage areas of the brain associated with language, cognition, and memory. Here, VR environment window **3805** shows an exemplary screenshot **3810** of a first-person (player) perspective of a three-dimensional game during game play and a corresponding map **3820** representing available pathways during gameplay. In this embodiment, map **3820** represents an internal logical of the game and is not visible to player. Pathways within map **3821** are represented visually to player as roads **3811** in screenshot **3810**. Screenshot **3810** further shows terrain comprising hills **3812**, a series of signposts **3830** located at road **3811** intersections containing (in this embodiment) mathematical equations to be solved by player.

The maze and pathways activate brain areas related to memory, specifically the individual's ability to remember pathways. As map **3820** is not visible to the player, player must remember which series of turns and pathways will lead player from the beginning to the end of map. Correct turns may be rewarded and wrong turns may be penalized (e.g., with points or scores) or the total time to complete map **3820** from beginning to end may be used as the scoring or measurement variable. Time to complete map may be tracked from one play to another (or alternately, player map be shown map **3820** for a period of time at the beginning of game play, and instructed to remember the pathway from beginning to end). Thus, the combination of low exercise intensity with memory-related secondary activities of the VR environment provides targeted neurogenesis and neuroplasticity for memory-related areas of the brain as indicated in brain functions targeted window **3803**.

The VR environment also contains elements related to cognitive function associated with math and problem-solving. Player may be instructed to solve the brief mathematical equations on signposts **3830** during game play, whereby solving an equation correctly provides an indication as to the proper path to take at each intersection. In this example, the instructions **3806** tell the individual to "take the path to the left when the answer is even," which requires the individual to process and understand language associated with the cognitive task. While mathematical equations are shown on signposts **3830** in this embodiment, any type of problem, puzzle, brain-teaser, or cognitive task may be used to measure aspects of cognitive function. Thus, the combination of low exercise intensity with cognitive-related and language-related secondary activities of the VR environment provides targeted neurogenesis and neuroplasticity for cognitive-related and language-related areas of the brain as indicated in brain functions targeted window **3803**.

FIG. **39** is a diagram showing an exemplary application of dual-task facilitation of neurogenesis and neuroplasticity for treatment of visual, sensory, and fine motor skill brain functions. Here, a series of computer display information windows **3901** are shown containing information about intensity of exercise **3902**, brain function(s) targeted **3903**, and history and improvement statistics **3904** of an individual undergoing evaluation or treatment. Another computer display window shows an exemplary virtual reality (VR) environment window **3905** inside of a virtual reality headset. Depending on configuration, information windows **3901** may be visible inside the VR headset or may be displayed on a separate screen outside of the VR headset.

Brain functions targeted as shown in brain functions target window **3903** are a combination of an intensity of exercise of the individual and a secondary (mental) activity performed by the individual in the virtual reality environment as shown in virtual reality (VR) window **3905** inside of a virtual reality headset. In this example, intensity of exercise indicator **3902** indicates that the individual undergoing evaluation or treatment is engaged in a moderate intensity exercise such as cycling as suggested or dictated by the virtual reality environment of VR environment window **3905**. This moderate intensity exercise has been chosen to facilitate activation areas of the brain associated with fine motor skills, sensory processing, and visual processing, as shown in brain functions targeted window **3903**. The moderate intensity exercise may be reflected in the VR environment as, for example, movement along pathways in the VR environment. The cycling may be performed, for example, on a stationary bicycle, and the pedaling motion on the stationary bicycle may be translated into movement in the VR environment. In this case, cycling on the stationary bicycle is translated into power to a propeller on a small aircraft **3908** in which the individual's avatar sits.

Correspondingly, a cognitive flying game has been chosen for the VR environment of **3905** to further engage areas of the brain associated with fine motor skills, sensory/spatial processing, and visual processing. Here, VR environment window **3905** shows an exemplary screenshot **3910** of a third-person (non-player) perspective of a three-dimensional game during game play having a small aircraft **3908** in which the individual's avatar sits. As the individual undergoing treatment cycles on the stationary bicycle, the pedaling motion is translated into power to a propeller on small aircraft **3908**. The individual receives instructions **3906** to "pop all balloons that are shades of purple." Purple balloons are indicated in this diagram as white (unpatterned) balloons. Identification of the colored balloons activates brain areas associated with visual processing. The experience of "flying" in the virtual reality environment activates areas of the brain associated with sensory and spatial input. Areas of the brain associated with fine motor skills are activated by the individual's operation of a game controller with his or her hands which directs the flight path of small aircraft **3908**. Thus, the combination of moderate exercise intensity with visually-related, sensory/spatially-related, and fine motor skills-related secondary activities of the VR environment provides targeted neurogenesis and neuroplasticity for fine motor skills, sensory/spatial processing, and visual processing areas of the brain as indicated in brain functions targeted window **3903**.

FIG. **40** is a diagram showing an exemplary application of dual-task facilitation of neurogenesis and neuroplasticity for treatment of gross motor skills, balance, and emotion brain functions. Here, a series of computer display information windows **4001** are shown containing information about intensity of exercise **4002**, brain function(s) targeted **4003**, and history and improvement statistics **4004** of an individual undergoing evaluation or treatment. Another computer display window shows an exemplary virtual reality (VR) environment window **4005** inside of a virtual reality headset. Depending on configuration, information windows **4001** may be visible inside the VR headset or may be displayed on a separate screen outside of the VR headset.

Brain functions targeted as shown in brain functions target window **4003** are a combination of an intensity of exercise of the individual and a secondary (mental) activity performed by the individual in the virtual reality environment as shown in virtual reality (VR) window **4005** inside of a virtual reality headset. In this example, intensity of exercise indicator **4002** indicates that the individual undergoing evaluation or treatment is engaged in a high intensity exercise such as running as suggested or dictated by the virtual reality environment of VR environment window **4005**. This high intensity exercise has been chosen to facilitate activation areas of the brain associated with gross motor skills, balance, and emotional brain functions, as shown in brain functions targeted window **4003**. The high intensity exercise may be reflected in the VR environment as, for example, movement along pathways in the VR environment. The running may be performed, for example, on a treadmill, and the running motion on the treadmill may be translated into movement in the VR environment. In this case, running on the treadmill is translated into running of the individual's avatar along a balance beam **4007** (which may be continuous or infinite to facilitate longer running periods).

Correspondingly, a balance beam game has been chosen for the VR environment of **4005** to further engage areas of the brain associated with gross motor skills, balance, and emotional processing. Here, VR environment window **4005** shows an exemplary screenshot of a third-person (non-player) perspective of a balance beam running game wherein the individual's avatar is shown running along a balance beam **4007** while trying to avoid falling off the balance beam. As the individual undergoing treatment runs on the treadmill, the running motion is translated into forward motion of the avatar on the balance beam **4007**, and the individual's posture or position on the treadmill (e.g., leaning left or right, running on the left or right edges of the treadmill) is translated into balance movements of the avatar. The individual receives instructions **4006** to "run across the beam, but don't fall!" One or more emotionally-related consequences of falling off the beam are shown, such as a pit of hungry alligators **4008**. The need to balance while running activates areas of the brain associated with gross motor skills and balance, while the emotionally-related consequences such as the pit of alligators activate areas of the brain associated with emotion (in this case fear of being eaten by alligators). Thus, the combination of high exercise intensity with gross-motor-skill-related, balance-related, and emotionally-related secondary activities of the VR environment provides targeted neurogenesis and neuroplasticity for gross motor skills, balance, and emotional brain functions areas of the brain as indicated in brain functions targeted window **4003**.

Hardware Architecture

Generally, the techniques disclosed herein may be implemented on hardware or a combination of software and hardware. For example, they may be implemented in an operating system kernel, in a separate user process, in a library package bound into network applications, on a specially constructed machine, on an application-specific integrated circuit (ASIC), or on a network interface card.

Software/hardware hybrid implementations of at least some of the embodiments disclosed herein may be implemented on a programmable network-resident machine (which should be understood to include intermittently connected network-aware machines) selectively activated or reconfigured by a computer program stored in memory. Such network devices may have multiple network interfaces that may be configured or designed to utilize different types of network communication protocols. A general architecture for some of these machines may be described herein in order to illustrate one or more exemplary means by which a given unit of functionality may be implemented. According to specific embodiments, at least some of the features or functionalities of the various embodiments disclosed herein may be implemented on one or more general-purpose computers associated with one or more networks, such as for example an end-user computer system, a client computer, a network server or other server system, a mobile computing device (e.g., tablet computing device, mobile phone, smartphone, laptop, or other appropriate computing device), a consumer electronic device, a music player, or any other suitable electronic device, router, switch, or other suitable device, or any combination thereof. In at least some embodiments, at least some of the features or functionalities of the various embodiments disclosed herein may be implemented in one or more virtualized computing environments (e.g., network computing clouds, virtual machines hosted on one or more physical computing machines, or other appropriate virtual environments).

Referring now to FIG. **13**, there is shown a block diagram depicting an exemplary computing device **10** suitable for implementing at least a portion of the features or functionalities disclosed herein. Computing device **10** may be, for example, any one of the computing machines listed in the previous paragraph, or indeed any other electronic device capable of executing software- or hardware-based instructions according to one or more programs stored in memory. Computing device **10** may be configured to communicate with a plurality of other computing devices, such as clients or servers, over communications networks such as a wide area network a metropolitan area network, a local area network, a wireless network, the Internet, or any other network, using known protocols for such communication, whether wireless or wired.

In one embodiment, computing device **10** includes one or more central processing units (CPU) **12**, one or more interfaces **15**, and one or more busses **14** (such as a peripheral component interconnect (PCI) bus). When acting under the control of appropriate software or firmware, CPU **12** may be responsible for implementing specific functions associated with the functions of a specifically configured computing device or machine. For example, in at least one embodiment, a computing device **10** may be configured or designed to function as a server system utilizing CPU **12**, local memory **11** and/or remote memory **16**, and interface(s) **15**. In at least one embodiment, CPU **12** may be caused to perform one or more of the different types of functions and/or operations under the control of software modules or components, which for example, may include an operating system and any appropriate applications software, drivers, and the like.

CPU **12** may include one or more processors **13** such as, for example, a processor from one of the Intel, ARM, Qualcomm, and AMD families of microprocessors. In some embodiments, processors **13** may include specially designed hardware such as application-specific integrated circuits (ASIC s), electrically erasable programmable read-only memories (EEPROMs), field-programmable gate arrays (FPGAs), and so forth, for controlling operations of computing device **10**. In a specific embodiment, a local memory **11** (such as non-volatile random access memory (RAM) and/or read-only memory (ROM), including for example one or more levels of cached memory) may also form part of CPU **12**. However, there are many different ways in which memory may be coupled to system **10**. Memory **11** may be used for a variety of purposes such as, for example, caching and/or storing data, programming instructions, and the like. It should be further appreciated that CPU **12** may be one of a variety of system-on-a-chip (SOC) type hardware that may include additional hardware such as memory or graphics processing chips, such as a QUALCOMM SNAPDRAGON™ or SAMSUNG EXYNOS™ CPU as are becoming increasingly common in the art, such as for use in mobile devices or integrated devices.

As used herein, the term "processor" is not limited merely to those integrated circuits referred to in the art as a processor, a mobile processor, or a microprocessor, but broadly refers to a microcontroller, a microcomputer, a programmable logic controller, an application-specific integrated circuit, and any other programmable circuit.

In one embodiment, interfaces **15** are provided as network interface cards (NICs). Generally, NICs control the sending and receiving of data packets over a computer network; other types of interfaces **15** may for example support other peripherals used with computing device **10**. Among the interfaces that may be provided are Ethernet interfaces, frame relay interfaces, cable interfaces, DSL interfaces, token ring interfaces, graphics interfaces, and the like. In addition, various types of interfaces may be provided such as, for example, universal serial bus (USB), Serial, Ethernet, FIREWIRE™, THUNDERBOLT™, PCI, parallel, radio frequency (RF), BLUETOOTH™, near-field communications (e.g., using near-field magnetics), 802.11 (WiFi), frame relay, TCP/IP, ISDN, fast Ethernet interfaces, Gigabit Ethernet interfaces, Serial ATA (SATA) or external SATA (ESATA) interfaces, high-definition multimedia interface (HDMI), digital visual interface (DVI), analog or digital audio interfaces, asynchronous transfer mode (ATM) interfaces, high-speed serial interface (HSSI) interfaces, Point of Sale (POS) interfaces, fiber data distributed interfaces (FDDIs), and the like. Generally, such interfaces **15** may include physical ports appropriate for communication with appropriate media. In some cases, they may also include an independent processor (such as a dedicated audio or video processor, as is common in the art for high-fidelity A/V hardware interfaces) and, in some instances, volatile and/or non-volatile memory (e.g., RAM).

Although the system shown in FIG. **13** illustrates one specific architecture for a computing device **10** for implementing one or more of the inventions described herein, it is by no means the only device architecture on which at least a portion of the features and techniques described herein may be implemented. For example, architectures having one or any number of processors **13** may be used, and such processors **13** may be present in a single device or distributed among any number of devices. In one embodiment, a single processor **13** handles communications as well as routing computations, while in other embodiments a separate dedicated communications processor may be provided. In various embodiments, different types of features or functionalities may be implemented in a system according to the invention that includes a client device (such as a tablet device or smartphone running client software) and server systems (such as a server system described in more detail below).

Regardless of network device configuration, the system of the present invention may employ one or more memories or memory modules (such as, for example, remote memory block **16** and local memory **11**) configured to store data, program instructions for the general-purpose network operations, or other information relating to the functionality of the embodiments described herein (or any combinations of the above). Program instructions may control execution of or comprise an operating system and/or one or more applications, for example. Memory **16** or memories **11**, **16** may also be configured to store data structures, configuration data, encryption data, historical system operations information, or any other specific or generic non-program information described herein.

Because such information and program instructions may be employed to implement one or more systems or methods described herein, at least some network device embodiments may include nontransitory machine-readable storage media, which, for example, may be configured or designed to store program instructions, state information, and the like for performing various operations described herein. Examples of such nontransitory machine-readable storage media include, but are not limited to, magnetic media such as hard disks, floppy disks, and magnetic tape; optical media such as CD-ROM disks; magneto-optical media such as optical disks, and hardware devices that are specially configured to store and perform program instructions, such as read-only memory devices (ROM), flash memory (as is common in mobile devices and integrated systems), solid state drives (SSD) and "hybrid SSD" storage drives that may combine physical components of solid state and hard disk drives in a single hardware device (as are becoming increasingly common in the art with regard to personal computers), memristor memory, random access memory (RAM), and the like. It should be appreciated that such storage means may be integral and non-removable (such as RAM hardware modules that may be soldered onto a motherboard or otherwise integrated into an electronic device), or they may be removable such as swappable flash memory modules (such as "thumb drives" or other removable media designed for rapidly exchanging physical storage devices), "hot-swappable" hard disk drives or solid state drives, removable optical storage discs, or other such removable media, and that such integral and removable storage media may be utilized interchangeably. Examples of program instructions include both object code, such as may be produced by a compiler, machine code, such as may be produced by an assembler or a linker, byte code, such as may be generated by for example a JAVA™ compiler and may be executed using a Java virtual machine or equivalent, or files containing higher level code that may be executed by the computer using an interpreter (for example, scripts written in Python, Perl, Ruby, Groovy, or any other scripting language).

In some embodiments, systems according to the present invention may be implemented on a standalone computing system. Referring now to FIG. **14**, there is shown a block diagram depicting a typical exemplary architecture of one or more embodiments or components thereof on a standalone computing system. Computing device **20** includes processors **21** that may run software that carry out one or more functions or applications of embodiments of the invention, such as for example a client application **24**. Processors **21** may carry out computing instructions under control of an operating system **22** such as, for example, a version of MICROSOFT WINDOWS™ operating system, APPLE MACOS™ or iOS™ operating systems, some variety of the Linux operating system, ANDROID™ operating system, or the like. In many cases, one or more shared services **23** may be operable in system **20**, and may be useful for providing common services to client applications **24**. Services **23** may for example be WINDOWS™ services, user-space common services in a Linux environment, or any other type of common service architecture used with operating system **21**. Input devices **28** may be of any type suitable for receiving user input, including for example a keyboard, touchscreen, microphone (for example, for voice input), mouse, touchpad, trackball, or any combination thereof. Output devices **27** may be of any type suitable for providing output to one or more users, whether remote or local to system **20**, and may include for example one or more screens for visual output, speakers, printers, or any combination thereof. Memory **25** may be random-access memory having any structure and architecture known in the art, for use by processors **21**, for example to run software. Storage devices **26** may be any magnetic, optical, mechanical, memristor, or electrical storage device for storage of data in digital form (such as those described above, referring to FIG. **13**). Examples of storage devices **26** include flash memory, magnetic hard drive, CD-ROM, and/or the like.

In some embodiments, systems of the present invention may be implemented on a distributed computing network, such as one having any number of clients and/or servers. Referring now to FIG. **15**, there is shown a block diagram depicting an exemplary architecture **30** for implementing at least a portion of a system according to an embodiment of the invention on a distributed computing network. According to the embodiment, any number of clients **33** may be provided. Each client **33** may run software for implementing client-side portions of the present invention; clients may comprise a system **20** such as that illustrated in FIG. **14**. In addition, any number of servers **32** may be provided for handling requests received from one or more clients **33**. Clients **33** and servers **32** may communicate with one another via one or more electronic networks **31**, which may be in various embodiments any of the Internet, a wide area network, a mobile telephony network (such as CDMA or GSM cellular networks), a wireless network (such as WiFi, WiMAX, LTE, and so forth), or a local area network (or indeed any network topology known in the art; the invention does not prefer any one network topology over any other). Networks **31** may be implemented using any known network protocols, including for example wired and/or wireless protocols.

In addition, in some embodiments, servers **32** may call external services **37** when needed to obtain additional information, or to refer to additional data concerning a particular call. Communications with external services **37** may take place, for example, via one or more networks **31**. In various embodiments, external services **37** may comprise web-enabled services or functionality related to or installed on the hardware device itself. For example, in an embodiment where client applications **24** are implemented on a smartphone or other electronic device, client applications **24** may obtain information stored in a server system **32** in the cloud or on an external service **37** deployed on one or more of a particular enterprise's or user's premises.

In some embodiments of the invention, clients **33** or servers **32** (or both) may make use of one or more specialized services or appliances that may be deployed locally or remotely across one or more networks **31**. For example, one or more databases **34** may be used or referred to by one or more embodiments of the invention. It should be understood by one having ordinary skill in the art that databases **34** may be arranged in a wide variety of architectures and using a wide variety of data access and manipulation means. For example, in various embodiments one or more databases **34** may comprise a relational database system using a structured query language (SQL), while others may comprise an alternative data storage technology such as those referred to in the art as "NoSQL" (for example, HADOOP CAS SANDRA™, GOOGLE BIGTABLE™, and so forth). In some embodiments, variant database architectures such as column-oriented databases, in-memory databases, clustered databases, distributed databases, or even flat file data repositories may be used according to the invention. It will be appreciated by one having ordinary skill in the art that any combination of known or future database technologies may be used as appropriate, unless a specific database technology or a specific arrangement of components is specified for a particular embodiment herein. Moreover, it should be appreciated that the term "database" as used herein may refer to a physical database machine, a cluster of machines acting as a single database system, or a logical database within an overall database management system. Unless a specific meaning is specified for a given use of the term "database", it should be construed to mean any of these senses of the word, all of which are understood as a plain meaning of the term "database" by those having ordinary skill in the art.

Similarly, most embodiments of the invention may make use of one or more security systems **36** and configuration systems **35**. Security and configuration management are common information technology (IT) and web functions, and some amount of each are generally associated with any IT or web systems. It should be understood by one having ordinary skill in the art that any configuration or security subsystems known in the art now or in the future may be used in conjunction with embodiments of the invention without limitation, unless a specific security **36** or configuration system **35** or approach is specifically required by the description of any specific embodiment.

FIG. **16** shows an exemplary overview of a computer system **40** as may be used in any of the various locations throughout the system. It is exemplary of any computer that may execute code to process data. Various modifications and changes may be made to computer system **40** without departing from the broader scope of the system and method disclosed herein. Central processor unit (CPU) **41** is connected to bus **42**, to which bus is also connected memory **43**, nonvolatile memory **44**, display **47**, input/output (I/O) unit **48**, and network interface card (NIC) **53**. I/O unit **48** may, typically, be connected to keyboard **49**, pointing device **50**, hard disk **52**, and real-time clock **51**. NIC **53** connects to network **54**, which may be the Internet or a local network, which local network may or may not have connections to the Internet. Also shown as part of system **40** is power supply unit **45** connected, in this example, to a main alternating current (AC) supply **46**. Not shown are batteries that could be present, and many other devices and modifications that are well known but are not applicable to the specific novel functions of the current system and method disclosed herein. It should be appreciated that some or all components illustrated may be combined, such as in various integrated applications, for example Qualcomm or Samsung system-on-a-chip (SOC) devices, or whenever it may be appropriate to combine multiple capabilities or functions into a single hardware device (for instance, in mobile devices such as smartphones, video game consoles, in-vehicle computer systems such as navigation or multimedia systems in automobiles, or other integrated hardware devices).

FIG. **41** is a block diagram showing an exemplary architecture for a cloud-based gaming platform with integrated healthcare diagnostics and treatment. The exemplary architecture shown in this diagram is a client-server architecture shown in the context of a computer gaming platform, although the architecture shown can be applied to any multi-person virtual experience. Cloud-based gaming platform with integrated healthcare diagnostics and treatment **4100** comprises frontend services **4110**, backend services **4120**, and a healthcare diagnostics and treatment module **4200**. The cloud-based gaming platform with integrated healthcare diagnostics and treatment **4100** of this embodiment may be accessed by any number of game clients **4130** each operating on a particular computing device and through which the user of the computing device accesses the virtual experience (in this example, a multi-player computer game).

Frontend services **4110** are accessible via a network connection (e.g., via the Internet) by game clients **4130**, and comprise game platform services **4111** and game servers **4112**. Game platform services **4111** provide services outside of game play such as leaderboards, group formation, lobbies, chat rooms, inventory management, and account management. Game servers **4112** are the hardware (real or virtual) on which the games are run. Game servers **4112** may be real or virtual, and dedicated or not, and distributed or not. The exemplary architecture of this diagram shows dedicated, non-distributed game servers (which may still be either real machines or virtual machines). Each game server **4112** of this embodiment further comprises a brainwave entrainment manager **4141** with functionality similar to brainwave entrainment manager **200** of earlier-described embodiments. In this embodiment, however, brainwave entrainment manager **4141** manages brainwave entrainment routines for a plurality of users engaging in a shared virtual experience. In addition to receiving game data, frontend **4110** may be configured to receive confidential health-related data via a Health Information Patient Protection Act (HIPPA) compliant health data tunnel such as a virtual private network (VPN), which health-related data may be passed to healthcare diagnostics and treatment module **4200** via a HIPPA-compliant security gateway **4210**. In an alternate configuration, healthcare diagnostics and treatment module **4200** receives health-related data direct from game clients **4130** through a HIPPA-compliant health data tunnel such as a VPN via HIPPA-compliant security gateway **4210**. In either configuration, the health-related data is secured against receipt by third parties.

Backend services **4120** interface only with frontend services **4110** and are not directly accessible by game clients **4130**. Backend services **4120** provide the storage and administrative functionality for the game and comprise an analytics stack **4121** and a game database **4122**. Analytics stack **4121** allows for queries about game operations and statistics by gameplay programmers and designers, business managers, and customer service representatives. Game database **4122** stores all information about the game necessary for its operation including, but not limited to, locations, maps, activities of player characters and non-player characters, game states, and game events.

Game clients **4130** are software applications or web browsers operating on end-user computing devices which access cloud-based gaming platform **4100** to send and retrieve game-related data. Game clients **4100** may operate on any end-user device, three examples of which are smartphones **4131**, personal computers (PCs) **4132**, and gaming consoles or handheld devices **4133**. Game clients **4130** access cloud-based gaming platform **4100** via a network connection (e.g., via the Internet or a local area network) exchange game-related data with frontend **4110** of cloud-based gaming platform **4100** and display the user's perspective of the game to the user of the device on which the game client is operating. Thus, each user sees a different perspective of the game from each other user. As will be further described below, in certain configurations this allows for a shared virtual experience or common activity along with a different gameplay feel or intensity and/or different brainwave entrainments for each user.

Healthcare diagnostics and treatment module **4200** comprises a HIPPA-compliant security gateway **4210**, an artificial intelligence (AI) assisted healthcare diagnostics module **4220**, and an alerts and treatments module **4230**. In this embodiment, healthcare diagnostics and treatment module **4200** is shown as being located on a common platform. However, being a cloud-based system, a variety of network configurations may be used. The components of frontend **4110**, backend **4120**, and healthcare diagnostics and treatment module **4200** may be hosted on the same server, across a network of servers, or across a network of cloud-based services and microservices (Amazon Web Services (AWS)™ being one example of such a cloud-based service).

Healthcare diagnostics and treatment module **4200** receives a variety of health-related data such as dual-task data (data regarding physical and mental performance of a user while engaged in combinations of physical and mental activity), social game data (such as audio recordings or chat logs between players during gameplay), and certain types of game play data (including, for example, reaction times to events in the game). As will be described below, healthcare diagnostics and treatment module **4200** processes the received data through a variety of analyzers to make diagnostic predictions about the physical and mental health of the user. One or more trained machine learning algorithms may be used to identify patterns and correlations across disparate types of data to make predictions and diagnoses about the physical and mental health of the user.

FIG. **42** is a block diagram showing an exemplary architecture for an AI-assisted healthcare diagnostics module aspect of a cloud-based gaming platform with integrated healthcare diagnostics and treatment. AI-assisted healthcare diagnostics module **4220** of this embodiment comprises a patient database **4221**, a physical function analysis module **4222**, a neurological function analysis module **4223**, a reaction time analysis module **4224**, a mental health analysis module **4225**, and one or more machine learning algorithms **4226**.

AI-assisted healthcare diagnostics module **4220** receives and transmits data via a HIPPA-compliant security gateway **4210** which is configured to secure protected health information to, from, and within healthcare diagnostics and treatment module **4200** to prevent unauthorized access to protected health information of persons using the system. HIPPA-compliant security gateway **4210** uses information technology (IT) security best practices which include, but are not limited to, data encryption, access restrictions, permission controls, user authentications, and audit controls. HIPPA-compliant security gateway **4210** may use a HIPPA-compliant data health data tunnel **4102** which secures transmitted data end-to-end from game clients **4130** to healthcare diagnostics and treatment module **4200**. HIPPA-compliant data health data tunnel **4102** may, as one example, use a virtual private network (VPN) established between game clients **4130** and healthcare diagnostics and treatment module **4200**, which may, depending on configuration, run through frontend **4110**. HIPPA-compliant data health data tunnel **4102** may be a separate connection from the connections between game clients **4130** and frontend **4110** through which game data **4110** is transmitted, in which case game data **4110** may not need to be encrypted. In some embodiments, HIPPA-compliant data health data tunnel **4102** and game data **4101** may be the same connection, but with HIPPA-compliant data health data encrypted or otherwise secured within the connection. For simplicity, HIPPA-compliant data health data tunnel **4102** and game data **4101** connections are shown as being singular connections, but in practice those connections will be made separately to each individual game client.

Patient database **4221** stores all incoming and outgoing patient data including, but not limited to, health-related data, game data, and healthcare diagnoses. Healthcare related data may include patient data such as medical records, medical evaluations, diagnoses, physical and mental evaluations, and dual-task data comprising physical and mental data acquired during dual-task testing. Game data may include patient data acquired during game play or related to game play such as types of games played, game scores, time spent playing, in-game performance data such as reaction times to certain events, trends in gameplay such as improvements or degradation over time, and social game data such as audio recordings or chat logs between players during gameplay. Diagnoses may include any sort of evaluation and recommendation regarding physical function or mental function including both high and low functionality or ability such as evaluations of cognition, speech, auditory processing, vision processing, motor skills, emotions, and memory. Diagnoses may include, but are not limited to, diagnoses of medical conditions, suspicions or indications of disease, suspicions or indications of chronic physical conditions, suspicions or indications of chronic mental or neurological conditions, and recommendations for treatment, rehabilitation, or therapy of any of the above.

Physical function analysis module **4222** receives and analyzes data associated with a physical task of a dual-task functional analysis. This will typically be the primary task of a dual-task functional analysis. Physical function analysis module **4222** may be configured to evaluate a variety of aspects of a patient's physical function including, but not limited to, range of motion, movement profiles, and gait analysis, each of which is described in more detail herein below. Analyses produced by physical function analysis module **4222** are output to machine learning algorithm **4226** as data for inclusion in determination of diagnoses by machine learning algorithm **4226**. Physical function analysis module **4222** and neurological function analysis module **4223** may coordinate with one another to act as a dual-task functional analysis system as further described herein below.

Neurological function analysis module **4223** receives and analyzes data associated with mental task of a dual-task functional analysis. This will typically be the associative activity task of a dual-task functional analysis. Neurological function analysis module **4223** may be configured to evaluate a variety of aspects of a patient's mental or neurological function including both high and low functionality or ability such as evaluations of cognition, speech, auditory processing, vision processing, motor skills, emotions, and memory, each of which is described in more detail herein below. Analyses produced by physical function analysis module **4222** are output to machine learning algorithm **4226** as data for inclusion in determination of diagnoses by machine learning algorithm **4226**. Physical function analysis module **4222** and neurological function analysis module **4223** may coordinate with one another to act as a dual-task functional analysis system as further described herein below.

In-game performance analysis module **4224** receives and analyzes data associated with game play including, but not limited to, in-game performance data. In-game performance data may be separate from associative activity task data received by neurological function analysis module **4223**, although there may be some overlap, depending on the game or software in operation. As a general rule, in-game performance data will be ancillary game play data such as shots per kill (in first person shooter games), number of tiles removed (in tile-based games like Mahjong), track records and lap times (in racing games), time spent playing each game, and similar data. This data may be tracked over time to identify deviances from expectation that may indicate some physical or cognitive change. For example, if a player plays a certain game on a daily basis, the player's in-game performance data would be expected to improve over time. A decrease in performance despite continued play, especially a sudden or dramatic decrease, may suggest a physical or neurological deficiency. Analyses produced by in-game performance analysis module **4224** are output to machine learning algorithm **4226** as data for inclusion in determination of diagnoses by machine learning algorithm **4226**.

Mental health analysis module **4225** receives and analyzes game-related social data such as chat sessions, audio from team or group gameplay, lobby interactions, game-joining or partnering preferences and trends, game-related group activities, and membership in game-related clubs, factions, or guilds. Game-related social data may be separate from associative activity task data received by neurological function analysis module **4223**, although there may be some overlap, depending on the game or software in operation. This data may be analyzed to identify potential mental health issues such as depression, anxiety, anger, sadness, frustration, and the like. One method of analysis that may be used by mental health analysis module **4225** to identify potential mental health issues is use of a trained machine learning algorithm to recognize patterns of social interaction similar to those that have led to such issues in a larger group comprising the training data for the machine learning algorithm. For example, a trend in the player's social data toward the use of words and phrases like "tired," "exhausted," "lack of energy," in chats and audio with friends and teammates, combined with a withdrawal from or reduced activity in group activities, may suggest that the player is suffering from depression. A machine learning algorithm trained on a large database of similar social data and associated mental health issues would be able to recognize complex patterns in the social data suggestive of various mental health issues. Analyses produced by mental health analysis module **4225** are output to machine learning algorithm **4226** as data for inclusion in determination of diagnoses by machine learning algorithm **4226**.

Machine learning algorithm **4226** is one or more machine learning algorithms configured to identify patterns in data. Machine learning algorithm **4226** is configured to make health-related diagnoses by identifying patterns in complex data. Machine learning algorithm **4225** may receive either data input to the various modules **4222-4225**, or the outputs of the various modules **4222-4225**, or a combination of the two. Based on this complex data set comprising many factors associated with a person's physical function and mental function as described above for the various modules **4222-4225**, machine learning algorithm **4226** makes one or more diagnoses regarding one or more aspects of the person's physical or mental function. Diagnoses may include any sort of evaluation and recommendation regarding physical function or mental function including both high and low functionality or ability such as evaluations of cognition, speech, auditory processing, vision processing, motor skills, emotions, and memory. Diagnoses may include, but are not limited to, diagnoses of medical conditions, suspicions or indications of disease, suspicions or indications of chronic physical conditions, suspicions or indications of chronic mental or neurological conditions, and recommendations for treatment, rehabilitation, or therapy of any of the above. Diagnoses may be combined into a composite functioning score spatial map showing ability in a plurality of functional areas, as further described below.

Machine learning algorithms excel at finding patterns in complex data or exploring the outcomes of large numbers of potential options. There are three primary categories of machine learning algorithms, supervised machine learning algorithms, unsupervised machine learning algorithms, and reinforcement machine learning algorithms. Supervised machine learning algorithms are trained to recognize patterns by training them with labeled training data. For example, a supervised machine learning algorithm may be fed pictures of oranges with the label "orange" and pictures of basketballs with the label basketball. The supervised machine learning algorithm will identify similarities (e.g., orange color, round shape, bumpy surface texture) and differences (e.g., black lines on basketball, regular dot pattern texture on basketball versus random texture on oranges) among the pictures to teach itself how to properly classify unlabeled pictures input after training. An unsupervised machine learning algorithm learns from the data itself by association, clustering, or dimensionality reduction, rather than having been pre-trained to discriminate between labeled input data. Unsupervised machine learning algorithms are ideal for identifying previously-unknown patterns within data. Reinforcement machine learning algorithms learn from repeated iterations of outcomes based on probabilities with successful outcomes being rewarded. Reinforcement machine learning algorithms are ideal for exploring large number of possible outcomes such as possible outcomes from different moves on a chess board. Within each primary category of machine learning algorithms, there are many different types or implementations of such algorithms (e.g., a non-exhaustive list of unsupervised machine learning algorithms includes k-means clustering algorithms, hierarchical clustering algorithms, anomaly detection algorithms, principle component analysis algorithms, and neural networks. The category and type of machine learning algorithm chosen will depend on many factors such as the type of problem to be solved (e.g., classification of objects versus exploration of possible outcomes), the need for insight into how the machine learning algorithm is making its decisions (most machine learning algorithms operate as black boxes wherein their decision-making is opaque to users), accuracy required of predictions, availability of labeled training data sets, and knowledge or lack of knowledge regarding patterns or expected patterns within the data.

FIG. **43** is a block diagram showing an exemplary architecture for an alerts and treatments module aspect of a cloud-based gaming platform with integrated healthcare diagnostics and treatment. Alerts and treatments module **4230** of this embodiment comprises a patient database **4221**, a brainwave entrainment manager **4231**, an alert generator **4232**, a physical activity routine generator **4233**, a mental activity routine generator **4234**, and a treatment control signal generator **4235**.

Alerts and treatments module **4230** receives and transmits data via a HIPPA-compliant security gateway **4210** which is configured to secure protected health information to, from, and within healthcare diagnostics and treatment module **4200** to prevent unauthorized access to protected health information of persons using the system. HIPPA-compliant security gateway **4210** uses information technology (IT) security best practices which include, but are not limited to, data encryption, access restrictions, permission controls, user authentications, and audit controls. HIPPA-compliant security gateway **4210** may use a HIPPA-compliant data health data tunnel **4102** which secures transmitted data end-to-end from game clients **4130** to healthcare diagnostics and treatment module **4200**. HIPPA-compliant data health data tunnel **4102** may, as one example, use a virtual private network (VPN) established between game clients **4130** and healthcare diagnostics and treatment module **4200**, which may, depending on configuration, run through frontend **4110**. HIPPA-compliant data health data tunnel **4102** may be a separate connection from the connections between game clients **4130** and frontend **4110** through which game data **4110** is transmitted, in which case game data **4110** may not need to be encrypted. In some embodiments, HIPPA-compliant data health data tunnel **4102** and game data **4101** may be the same connection, but with HIPPA-compliant data health data encrypted or otherwise secured within the connection. For simplicity, HIPPA-compliant data health data tunnel **4102** and game data **4101** connections are shown as being singular connections, but in practice those connections will be made separately to each individual game client.

Patient database **4221** stores all incoming and outgoing patient data including, but not limited to, health-related data, game data, and healthcare diagnoses. Healthcare related data may include patient data such as medical records, medical evaluations, diagnoses, physical and mental evaluations, and dual-task data comprising physical and mental data acquired during dual-task testing. Game data may include patient data acquired during game play or related to game play such as types of games played, game scores, time spent playing, in-game performance data such as reaction times to certain events, trends in gameplay such as improvements or degradation over time, and social game data such as audio recordings or chat logs between players during gameplay. Diagnoses may include any sort of evaluation and recommendation regarding physical function or mental function including both high and low functionality or ability such as evaluations of cognition, speech, auditory processing, vision processing, motor skills, emotions, and memory. Diagnoses may include, but are not limited to, diagnoses of medical conditions, suspicions or indications of disease, suspicions or indications of chronic physical conditions, suspicions or indications of chronic mental or neurological conditions, and recommendations for treatment, rehabilitation, or therapy of any of the above.

Brainwave entrainment manager **4231** functions in a manner similar to brainwave entrainment manager of other embodiments previously described (e.g., brainwave entrainment manager **200**, brainwave entrainment manager **1313**). Upon receipt of one or more diagnoses for a person, brainwave entrainment manager **4231** selects an appropriate brainwave entrainment routine, and passes the selection on to a treatment control signal generator **4235**, which generates the appropriate game instructions or controls to implement the brainwave entrainment routine selected within a current or future game.

Alert generator **4232** generates alerts for a player based on lists or databases of diagnoses and information relevant thereto. Upon receipt of one or more diagnoses for a person, alert generator retrieves relevant information about the diagnoses from one or more lists or databases, identifies issues of concern associated with each diagnosis, and generates alerts based on rule sets. The issues of concern may be things such as health warnings, side effect warnings, medication interaction warnings, physical exertion warnings, and the like. The issue of concern may include thresholds for action or generation of alerts such as warnings not to exceed a certain heartrate or time of exercise.

While not show in this diagram, alert generator **4232** may further generate alerts based on data such as that described above is being received by the various modules **4222-4225** of AI-assisted healthcare diagnostics module **4220** (dual-task data, in-game performance data, game-related social data). For example, alert generator **4232** may receive real-time data regarding a person's posture and balance and generate immediate in-game or on-screen warnings of an imminent fall. Such real-time warnings of falls are particularly useful in certain circumstances such as use of exercise machines my elderly persons who may be prone to falls or by persons wearing virtual reality (VR) headgear who are unable to visually perceive the real-world environment in which they are gaming.

Alerts may be in-game alerts such as pop-up windows or text on the screen or flashing of the screen or elements on the screen in a particular color or pattern (e.g., three flashes of the full screen in red to indicate serious alerts or in yellow to indicate moderate alerts). Alerts may be outside-of-game alerts such as via text or email, warning or advising of concerns identified while gaming. Outside-of-game alerts are appropriate where immediate action does not need to be taken, but the person should be advised of some information or concern.

Physical activity routine generator **4233** receives diagnoses and selects appropriate physical exercise routines for the person diagnosed to perform based on known or suspected associations between physical exercise and the diagnosis. For example, walking is considered to be an effective form of physical rehabilitation for patients with various chronic conditions such as low back pain, strokes, and peripheral artery disease. Walking can be prescribed at various speeds and lengths of time and can be adjusted based on feedback either from the patient or from analyses from a dual-task functional analysis system comprising an exercise device and sensors for providing feedback from physical activity. Likewise, cardiovascular exercise such as running and cycling can be an effective form of medical treatment for chronic diseases such as obesity, high blood pressure, heart disease, and diabetes. It can be used preventatively to reduce the risk of other diseases such as thrombosis. Studies have also shown that cardiovascular exercise may be an effective treatment for neurological diseases such as Alzheimer's disease, age-related memory loss, and age-related dementia. Strength training such as weight lifting resistance-based exercise machines can prevent and treat muscle loss and bone density loss, particularly in elderly patients. Using databases of medical knowledge comprising known and suspected uses of physical activity to prevent and/or treat diseases and medical conditions, physical activity routine generator **4233** generates a physical activity routine appropriate to the received diagnosis, and passes the selection on to a treatment control signal generator **4235**, which generates the appropriate game instructions or controls to implement the brainwave entrainment routine selected within a current or future game. In some embodiments, physical activity routine generator **4233** may be the primary task generator for a dual-task functional analysis system.

Mental activity routine generator **4234** receives diagnoses and selects appropriate mental activity routines for the person diagnosed to perform based on known or suspected associations between mental exercise and the diagnosis. Brain-training games such as memory games, reasoning games, math-related games, puzzles, concept-association games, knowledge games, and the like, have been shown to be an effective form of mental rehabilitation and treatment for patients with neurological conditions such as age-related dementia and rehabilitation from brain injuries incurred from accidents and strokes. Different types of brain games encourage thinking and engage different portions of the brain, so different brain games can be prescribed as part of a mental activity routine for targeting treatment of different types of brain injuries, different brain locations, or different types of brain deficiencies. Using databases of medical knowledge comprising known and suspected uses of brain games to prevent and/or treat diseases and medical conditions, mental activity routine generator **4234** generates a mental activity routine appropriate to the received diagnosis, and passes the selection on to a treatment control signal generator **4235**, which generates the appropriate game instructions or controls to implement the brainwave entrainment routine selected within a current or future game. In some embodiments, mental activity routine generator **4234** may be the associative activity task generator for a dual-task functional analysis system.

In various embodiments, functionality for implementing systems or methods of the present invention may be distributed among any number of client and/or server components. For example, various software modules may be implemented for performing various functions in connection with the present invention, and such modules may be variously implemented to run on server and/or client components.

The skilled person will be aware of a range of possible modifications of the various embodiments described above. Accordingly, the present invention is defined by the claims and their equivalents.

What is claimed is:

1. A system for treatment of neurological conditions, comprising:

a physical activity data capture device, configured to capture movement data associated with physical activity of persons exercising thereon;

a computing device comprising a processor and a memory;

a computer display configured to display outputs from a software application to persons undergoing treatment;

the software application, comprising a first plurality of programming instructions stored in a memory which, when operating on the processor, causes the computing device to:

receive a neurological condition of a person for treatment;

select a primary activity for performance by the person, the primary activity comprising a physical activity associated with activation of portions of the human brain associated with the neurological condition and a recommended level of intensity of the physical activity also associated with activation of portions of the human brain associated with the neurological condition;

select a secondary activity for performance by the person, the secondary activity being a mental activity to be engaged in by the person within a virtual reality environment controlled in whole or in part by the software application, the secondary activity also being associated with activation of portions of the human brain associated with the neurological condition;

operate the secondary activity within the virtual reality environment;

while the person engages in the primary task, record movement data associated with the physical activity using the physical activity data capture device;

convert the movement data into movements in the virtual reality environment;

display the virtual reality environment comprising the secondary activity and the movements to the person via the display;

determine from the movement data an actual level of intensity of the physical activity, and either:

provide feedback to the person via the display as to the actual level of intensity relative to the recommended level of intensity, such that the person can self-regulate the actual level of intensity to be similar to the recommended level of intensity; or

adjust the operation of the physical activity data capture device to cause the actual level of intensity to more closely correspond with the recommended level of intensity; and

record activity data associated with the person's engagement in the secondary activity within the virtual reality environment; and

a second plurality of programming instructions stored in the memory which, when operating on the processor, causes the computing device to:

capture and analyze the movement data while the secondary task is stopped;

capture and analyze the activity data while the primary activity is stopped;

restart operation of the primary activity;

capture and analyze the movement data and the activity data while the primary activity and secondary activity are performed simultaneously; and

calculate a composite functioning score for the person, the composite functioning score comprising an indication of the level of functionality of an aspect of the person's nervous system based on differences in performance between the separate performance of the primary activity and secondary activity relative to the combined performance of the primary activity and secondary activity; and

generate a neurological condition profile of the person comprising an indication of relative functioning of the aspect of the person's nervous system.

2. The system of claim **1**, wherein one or more primary tasks and one or more secondary activities are assigned to the person, a plurality of composite functioning scores are calculated comprising indications of the level of functionality of a plurality of aspects of the person's nervous system, and the neurological functioning profile comprises indications of relative functioning of the plurality of aspects of the person's nervous system.

3. The system of claim **1**, wherein the neurological functioning analyzer creates a composite functioning score spatial map based on the neurological functioning profile.

4. The system of claim **1**, wherein the software application changes the primary activity or secondary activity or selects a different primary activity or secondary activity based on changes to the person's composite function score while undergoing treatment.

5. The system of claim **1**, wherein the neurological functioning analyzer determines an effectiveness of the treatment of the neurological condition based on changes to the person's neurological condition profile over time.

6. The system of claim **1**, wherein the composite functioning score is compared to statistical composite functioning score data for a larger population to determine the person's relative position in the larger population with respect to the composite functioning score.

7. The system of claim **1**, wherein the primary activity is exercise on a treadmill, elliptical machine, or stationary bicycle.

8. The system of claim **1**, wherein the secondary activity involves overcoming a computer or video game-like challenge or a computer simulation of a life-like scenario or challenge.

9. The system of claim **8**, wherein the system selects the primary activity and secondary activity based on changes to the person's neurological condition profile over time that suggest improvements in the person's performance.

10. A method for treatment of neurological conditions, comprising the steps of:

a computer display configured to display outputs from a software application to persons undergoing treatment;

using a software application operating on a computing device comprising a processor and a memory to:

receive a neurological condition of a person for treatment;

select a primary activity for performance by the person, the primary activity comprising a physical activity associated with activation of portions of the human brain associated with the neurological condition and a recommended level of intensity of the physical activity also associated with activation of portions of the human brain associated with the neurological condition;

select a secondary activity for performance by the person, the secondary activity being a mental activity to be engaged in by the person within a virtual reality environment controlled in whole or in part by the software application, the secondary activity also being associated with activation of portions of the human brain associated with the neurological condition;

operate the secondary activity within the virtual reality environment;

while the person engages in the primary task, record movement data associated with the physical activity using a physical activity data capture device, configured to capture movement data associated with physical activity of persons exercising thereon;

convert the movement data into movements in the virtual reality environment;

display the virtual reality environment comprising the secondary activity and the movements to the person via a computer display configured to display outputs from the software application to the person undergoing treatment;

determine from the movement data an actual level of intensity of the physical activity, and either:

provide feedback to the person via the display as to the actual level of intensity relative to the recommended level of intensity, such that the person can self-regulate the actual level of intensity to be similar to the recommended level of intensity; or

adjust the operation of the physical activity data capture device to cause the actual level of intensity to more closely correspond with the recommended level of intensity; and

record activity data associated with the person's engagement in the secondary activity within the virtual reality environment; and using a neurological functioning analyzer operating on the computing device to:

capture and analyze the movement data while the secondary task is stopped;

capture and analyze the activity data while the primary activity is stopped;

restart operation of the primary activity;

capture and analyze the movement data and the activity data while the primary activity and secondary activity are performed simultaneously; and

calculate a composite functioning score for the person, the composite functioning score comprising an indication of the level of functionality of an aspect of the person's nervous system based on differences in performance between the separate performance of the primary activity and secondary activity relative to the combined performance of the primary activity and secondary activity; and

generate a neurological condition profile of the person comprising an indication of relative functioning of the aspect of the person's nervous system.

11. The method of claim **10**, wherein one or more primary tasks and one or more secondary activities are assigned to the person, a plurality of composite functioning scores are calculated comprising indications of the level of functionality of a plurality of aspects of the person's nervous system, and the neurological functioning profile comprises indications of relative functioning of the plurality of aspects of the person's nervous system.

12. The method of claim **10**, wherein the neurological functioning analyzer creates a composite functioning score spatial map based on the neurological functioning profile.

13. The method of claim **10**, wherein the software application changes the primary activity or secondary activity or selects a different primary activity or secondary activity based on changes to the person's composite function score while undergoing treatment.

14. The method of claim **10**, wherein the neurological functioning analyzer determines an effectiveness of the treatment of the neurological condition based on changes to the person's neurological condition profile over time.

15. The method of claim **10**, wherein the composite functioning score is compared to statistical composite functioning score data for a larger population to determine the person's relative position in the larger population with respect to the composite functioning score.

16. The method of claim **10**, wherein the primary activity is exercise on a treadmill, elliptical machine, or stationary bicycle.

17. The method of claim **10**, wherein the secondary activity involves overcoming a computer or video game-like challenge or a computer simulation of a life-like scenario or challenge.

18. The method of claim **17**, wherein the method selects the primary activity and secondary activity based on changes to the person's neurological condition profile over time that suggest improvements in the person's performance.

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